BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Characteristic of participants who take up screening tests for diabetes and lipid disorders: a systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-055764
Article Type:	Original research
Date Submitted by the Author:	22-Jul-2021
Complete List of Authors:	Ding, Hanyue; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care Huang, Junjie; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care Deng, Yunyang; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care Tin, Pamela; Our Hong Kong Foundation Wong, Martin Chi-Sang; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care; Peking University, School of Public Health Yeoh, EK; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care; The Chinese University of Hong Kong, Centre for Health Systems and Policy Research
Keywords:	PREVENTIVE MEDICINE, Lipid disorders < DIABETES & ENDOCRINOLOGY, General diabetes < DIABETES & ENDOCRINOLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Characteristic of participants who take up screening tests for diabetes and lipid disorders: a systematic review

Hanyue Ding[1], Junjie Huang[1], Yunyang Deng[1], Pamela Sze Pui Tin[2], Martin CS Wong[1,3,4], Eng-kiong Yeoh[1,5*]

1. JC School of Public Health and Primary Care, Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong Special Administrative Region, China
2. Our Hong Kong Foundation, Hong Kong Special Administrative Region, China
3. School of Public Health, Peking University, Beijing, China
4. School of Public Health, Chinese Academy of Medical Sciences and Peking Union Hospitals, Beijing, China
5. Centre for Health Systems and Policy Research, The Chinese University of Hong Kong, Hong Kong Special Administrative Region, China

*** Correspondence author**

Professor Eng-kiong Yeoh, GBS, OBE, JP

Director, Centre for Health Systems and Policy Research, JC School of Public Health and Primary Care, Faculty of Medicine, The Chinese University of Hong Kong

Email: yeoh_ek@cuhk.edu.hk; **Telephone number:** 2252-8403; **Fax number:** 2145-8517

Address: Room 202A, 2/F, School of Public Health, Prince of Wales Hospital

(Word count: 3805 words)

For peer review only

ABSTRACT

Objective: To perform a systematic review on the characteristics of participants who attended screening programs with blood glucose test, lipid profile, or a combination of them, respectively.

Methods: We followed the PRISMA guideline and searched the PubMed and Medline database for English literature from 1st Jan 2000 to 1st April 2020. Original observational studies that reported baseline characteristics of apparently healthy, adult screening participants for diabetes and lipid disorders were included in this review. We examined their sociodemographic characteristics, including age, gender, Body Mass Index (BMI), and lifestyle habits. The quality of the included articles was evaluated by the Appraisal of Cross-sectional Studies.

Results: A total of 33 articles involving 38 studies in 22 countries were included and analyzed in this systematic review. Overall, there was a higher participation rate among subjects who were female in all screening modalities (female vs. male: 46.6-63.9% vs. 36.1-53.4% in programs for diabetes screening; 48.8-58.4% vs. 41.6-51.2% lipid screening; 36.4-76.8% vs. 23.2-63.6% screening offering both). Compared with the BMI standard from the World Health Organization, participants in lipid screening had lower BMI. Furthermore, it is less likely for individuals of lower socioeconomic status to participate in diabetes or lipid screening in developed areas. Most studies reported a lower proportion of participants who were single, divorced, or widowed compared with participants who were married.

Conclusion: We identified individuals from lower socioeconomic groups who were less likely to take up programs for diabetes and/or lipid screening in developed areas. These populations are also likely to be at higher risk for non-communicable diseases. Future studies should investigate the barriers and facilitators of screening among non-

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

participants, where targeted interventions to enhance their screening uptake are warranted.

(Word count: 278 words)

Keywords: Diabetes Mellitus; Lipid Metabolism Disorders; Mass Screening; Population Characteristics; Systematic Review

Strengths and limitations of this study

- This is the first systematic review that presented the characteristics of screening participants in diabetes and lipid screening programs.
- Due to the descriptive manner, we cannot quantify the association between patients' characteristics and screening participation.
- We included articles that used different screening tests for diabetes, and this might pose challenges on comparison across studies.
- There are differences in the organization of different screening programs in terms of the tests offered, subsidy amount, and accessibility of the screening centers, and these could influence the screening uptake rate.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

Cardiovascular disease (CVD) represents the major cause of mortality, accounting for 17.9 million deaths or 31% of all global deaths in 2016.[1] It caused more than half of all deaths across the European region, and the death rates were higher in the Russian Federation, Bulgaria, and Romania than in other European region.[2,3] Most Asian countries had a higher age-adjusted mortality than Western countries.[4] Patients with CVD have a high risk of complications, including ischemic heart disease, stroke, and chronic kidney diseases.[5] The modifiable risk factors for CVD included smoking, inadequate physical activity, elevated blood pressure, increased body weight, increased plasma lipid (cholesterol, triglyceride), as well as diabetes.[6] Diabetes is not only the risk factor for CVD but is also associated with significant morbidity and mortality. About 1.6 million deaths were directly caused by diabetes mellitus (DM) in 2016. In addition, diabetes may lead to long-term complications, such as kidney failure, lower limb amputation, blindness, heart attacks, and stroke.[7,8] Worldwide, there were 108 million diabetes patients in 1980, and the number has increased to 422 million in 2014 [8]. The global prevalence of diabetes has been predicted to rise to 10.2% (578 million) by 2030 and 10.9% (700 million) by 2045.[9]

Globally, ischemic heart disease and stroke ranked the first and second causes of disability-adjusted life years (DALYs) in 2016, which were responsible for 12.8% of total DALYs lost.[10] Diabetes caused 65.7 million DALYs lost and 2.5% of the total global DALYs lost in 2016.[10] In addition, dyslipidemia also poses a serious threat to the health of populations. Overall, elevated cholesterol was estimated to contribute to 4.5% of total deaths and 2.0% of total DALYs.[11] Disease screening is regarded as an effective approach for early detection and prevention of diabetes and dyslipidemia. It

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

was demonstrated that cholesterol screening can save 14,300 quality-adjusted life years (QALYs) over 100,000 persons' lifetimes incurring a cost of $33,800 per QALY in US.[12] Diabetes screening has been shown to be more cost-effective for individuals aged 55 to 75 years as compared to younger subjects.[13]

According to international guidelines for screening **(Supplementary table 1)**, in the United States (US),[14] the United Kingdom,[15] Australia,[16] Canada,[17] and Singapore [18] cholesterol screening is recommended for adults, and the proposed age of onset for screening ranged from 35 to 45 years - or earlier for patients with CVD risk factors. Furthermore, Australia,[16] Canada,[19] and Singapore [20] suggested diabetes screening for average-risk adults above 40 years old, and the US recommended screening for adults aged 40-70 years who are overweight or obese.[21]

However, previous studies have shown that the uptake rate for diabetes and lipid screening was suboptimal. According to a cross-sectional study performed in the US between 2005 and 2012, the diabetes screening rates for "screening-recommended" and "screening-not-recommended" participants were 46.2% and 29.6% respectively.[22] Participation rates of the health examinations for CVD and diabetes ranged between 48% and 67% in Sweden.[23]. Although the attendance rate for National Health Service (NHS) health check in England quadrupled from 5.8% in 2010 to 30.1% in 2012, the uptake rates were still relatively low when compared with findings from other studies.[24] To our knowledge, there are no systematic reviews that explored the socioeconomic and demographic characteristics of participants who take up screening programs for diabetes and lipid disorders. Knowing their characteristics could help to target individuals who were less likely to receive screening and improve their

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

participation. These results may help to provide a basis for future studies to examine the enablers and barriers for diabetes and lipid screening. Therefore, we performed a systematic review of the existing literature to evaluate the characteristics of participants who received screening tests for diabetes and lipid disorders.

METHODS

Search strategy

We followed the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) statement.[25] We searched the PubMed and Medline databases for English literature from 1st Jan 2000 to 1st April 2020. The following keywords were used for the search: 1) AND 2) NOT 3): 1) "fasting plasma glucose test" OR "postprandial plasma glucose" OR "oral glucose tolerance test" OR "random plasma glucose test" OR "glycated hemoglobin" OR "lipid test" OR "cholesterol test" OR "triglyceride test" OR "lipid panel" OR "lipid profile" 2) "uptake" OR "participate" 3) "pediatric" OR "gestational".

Inclusion and exclusion criteria

Studies were included if they: 1) were cross-sectional or reported baseline characteristics of participants in cohort studies; 2) were conducted among adults without a known history of diabetes, prediabetes or CVD; 3) adopted finger stick or blood glucose (including fasting plasma glucose (FPG), oral glucose tolerance tests (OGTT), glycated hemoglobin (HbA1c), random plasma glucose (RBG)) or blood lipid tests as screening tools; and 4) reported participants' characteristics. The exclusion criteria included: 1) conference abstracts, systematic reviews, or studies published in languages other than English; 2) screening programs for pregnant women, the pediatric

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

population, and/or patients who were previously diagnosed with diabetes, prediabetes or CVD; 3) studies whose target population focused on obese or overweight subjects, people with a family history of diabetes or CVD; and 4) studies performed before 2000.

Data extraction and quality assessment

Two reviewers (HD and VL) independently searched and screened each title and abstract. Disagreements were solved by discussion. Once the title and abstract of each citation were assessed as eligible, two reviewers reviewed and appraised the full text. We used the Appraisal of Cross-sectional Studies (AXIS) to assess the quality of included articles, which consists of 20 questions and answers recorded "yes", "no" or "don't know".[26] AXIS does not involve a numerical scale that can be used to calculate to produce a quality assessment score. We extracted the following data from all included articles: the first author's name, publication year, country, region/city, project period, study type, subsidization, the age range of the population, number of participants; the screening tests used (finger prick, FPG, OGTT, HbA1c, RBG, fasting lipid profile, waist circumference, hip circumference, blood pressure and other relevant tests); participant characteristics: gender, age, body mass index (BMI), marital status, educational status, occupational status, other indicators of socioeconomic status (SES) (including Household wealth index, social deprivation index, and household income), family history, cigarette smoking, alcohol consumption, physical activity, and the reason of non-participation.

Statistical analysis

We summarized and descriptively reported the socioeconomics and demographic characteristics of the populations attending screening in the included studies. Based on

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

our findings in the initial literature review, we also selected gender as a subgroup to explore the difference in age and BMI of the study participants. Means ± standard deviations of these parameters were obtained from included studies. Figures were plotted using R software version x64 3.6.3 with ggplot package.

Patient and Public Involvement

It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

RESULTS

From the literature search, 6,407 citations were identified, of which 5,689 were from PubMed and 749 were from Medline **(Supplementary figure 1)**. There were 5,723 citations after the removal of duplicates. We retrieved 122 full-text articles to assess for eligibility after 5,536 citations were excluded during title or abstract screening with the pre-determined criteria. We excluded 89 articles which were duplicate studies (n=1); performed before the year 2000 (n=9); absence of full text or lack of sufficient data for extraction (n=22); presence of known diabetes, prediabetes or CVD (n=44); or target population whose subjects reported family history of diabetes, prediabetes or elevated BMI (n=13). Finally, we included 38 studies in 33 articles in the present systematic review.[27,28,37–46,29,47–56,30,57–59,31–36]

Regarding the AXIS for evaluating each included study, most studies presented with the following limitations: the sample size not being justified (23/33); absence of characterization and categorization of non-participants (29/33); a lack of control for non-response bias (30/33); and absence of description of non-participants (27/33).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Overall, all studies (33/33) met the following criteria: appropriate objectives and study design; appropriate risk factors and outcome variables; clear and sufficient description of methods; adequate description of the participants' characteristics; and internally consistent presentation of results (**Supplementary table 2**).

The characteristics of all eligible studies were presented in **Table 1**. Among these thirty-three manuscripts: fifteen were from Asia (China, Vietnam, Sri Lanka, Pakistan, Palestine, Thailand), seven were from Europe (Spain, England, Ireland, Netherlands, Lithuania, Sweden, Portugal), six were from Africa (Mauritius, Cameroon, Uganda, Nigeria, Tunisia, South African), four were from Americas (USA, Ecuador), and one was from Oceania (New Zealand). Ten studies were from developed countries, and the other twenty-three studies were from developing countries. According to screening items, eleven studies only implemented diabetes screening, four studies only conducted lipid tests, the rest of the eighteen studies performed both diabetes and lipid screening.

Diabetes Screening

Eleven studies adopted OGTT, HbA1c, FPG, or RBG to screening for diabetes.[29,32,59,40,44,47,48,50,51,53,55] OGTT and HbA1c were the commonly used modalities (in seven articles), followed by FPG (in five articles) and RBG (in one article). Two of these papers evaluated combinations of FPG, OGTT, and HbA1c. A total of four studies simultaneously took blood pressure during screening.[48,51,55,59] Except for three studies,[40,44,53] all the others included anthropometry measurements (body weight, height, waist and hip circumference). Nearly half of the studies were from developed countries (five out of eleven).[29,47,50,53,59] Among these eleven articles, one reported findings from workplace screening insured by employers,[29] two

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

were national health surveys and fully subsidized by government organizations,[47,59] five were funded by research institutes or funding agencies,[32,40,48,51,53] one was supported by a pharmaceutical company for free meters and strips,[50] and two did not mention their financial supports.[44,55] (**Table 1**)

The range of participation rate of female and male subjects in the diabetes screening programs was 46.6-63.9% and 36.1-53.4%, respectively. **(Supplementary figure 2)** More female participants participated in screening than males in sixteen studies. The mean age of the participants ranged from 38.5 to 48.0 years in all included studies. **(Figure 1)** The studies varied in terms of participant age, where the average age of most studies were more than 40 years. We extracted the national mean BMI of adults in various jurisdictions reported by the World Health Organization (WHO),[60] and compared these with the participants' average BMI. **(Figure 2)** The average BMI of participants tended to be higher than the national mean, in particular from studies performed in the Netherlands (26.1 kg/m^2 vs. 25.4 kg/m^2, $p<0.01$),[53] Thailand (24.3 kg/m^2 vs. 24.1 kg/m^2, $p=0.39$),[32] and the US (30.2 kg/m^2 vs. 28.8 kg/m^2, $p<0.01$).[59]

The sociodemographic information of the screening participants was shown in **Table 2**. For educational status, different studies adopted different classification methods for educational levels. Mayega *et al* reported that 39.6% of participants had a higher primary level, followed by a lower primary (21.9%), secondary (16.5%), and tertiary (4.5%) educational level in Uganda.[48] Zafar *et al* showed more than half participants had matriculation level or above (same as the undergraduate degree) (52.5%) and only 4.2% acquired conventional level (same as the postgraduate degree).[55] He also investigated the household income and found that 72.5% of the participants had

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

<30,000 PKR income (approximately 300 USD in 2014). Mainous *et al* reported the trend (2003, 2006, 2009, 2011) of the social deprivation index, which is a composite index based on income, employment, health deprivation and disability, education, skills and training, barriers to housing and services, crime and disorder, and living environment.[47] The most deprived quintile had the lowest participation percentages in all the years (16.5-17.7%), and the other four levels of less deprivation quantities presented similar proportions (18.1-23.1%). In addition, the proportions of single, divorced, or widowed status were also lower than that of married participants.[55]

Apart from age, gender, BMI, and sociodemographic information, we also extracted data on family history and lifestyle habits. **(Table 3)** A study from the Netherlands reported that around 70% of participants had a family history of type 2 diabetes,[53] whereas this uptake rate was under 40% in diabetes screening in Pakistan.[55] Of the two studies including smoking status, Sabir *et al* and Zafar *et al* reported low rates of participation among current smokers (9.7% and 13.4%, respectively).[51,55] Sabir *et al* also reported a very low proportion of screening uptake among alcohol drinkers (0.3%) in Nigeria.[51]

Lipid screening

A total of four studies offered lipid screening;[34,41,42,57] one cross-sectional screening survey was covered by research funding,[34] one study was workplace-based screening and insured by employers,[41] the other two studies were for national screening programs supported by government organizations.[42,57] Only one study was from a developed country.[41] **(Table 1)**

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

268 **Table 1 Characteristic of included studies**

Author	Published Year	Country	Region/City	Project Period	Study design	Subsidization
Only blood glucose tests						
Bali, V.	2018	USA		Jan 2012-Dec 2014	Workplace screening (Health Advocate)	Self-insured employers
Bumrerraj, S.	2012	Thailand	Khon Kaen	June-Dec 2009	Cross-sectional study	Research funding
Katulanda, G. W.	2019	Sri Lanka		Aug 2005-Sep 2006	Cross-sectional study	Research funding
Lin, S.	2014	China	Guangzhou	Oct 2010-Jan 2011	Cross-sectional survey	Not mentioned
Mainous, A. G.	2014	England		2003, 2006, 2009, 2011	Health Survey for England	Information Centre for Health and Social Care and the Department of Health
Mayega, R. W.	2014	Uganda	Iganga-Mayuge		Cross-sectional survey	Research funding
Rush, E.	2008	New Zealand	Waikato District	May 2004-Mar 2006	Te Wai o Rona Diabetes Prevention Strategy (Cohort)	Roche company and research funding
Sabir, A.	2013	Nigeria	Gumbi&Wamakko		Cross-sectional study	Research funding
Valkengoed, I. G.	2015	Netherlands	Hague	June-Dec 2009 April-Nov 2010	DHIAAN screening	Research funding
Zafar, J.	2016	Pakistan	Rawalpindi	May-Sep 2014	Cross-sectional survey	Not mentioned
Ziemer, D. C.	2010	USA		Jan 2005-Mar 2008	Third National Health and Nutrition Examination Survey	Centers for Disease Control and Prevention
Only blood lipid tests						
Deng, B.	2012	China	Yunnan	July-Oct 2010	Cross-sectional survey	Research funding
Koyama, A. K.	2018	USA		Jan 2012-Dec 2014	Workplace screening (Health Advocate)	Self-insured employers
Kutkiene, S.	2018	Lithuania		2009-2016	Lithuanian High Cardiovascular Risk primary prevention programme	Ministry of Health
Zhao, W. H.	2007	China		2002	Chinese national nutrition and health survey	Chinese Center for Disease Control and Prevention
Combination tests of blood glucose and lipid						
Ali, I.	2019	Palestine	Nablus municipality	Aug 2017-Feb 2018	Cross-sectional study	Not mentioned
Andersson, S.	2016	Sweden		2013	Qualitative and quantitative study	Not mentioned
Bao, Y.	2010	China	Shanghai	May 2007-Aug 2008	Cross-sectional survey	Research funding
Belfki, H.	2013	Tunisia		Apr 2004-Sep 2005	Transition and Health Impact in North Africa project	Research funding
Cuong, T. Q.	2007	Vietnam	Ho Chi Minh City	Mar-Apr 2004	Cross-sectional survey	Not mentioned
Falguera, M.	2020	Spain	Mollerussa	Mar 2011-July 2014	Cohort study	Research funding
Gao, W.	2008	China	Qingdao	Apr-July 2002	Cross-sectional survey	Research funding
Hare, M. J.	2013	Mauritius	Main island& Rodrigues		Mauritius and Rodrigues Non-Communicable Disease Surveys	Ministry of Health and Quality of Life, Mauritius
Hidalgo, L. A.	2006	Ecuador		Dec 2011-June 2012	Cross-sectional study	Research funding
Hwu, C. M.	2008	China	Taiwan		Cross-sectional study	Research funding
Li, L. J.	2012	China	Hainan	May 2010-Aug 2011	Health check up	Not mentioned
Lissock, C. N.	2011	Cameroon	Ngambe&Littoral		Cross-sectional study	Not mentioned
Liu, Y.	2016	China	Changchun	2011-2012	Cohort baseline survey	Not mentioned
Nunes, B.	2019	Portugal		2015	Portuguese National Health Examination Survey	Organizations of Ministry of Health
Sinnott, M.	2015	Ireland	Dublin	Jan 2009-Dec 2012	The Diabetes Mellitus and Vascular health initiative (Cohort study)	Private health insurance provider (for members)
Wang, S.	2020	China	Shanghai	Mar-Aug 2010	Cohort baseline survey	Research funding
Zemlin, A. E.	2015	South African	Cape town	2008	Cross-sectional survey	Research funding
Zhou, X.	2018	China	Shanghai	Jan-July 2013	Cohort baseline survey	Research funding

269
270

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1 Characteristic of included studies (Continued)

Author	Age range	Participants number	Test										
			FPG	OGTT	HbA1c	RBG	LP	BP	Weight	Height	WC	HC	Others
Only blood glucose tests													
Bali, V.		22,790	√	-	√	-	-	-	√	√	-	-	-
Bumrerraj, S.		874	-	√	-	-	-	-	√	√	√	-	-
Katulanda, G. W.		4,014	-	√	-	-	-	-	-	-	-	-	-
Lin, S.	>=50	1,312	√	√	√	-	-	-	-	-	-	-	-
Mainous, A. G.	>=16	20,139	-	-	√	-	-	-	√	√	-	-	-
Mayega, R. W.	35-60	795	√	-	√	-	-	√	√	√	-	-	-
Rush, E.	>=28	5,264	-	√	-	-	-	-	√	√	√	-	Point-of-care testing
Sabir, A.		393	√	√	-	√	-	√	√	√	√	√	-
Valkengoed, I. G.	18-60	1,092	-	√	√	-	-	-	-	-	-	-	-
Zafar, J.	18-65	404	-	-	√	-	-	√	√	√	√	√	-
Ziemer, D. C.	18-87	1,581	√	√	√	-	-	√	√	√	√	-	-
Only blood lipid tests													
Deng, B.	22-84	1,415	-	-	-	-	√	-	√	√	√	√	Food frequency
Koyama, A. K.		18,993	-	-	-	-	√	-	√	√	-	-	-
Kutkiene, S.	men 40-54 women 50-64	92,373	-	-	-	-	√	-	-	-	-	-	--
Zhao, W. H.	>=18	49,252	-	-	-	-	√	-	-	-	-	-	-
Combination tests of blood glucose and lipid													
Ali, I.	24-60	140	√	-	-	-	√	√	√	√	-	-	--
Andersson, S.	30-75	43	-	√	√	-	√	√	√	√	√	-	-
Bao, Y.	20-79	4,886	√	√	√	-	√	√	√	√	√	-	-
Belfki, H.	35-74	4,654	√	-	-	-	√	√	√	√	√	-	--
Cuong, T. Q.	20-60	1,488	√	-	-	-	√	√	√	√	√	√	Skinfold thicknesses
Falguera, M.	>=25	583	√	-	√	-	√	√	√	√	√	-	Renal function
Gao, W.	20-74	1,856	-	√	-	-	√	√	√	√	√	√	-
Hare, M. J.	19-78	6,701	-	√	√	-	√	√	√	√	√	√	-
Hidalgo, L. A.	40-65	204	√	-	-	-	√	√	√	√	√	-	-
Hwu, C. M.		49	√	√	-	-	√	√	√	√	√	-	-
Li, L. J.	27-91	1,337	√	-	√	-	√	√	√	√	-	-	Uric acid/Liver Function
Lissock, C. N.	>=18	452	√	-	-	-	√	√	√	√	√	√	Resting heart rate
Liu, Y.	>=40	7,611	√	√	√	-	√	√	√	√	√	√	-
Nunes, B.	25-74	4,911	-	-	√	-	√	√	√	√	√	√	Blood count
Sinnott, M.	45-75	29,144	√	-	-	-	√	√	√	√	√	√	-
Wang, S.	>=40	9,389	√	√	√	-	√	√	√	√	√	-	-
Zemlin, A. E.		667	√	√	√	-	√	√	√	√	√	√	-
Zhou, X.	15-98	7,909	√	√	√	-	√	√	√	√	√	-	Heart rate

FPG: fasting plasma glucose, OGTT: oral glucose tolerance tests, HbA1c: glycated hemoglobin, RGB: random plasma glucose, LP: lipid screening, BP: blood pressure, WC: waist circumference, HC: hip circumference.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2 Sociodemographic characteristics of participants in all included studies

Author	Marital status	Educational status	Occupational status	Others
Blood glucose tests only				
Mainous, A. G.	-	-	-	Social deprivation index (2003, 2006, 2009, 2011) Quintile 1 21.6% 20.6% 18.9% 20.1% Quintile 2 20.8% 20.2% 22.7% 23.1% Quintile 3 20.7% 22.3% 21.6% 21.0% Quintile 4 19.8 % 20.4% 20.3% 18.1% Quintile 5 17.0% 16.5% 16.5% 17.7%
Mayega, R. W.	-	None 139 (17.5%) Lower primary 174 (21.9%) Higher primary 315 (39.6%) Secondary 131 (16.5%) Tertiary 36 (4.5%)	Subsistence farmers 500 (62.9%) Traders 164 (20.6%) Formal/salaried 39 (4.9%) Mechanics 92 (11.6%)	-
Valkengoed, I. G.	-	Primary or less 14.7% 16.5% Secondary 11.7% 13.6% Lower vocational 56.8% 53.5%[c] Higher vocational 16.8% 16.5%[c]	Paid work (OGTT, HbA1c) 74.1% 71.5%	-
Zafar, J.	Married 322 (79.7%) Unmarried 61 (15.1%) Widow/Divorced 21 (5.2%)	Illiterate 55 (13.6%) Primary (1-5 grade) 59 (14.6%)[a] Middle (6-9 grade) 61 (15.1%)[b] > Matric 212 (52.5%)[c] Conventional 17 (4.2%)[c]	Govt. Employee 61 (15.1%) Private Employee 61 (15.1%) Self-Employee 80 (19.8%) Un-employed 84 (20.8%) Laborer 12 (3.0%) Any other 106 (26.2%)	Income (PKR) <10000 118 (29.2%) 10,000 - 30,000 175 (43.3%) 31000 - 50,000 72 (17.8%) > 50,000 39 (9.7%)
Plasma lipid tests only				
Deng, B.	Single 94 (6.6%) Married 1,306 (92.3%) Divorced 8 (0.6%) Widow(er) 7 (0.5%)	Never 21 (1.5%) Elementary school 116 (8.2%)[a] Junior-middle school 348 (24.6%)[b] Senior-high school 386 (27.3%)[b] Higher 544 (38.4%)[c]	-	-
Koyama, A. K.	-	Education, median (IQR) Less than high school 9.5% (5.7%-15.1%) High school or equivalent[b] 27.0% (19.5%-33.9%) Some college[c] 29.1% (24.2%-33.5%) College degree[c] 29.2% (19.2%-43.2%)	-	Median income, median (IQR) US$57,622 (US$45,161-US$75,313)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2 Sociodemographic characteristics of participants in all included studies (Continued)

Author	Marital status	Educational status	Occupational status	Others
Programs offering both diabetes and lipid tests				
Belfki, H.	Single 117 (2.5%) Married 4,035 (86.7%) Widowed/divorced 502 (10.8%)	Illiterate 2,041(43.9%) Low (<=6yrs) 1,552 (33.3%)[a] Intermediate(7-13yrs) 805 (17.3%)[b] Higher(>=14yrs) 238 (5.1%)[c]	No working/retired 2,479 (56.6%) Employee/worker 1,205 (27.5%) Intermediate 202 (4.6%) Upper 494 (11.3%)	-
Cuong, T. Q.	-	No schooling (1.8%) Primary school (16.3%)[a] Junior high school (33.3%)[b] Senior high school (33.3%)[b] College/University (15.3%)[c]	Teacher, Professional (10.2%) Government officers (14.9%) Small business, Skilled workers (17.9%) Laborers, street or home traders (24.3%) Retired/home maker/students (21.3%) Others (8.5%) No Job (2.9%)	Household wealth index (Male, female, total) Lowest 19.9% 19.9% NA Second 19.8% 20.1% NA Middle 20.5% 19.5% NA Fourth 21.2% 18.8% NA Highest 18.6% 21.7% NA
Falguera, M.	-	High level (>=secondary high school education) 421 (72.2%)	-	-
Hare, M. J.	-	Primary or none 3,317 (49.8%) Secondary 2,832 (42.5%) Tertiary 515 (7.7%)	-	-
Hidalgo, L. A.	Married 93 (45.6%) No 111 (54.4%)	0–6 73 (35.8%)[a] 7–12 68 (33.3%)[b] 13 63 (30.9%)[c]	-	-
Wang, S.	-	High school education or more 6,037 (64.6%)	-	-

[a] same as the primary education, b same as secondary education, c same as tertiary education; OGTT: oral glucose tolerance tests, HbA1c: glycated hemoglobin.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 3 Family history and lifestyle habits and of participants

Author	Family history	Smoking	Drinking	Physical activity
Blood glucose tests only				
Sabir, A.		Total 38 (9.7%) Male 38 (18.1%) Female 0 (0%)	Total 1 (0.3%) Male 1 (0.5%) Female 0 (0%)	-
Valkengoed, I. G.	Type 2 diabetes mellitus 74.8% (75-g OGTT group) 68.4% (HbA1c group)			
Zafar, J.	Diabetes Parental history 161 (39.9%) Siblings history 88 (21.8%)	Smokers 54 (13.4%)	-	Exercise 105 (26.0%)
Plasma lipid tests only				
Deng, B.		Current smoking 476 (33.6%)	Current drinking 557 (39.4%)	>2/week /at least 30 mins 806 (57.0%)
Programs offering both diabetes and lipid tests				
Ali, I.	Dyslipidemia 27.9%	Smokers 37.1%	-	-
Andersson, S.		Smoking 19%	-	-
Belfki, H.	Cardiovascular disease 119 (2.6%)	Never 3,246 (71.4%) Smoker 1,301 (28.6%)	-	-
Cuong, T. Q.		Non/ex-smoker (69.3%) Current smoker (30.7%)	-	-
Falguera, M.	Diabetes 180 (30.9%)	Current smoker 148 (25.4%)	-	394 (67.6%)
Hare, M. J.		1,417 (21.2%)	None 3,217 (49.7%) Moderate 2,728 (42.1%) Excessive 531 (8.2%)	Sedentary 694 (10.6%) Insufficient (<150 min/week) 1,883 (28.8%) Sufficient (>=150 min/week) 3,951 (60.5%)
Hidalgo, L. A.		-	-	Sedentary Yes 69 (33.8%) No 135 (66.2%)
Sinnott, M.	Diabetes 9,301 (31.9%)	Ever smoked 11,648 (40.0%)	23,245 (79.8%)	>=5days/week 10,343 (35.5%)
Wang, S.	Diabetes 884 (9.4%)	Current smoking 1,926 (21.2%)	958 (10.5%)	>=600 MET-min/week 6,670 (71.2%)
Zhou, X.	Diabetes 847 (10.71%)	1,746 (22.1%)	1,079 (13.64%)	-

OGTT: oral glucose tolerance tests, HbA1c: glycated hemoglobin, MET: metabolic equivalent of task.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In these four articles, two articles reported body weight and height,[34,41] and one study stated both waist and hip circumference.[34] For participants, the proportion of females ranged from 48.8% to 58.4%. **(Supplementary figure 2)** The mean age of the screening participants was older than 40 years. **(Figure 1)** Turning to mean BMI, however, both male and female subjects were lower than the national mean BMI reported by WHO (males: 23.8 kg/m^2 vs. 24.2 kg/m^2, $p<0.01$; females: 22.3 kg/m^2 vs. 23.6 kg/m^2, $p<0.01$). **(Figure 2)** Studies from Deng *et al* and Koyama *et al* reported the highest proportion of individuals having a higher educational level in China and the US (38.4% and 58.3%, respectively).[34,41] **(Table 2)**

Deng and colleagues reported the smoking, drinking, and physical activity habits of their lipid screening participants in China.[34] A third of them were current smokers. Moreover, 39.4% of subjects reported that they were current drinkers. For physical activity, 57.0% of them took exercise or participated in physical labor activities more than twice per week for at least 30 minutes each. **(Table 3)**

Participants receiving both diabetes and lipid screening

The papers that presented the characteristics of screening participants in programs that offered a screening of both diabetes and lipid profiles were listed in **Table 1**. Fourteen of these papers adopted FPG to evaluate blood glucose.[27,30,52,54,56,58,31,33,35,38,39,43,45,46] Ten studies used HbA1c,[28,30,35,37,43,46,49,54,56,58], nine studies used OGTT,[28,30,36,37,39,46,54,56,58] while five of the included studies offered all three tests.[30,46,54,56,58] Only four studies were from developed countries.[28,35,49,52] Two were organized by the government,[37,49] and one was offered by a private health

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

insurance provider for their members.[52]

For comparison of diabetes and lipid screening participation between males and females, only one study reported having more male than female subjects joining the screening program.[43] **(Supplementary figure 2)** Five studies indicated that male participants were older than females,[45,46,52,56] while the other two studies reported female participants being older.[30,31] **(Figure 1)** Turning to the mean BMI of participants, most of the studies reported a higher number than the national BMI levels, including those conducted in China (24.7 kg/m^2 vs. 23.9 kg/m^2, p<0.0;[58] 25.1 kg/m^2 vs. 23.9 kg/m^2, p<0.01;[54] 24.9 kg/m^2 vs. 23.9 kg/m^2, p<0.01;[46] 24.8 kg/m^2 vs. 23.9 kg/m^2, p<0.01;[43] 25.4 kg/m^2 vs. 23.9 kg/m^2, p<0.01);[39] South Africa (29.1 kg/m^2 vs. 27.3 kg/m^2, p<0.01);[56] Sweden (29.0 kg/m^2 vs. 25.8 kg/m^2, p<0.01);[28] and Tunisia (26.8 kg/m^2 vs. 27.8 kg/m^2, p<0.01).[31] **(Figure 2)** Overall, females had higher BMI levels than their national average as reported in most studies, whilst only two studies showed a higher BMI among male participants when compared with the WHO's report.[45,46] Some other screening tests were also included in other literature, such as skinfold thicknesses,[33] renal function,[35] uric acid,[43] liver function,[43] blood count,[49] and heart rate.[45,58]

For educational status, we found a substantial difference in participants' educational status across different studies. Belfki *et al* showed that the 77.2% of participants had a low educational level in Tunisia.[31] However, Wang and colleagues reported that around 65% of the individuals had an educational level at higher school or above in Shanghai.[54] For occupational status, Belfki *et al* indicated more than half (56.6%) of the participants were retired or had no current jobs.[31] The corresponding proportion

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

was only 24.2% in Cuong's study.[33] Cuong and colleagues also measured the household wealth index of participants and showed similar proportions among males (18.6-21.2%) and female participants (18.8-21.7%) across all levels of the index.[33] **(Table 2)**

Of the six studies that reported family history, one study described that 27.9% of the participants had a family history of dyslipidemia,[27] another study indicated the proportion of a family history of CVD was 2.6%,[31] and the other four studies mentioned that 9.4-31.9% of the participants with a family history of diabetes joined the screening programs.[35,52,54,58] **(Table 3)**

Of the ten studies with information on the status of smoking, three reported high rates of current smoking,[27,31,33] which were 37.1%, 28.6%, and 30.7%, respectively. Four studies stated the participants' alcohol consumption, one reported a high rate (79.8%),[52] compared with that reported in two studies (10.5-13.64%).[54,58] Among the five studies which included physical activity, three of them reported that more than half of the participants performed adequate physical activity (60.5-71.2%).[35,37,54] **(Table 3)**

The reasons for non-participation

Four studies mentioned the reasons of non-participation. Two studies on diabetes screening assessed reasons for non-participation. Mayega *et al* reported 5.4% of subjects refused to participate due to travel distance and 4.2% of non-participants declined the blood tests.[48] Valkengoed *et al* described that "no time" and "no interest" were the major reasons among prospective eligible participants.[53] Regarding lipid

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

screening, Deng *et al* explained that 1.0% of subjects declined screening tests due to time constraints.[34] Nunes and colleagues showed that 43.9% of individuals participated in the first Portuguese National Health Examination Survey involving both diabetes and lipid screening.[49] Furthermore, they found that work-related issues (26.6%) and lack of time (26.6%) were the most frequently mentioned reasons of not joining.

DISCUSSION

This review provided socioeconomic and demographic characteristics of participants who received screening tests for diabetes and lipid disorders, including age, gender, BMI, sociodemographic status, family history, and lifestyle habits. We found a higher proportion of females among the screening participants when compared with the male gender in most studies, irrespective of the tests offered. Participants who joined diabetes screening tended to have higher BMI, whilst those who received lipid screening tended to have lower BMI. Regarding screening programs offering both diabetes and lipid tests, female participants had higher BMI and male had lower BMI than the national means reported by the WHO.

Sargeant and colleagues reported male gender was significantly associated with lower uptake of RBG testing, which is consistent with our findings on diabetes screening.[61] A serial study of cross-sectional data (1992-2008) evaluated the utilization of preventive health assessments in the UK, and found that the participation rate of lipid tests increased with age, but they reported an absence of association between gender and participation in lipid tests.[62] They also showed that smokers had significantly lower uptake of lipid tests. In our review, around one-third of participants were current

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

smokers in lipid tests and programs offering both diabetes and lipid tests. A longitudinal study indicated that current smokers had limited health literacy and lower cognitive ability than ever smokers,[63] which may explain their lower screening uptake rates. Interventions to address health literacy of prospective participants are needed to improve the screening participation, which at the same time could also help to change their lifestyle habits for prevention of chronic diseases.[64]

A postal survey investigated the participation rate of women's health check-ups, including identification of CVD risk factors, as well as examinations for type 2 diabetes and kidney disease in Germany in 2004.[65] Its adjusted model showed that single, divorced, separated or widowed women (OR=0.63, 95% CI: 0.57-0.71) were less likely to receive a preventive health check-up when compared with married women. The proportions of single, divorced, or widowed status were also lower than that of married participants in our review, except for a study performed in Italy. This may be due to peer effects where family members may exert a motivational influence for chronic diseases screening. It is well recognized that socioeconomic status (SES) influenced disease risk, health behavior, and health care of individuals. These health inequities have been affirmed by the WHO.[66] Researchers in Germany found that there was a significant association between women with lower SES (estimated based on educational status, occupational position and household income) (OR=0.82, 95% CI: 0.74-0.92) and lower participation rate of health screening attendance.[65] This is consistent with our finding that health inequities might exist as the most deprived population had the lowest participation rates in England, where only blood glucose tests were provided. Meanwhile, participants of studies in Spain and China (Shanghai) had higher educational status, whereas studies in Tunisia, Mauritius, and Ecuador recruited

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

more participants with lower educational level. Health inequities are more likely to be found in developed areas,[67] yet they also exist in developing countries due to differential health literacy and access to primary care services. For the non-participation of screening programs among those with lower educational levels, health literacy plays an important role in screening participation. The American Heart Association highlighted that low health literacy is a barrier to healthcare screening and basic services for diabetes and CVD.[68] A systematic review included studies that evaluated the effectiveness of health literacy interventions using pre- and post-program assessment, and concluded their positive impact on enhancing screening uptake.[64]

This is the first systematic review that presented the characteristics of screening participants in diabetes and lipid screening programs. The findings inform population groups where interventions to enhance screening uptake should be targeted. Nevertheless, there are several limitations. Firstly, this is a systematic review of literature presented in a descriptive manner. Therefore, we cannot quantify the association between patients' characteristics and screening participation. In addition, we included articles that used different screening tests for diabetes, and this might pose challenges on comparison across studies. The American Diabetes Association proposed that FPG is the best screening test and also a component of diagnostic test for diabetes.[69] FPG is more convenient and acceptable to patients since it is easier and faster to perform, and is less expensive than OGTT. There are differences in the organization of different screening programs in terms of the tests offered, subsidy amount, and accessibility of the screening centers, and these could influence the screening uptake rate. In addition, the prevalence of risk factors in screening participants who take up the tests might be different from those who do not.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Overall, the present study systematically summarized the characteristics of screening participants in diabetes and lipid screening programs. A higher proportion of female participants was found in diabetes and lipid screening programs than males in most studies. For diabetes screening, participants tended to have higher BMI, while it was the opposite for subjects who participated in lipid screening. Meanwhile, females tended to have higher BMI and males tended to have lower BMI in screening programs offering both diabetes and lipid tests. Participants with lower SES were less likely to undergo screening in developed areas. Around one-third of participants were current smokers in lipid screening and a combination of lipid and diabetes screening programs. The proportion of participants using alcohol and performing regular physical activity varied substantially among included studies. Our findings could inform future studies to investigate the enablers and barriers of screening among participants and non-participants based on their characteristics, and also interventions to enhance their screening uptake.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Acknowledgments: We appreciate the invaluable contributions of Ms. Xiao Chen and Ms. Veeleah Lok.

Competing Interests: Ms. Pamela Sze Pui Tin is from Our Hong Kong Foundation. The other authors declare no competing interests.

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability: All data generated or analysed during this study are included in this published article (and its Supplementary Information files).

Ethics declarations: N/A

Author contributions: HD: Formal analysis, Investigation, Data Curation, Writing - Original Draft; JH: Writing - Review & Editing; YD: Validation, Writing - Review & Editing; PT: Writing - Review & Editing; MCSW: Methodology, Supervision, Project administration; EKY: Conceptualization, Supervision.

FIGURE LEGENDS

Figure 1 The mean age of screening participants

Figure 2 The Body Mass Index (BMI) of screening participants

We compared the national mean BMI of adults, extracted from the World Health Organization (WHO) report, with the participants' average BMI.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

REFERENCE

1 World Health Organization. Cardiovascular diseases (CVDs). World Heal. Organ. 2017.https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds) (accessed 8 May 2020).

2 World Health Organization. WHO/Europe | Cardiovascular diseases - Data and statistics. 2020.http://www.euro.who.int/en/health-topics/noncommunicable-diseases/cardiovascular-diseases/data-and-statistics (accessed 13 May 2020).

3 American Heart Association. Statistical Fact Sheet - International Cardiovascular Disease Statistics. 2004.http://www.sld.cu/galerias/pdf/servicios/hta/international_cardiovascular_disease_statistics.pdf (accessed 13 May 2020).

4 Ohira T, Iso H. Cardiovascular disease epidemiology in Asia - An overview. *Circ J* 2013;**77**:1646–52. doi:10.1253/circj.CJ-13-0702

5 Mendis S, Puska P, Norrving B. Global atlas on cardiovascular disease prevention and control. 2011.

6 Francula-Zaninovic S, Nola IA. Management of Measurable Variable Cardiovascular Disease' Risk Factors. *Curr Cardiol Rev* 2018;**14**:153–63. doi:10.2174/1573403x14666180222102312

7 American Diabetes Association. Diagnosis and classification of diabetes mellitus. *Diabetes Care* 2014;**37**:S81–90. doi:10.2337/dc14-S081

8 World Health Organization. Diabetes. 2020.https://www.who.int/news-room/fact-sheets/detail/diabetes (accessed 11 May 2020).

9 Saeedi P, Petersohn I, Salpea P, *et al.* Global and regional diabetes prevalence estimates for 2019 and projections for 2030 and 2045: Results from the International Diabetes Federation Diabetes Atlas, 9th edition. *Diabetes Res Clin Pract* 2019;**157**:107843. doi:10.1016/j.diabres.2019.107843

10 World Health Organization. Disease burden and mortality estimates. 2020.https://www.who.int/healthinfo/global_burden_disease/estimates/en/index1.html (accessed 9 Jun 2020).

11 World Health Organization. WHO | Raised cholesterol. 2020.https://www.who.int/gho/ncd/risk_factors/cholesterol_text/en/ (accessed 7 Jul 2020).

12 Dehmer SP, Maciosek M V., LaFrance AB, *et al.* Health benefits and cost-effectiveness of asymptomatic screening for hypertension and high cholesterol and aspirin counseling for primary prevention. *Ann Fam Med* 2017;**15**:23–36. doi:10.1370/afm.2015

13 Hoerger TJ, Harris R, Hicks KA, *et al.* Screening for Type 2 Diabetes Mellitus: A Cost-Effectiveness Analysis. *Ann Intern Med* 2004;**140**:689. doi:10.7326/0003-4819-140-9-200405040-00008

14 Bibbins-Domingo K, Grossman DC, Curry SJ, *et al.* Statin use for the primary prevention of cardiovascular disease in adults: US preventive services task force recommendation statement. JAMA - J. Am. Med. Assoc. 2016;**316**:1997–2007. doi:10.1001/jama.2016.15450

15 NICE guideline. Cardiovascular disease: risk assessment and reduction, including lipid modification. Natl. Inst. Heal. Care Excell. 2016.https://www.nice.org.uk/guidance/cg181/resources/surveillance-report-2018-cardiovascular-disease-risk-assessment-and-reduction-including-lipid-modification-2014-nice-guideline-cg181-4724759773/chapter/Surveillance-decision?tab=evidence (accessed 7 Jul 2020).

16 The Royal Australian College of General Practitioners. Guidelines for preventive activities in general practice 9th edition. 2018.https://www.racgp.org.au/download/Documents/Guidelines/Redbook9/17048-Red-Book-9th-Edition.pdf (accessed 9 Jun 2020).

17 Allan GM, Lindblad AJ, Comeau A, *et al.* Simplified lipid guidelines: Prevention and management of cardiovascular disease in primary care. *Can Fam Physician* 2015;**61**:857–67, e439-50.

18 Ministry of Health. LIPIDS EXECUTIVE SUMMARY. 2016.https://www.moh.gov.sg/docs/librariesprovider4/guidelines/moh-lipids-cpg---executive-summary.pdf (accessed 10 Jun 2020).

19 Ekoe JM, Goldenberg R, Katz P. Screening for Diabetes in Adults. *Can J Diabetes* 2018;**42**:S16–9. doi:10.1016/j.jcjd.2017.10.004

20 Goh SY, Ang S Bin, Bee YM, *et al.* Ministry of Health clinical practice guidelines: Diabetes mellitus. *Singapore Med J* 2014;**55**:334–47. doi:10.11622/smedj.2014079

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

21 Siu AL. Screening for abnormal blood glucose and type 2 diabetes mellitus: U.S. preventive services task force recommendation statement. Ann. Intern. Med. 2015;**163**:861–8. doi:10.7326/M15-2345

22 Kiefer MM, Silverman JB, Young BA, *et al.* National Patterns in Diabetes Screening: Data from the National Health and Nutrition Examination Survey (NHANES) 2005–2012. *J Gen Intern Med* 2015;**30**:612–8. doi:10.1007/s11606-014-3147-8

23 Norberg M, Wall S, Boman K, *et al.* The Västerbotten Intervention Programme: background, design and implications. *Glob Health Action* 2010;**3**:4643. doi:10.3402/gha.v3i0.4643

24 Robson J, Dostal I, Sheikh A, *et al.* The NHS Health Check in England: an evaluation of the first 4 years. doi:10.1136/bmjopen-2015-008840

25 PRISMA. PRISMA - Transparent Reporting of Systematic Reviews and Meta-analyses. Checklist. 2009;:1–2.http://www.prisma-statement.org/documents/PRISMA 2009 checklist.doc

26 Downes MJ, Brennan ML, Williams HC, *et al.* Development of a critical appraisal tool to assess the quality of cross-sectional studies (AXIS). *BMJ Open* 2016;**6**. doi:10.1136/bmjopen-2016-011458

27 Ali I, Kharma A, Samara M, *et al.* Prevalence of Dyslipidemia in Undiagnosed Palestinian Men: A Cross-Sectional Study. *J Lipids* 2019;**2019**:1–6. doi:10.1155/2019/3473042

28 Andersson S, Karlsson V, Bennet L, *et al.* Attitudes Regarding Participation in a Diabetes Screening Test among an Assyrian Immigrant Population in Sweden. *Nurs Res Pract* 2016;**2016**:1–7. doi:10.1155/2016/1504530

29 Bali V, Yermilov I, Koyama A, *et al.* Secondary prevention of diabetes through workplace health screening. *Occup Med (Chic Ill)* 2018;**68**:610–6. doi:10.1093/OCCMED/KQY138

30 Bao Y, Ma X, Li H, *et al.* Glycated haemoglobin A1c for diagnosing diabetes in Chinese population: Cross sectional epidemiological survey. *BMJ* 2010;**340**:1178. doi:10.1136/bmj.c2249

31 Belfki H, Ben Ali S, Aounallah-Skhiri H, *et al.* Prevalence and determinants of the metabolic syndrome among Tunisian adults: results of the Transition and Health Impact in North Africa (TAHINA) project. *Public Heal Nutr* 2013;**16**:582–90. doi:10.1017/s1368980012003291

32 Bumrerraj S, Kaczorowski J, Kessomboon P, *et al.* Diagnostic performance of 2 h postprandial capillary and venous glucose as a screening test for abnormal glucose tolerance. *Prim Care Diabetes* 2012;**6**:207–11. doi:10.1016/j.pcd.2012.03.002

33 Cuong TQ, Dibley MJ, Bowe S, *et al.* Obesity in adults: An emerging problem in urban areas of Ho Chi Minh City, Vietnam. *Eur J Clin Nutr* 2007;**61**:673–81. doi:10.1038/sj.ejcn.1602563

34 Deng B, Luo T, Huang Y, *et al.* Prevalence and determinants of hyperlipidemia in moderate altitude areas of the yunnan-kweichow plateau in southwestern China. *High Alt Med Biol* 2012;**13**:13–21. doi:10.1089/ham.2011.1037

35 Falguera M, Vilanova MB, Alcubierre N, *et al.* Prevalence of pre-diabetes and undiagnosed diabetes in the Mollerussa prospective observational cohort study in a semi-rural area of Catalonia. *BMJ Open* 2020;**10**:e033332. doi:10.1136/bmjopen-2019-033332

36 Gao W, Dong Y, Nan H, *et al.* The likelihood of diabetes based on the proposed definitions for impaired fasting glucose. *Diabetes Res Clin Pract* 2008;**79**:151–5. doi:10.1016/j.diabres.2007.07.018

37 Hare MJL, Magliano DJ, Zimmet PZ, *et al.* Glucose-independent ethnic differences in HbA1c in people without known diabetes. *Diabetes Care* 2013;**36**:1534–40. doi:10.2337/dc12-1210

38 Hidalgo LA, Chedraui PA, Morocho N, *et al.* The metabolic syndrome among postmenopausal women in Ecuador. *Gynecol Endocrinol* 2006;**22**:447–54. doi:10.1080/09513590600890272

39 Hwu CM, Lin MW, Liou TL, *et al.* Fasting triglyceride is a major determinant of postprandial triglyceride response in postmenopausal women. *Menopause* 2008;**15**:150–6. doi:10.1097/gme.0b013e31800577856

40 Katulanda GW, Katulanda P, Dematapitiya C, *et al.* Plasma glucose in screening for diabetes and pre-diabetes: How much is too much? Analysis of fasting plasma glucose and oral glucose tolerance test in Sri Lankans. *BMC Endocr Disord* 2019;**19**:11. doi:10.1186/s12902-019-0343-x

41 Koyama AK, Bali V, Yermilov I, *et al.* Identification of Undiagnosed Hyperlipidemia: Do Work Site Screening Programs Work? *Am J Heal Promot* 2018;**32**:971–8. doi:10.1177/0890117116671537

42 Kutkiene S, Petrulioniene Z, Laucevicius A, *et al.* Lipid profile evaluation and severe hypercholesterolaemia screening in the middle-aged population according to nationwide primary prevention programme in Lithuania. *Atherosclerosis* 2018;**277**:267–72. doi:10.1016/j.atherosclerosis.2018.06.008

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

43 Li LJ, Zhou JX, Chen HT, *et al.* Effect of HbA 1c combined FPG on screening diabetes in health check-up. *Asian Pac J Trop Med* 2012;**5**:472–5. doi:10.1016/S1995-7645(12)60081-9

44 Lin S, Hu L, Li X, *et al.* Glycated haemoglobin A1c for diagnosing diabetes in Chinese subjects over 50 years old: A community-based cross-sectional study. *Clin Endocrinol (Oxf)* 2014;**80**:656–61. doi:10.1111/cen.12202

45 Lissock CNAA, Sobngwi E, Ngassam E, *et al.* Rural and urban differences in metabolic profiles in a Cameroonian population. *Pan Afr Med J* 2011;**10**:1. doi:10.4314/pamj.v10i0.72204

46 Liu Y, Xiao X, Sun C, *et al.* Ideal glycated hemoglobin cut-off points for screening diabetes and prediabetes in a Chinese population. *J Diabetes Investig* 2016;**7**:695–702. doi:10.1111/jdi.12498

47 Mainous AG, Tanner RJ, Baker R, *et al.* Prevalence of prediabetes in England from 2003 to 2011: population-based, cross-sectional study. *BMJ Open* 2014;**4**:e005002. doi:10.1136/bmjopen-2014-005002

48 Mayega RW, Guwatudde D, Makumbi FE, *et al.* Comparison of fasting plasma glucose and haemoglobin A1c point-of-care tests in screening for diabetes and abnormal glucose regulation in a rural low income setting. *Diabetes Res Clin Pr* 2014;**104**:112–20. doi:10.1016/j.diabres.2013.12.030

49 Nunes B, Barreto M, Gil AP, *et al.* The first Portuguese National Health Examination Survey (2015): design, planning and implementation. *J Public Health (Oxf)* 2019;**41**:511–7. doi:10.1093/pubmed/fdy150

50 Rush E, Crook N, Simmons D. Point-of-care testing as a tool for screening for diabetes and pre-diabetes. *Diabet Med* 2008;**25**:1070–5. doi:10.1111/j.1464-5491.2008.02526.x

51 Sabir A, Ohwovoriole A, Isezuo S, *et al.* Type 2 diabetes mellitus and its risk factors among the rural Fulanis of Northern Nigeria. *Ann Afr Med* 2013;**12**:217–22. doi:10.4103/1596-3519.122689

52 Sinnott M, Kinsley BT, Jackson AD, *et al.* Fasting plasma glucose as initial screening for diabetes and prediabetes in irish adults: The Diabetes Mellitus and Vascular health initiative (DMVhi). *PLoS One* 2015;**10**:e0122704. doi:10.1371/journal.pone.0122704

53 Valkengoed VIG, Vlaar EM, Nierkens V, *et al.* The Uptake of Screening for Type 2 Diabetes and Prediabetes by Means of Glycated Hemoglobin versus the Oral Glucose Tolerance Test among 18 to 60-Year-Old People of South Asian Origin: A Comparative Study. *PLoS One* 2015;**10**:e0136734. doi:10.1371/journal.pone.0136734

54 Wang S, Niu J, Zhao Z, *et al.* Detection of diabetes and prediabetes using glycosylated hemoglobin in Chinese adults living in Shanghai: A prospective analysis. *J Diabetes* Published Online First: 2020. doi:10.1111/1753-0407.13028

55 Zafar J, Nadeem D, Khan SA, *et al.* Prevalence of diabetes and its correlates in urban population of Pakistan: A Cross-sectional survey. *J Pak Med Assoc* 2016;**66**:922–7.

56 Zemlin AE, Matsha TE, Kengne AP, *et al.* Derivation and validation of an HbA1c optimal cutoff for diagnosing prediabetes in a South African mixed ancestry population. *Clin Chim Acta* 2015;**448**:215–9. doi:10.1016/j.cca.2015.06.019

57 Zhao WH, Zhang J, Zhai Y, *et al.* Blood lipid profile and prevalence of dyslipidemia in Chinese adults. *Biomed Env Sci* 2007;**20**:329–35.

58 Zhou X, Ruan X, Hao L, *et al.* Optimal hemoglobin A1C cutoff value for diabetes mellitus and pre-diabetes in Pudong New Area, Shanghai, China. *Prim Care Diabetes* 2018;**12**:238–44. doi:10.1016/j.pcd.2017.12.006

59 Ziemer DC, Kolm P, Weintraub WS, *et al.* Glucose-independent, black-white differences in hemoglobin A1c levels: a cross-sectional analysis of 2 studies. *Ann Intern Med* 2010;**152**:770–7. doi:10.7326/0003-4819-152-12-201006150-00004

60 Mendis S, Armstrong T, Bettcher D, *et al.* Global status report on noncommunicable diseases 2014. World Health Organisation. 2014. doi:ISBN 9789241564854

61 Sargeant LA, Simmons RK, Barling RS, *et al.* Who attends a UK diabetes screening programme? Findings from the ADDITION-Cambridge study. *Diabet Med* 2010;**27**:995–1003. doi:10.1111/j.1464-5491.2010.03056.x

62 Labeit A, Peinemann F, Baker R. Utilisation of preventative health check-ups in the UK: Findings from individual-level repeated cross-sectional data from 1992 to 2008. *BMJ Open* 2013;**3**. doi:10.1136/bmjopen-2013-003387

63 Fawns-Ritchie C, Starr JM, Deary IJ. Health literacy, cognitive ability and smoking: A cross-sectional analysis of the English Longitudinal Study of Ageing. *BMJ Open* 2018;**8**:e023929. doi:10.1136/bmjopen-2018-023929

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

64 Walters R, Leslie SJ, Polson R, *et al.* Establishing the efficacy of interventions to improve health literacy and health behaviours: A systematic review. *BMC Public Health* 2020;**20**:1040. doi:10.1186/s12889-020-08991-0

65 Schülein S, Taylor KJ, Schriefer D, *et al.* Participation in preventive health check-ups among 19,351 women in Germany. *Prev Med Reports* 2017;**6**:23–6. doi:10.1016/j.pmedr.2017.01.011

66 World Health Organization. Social determinants of health. World Heal. Organ. 2013.https://www.who.int/social_determinants/thecommission/finalreport/key_concepts/en/ (accessed 6 May 2020).

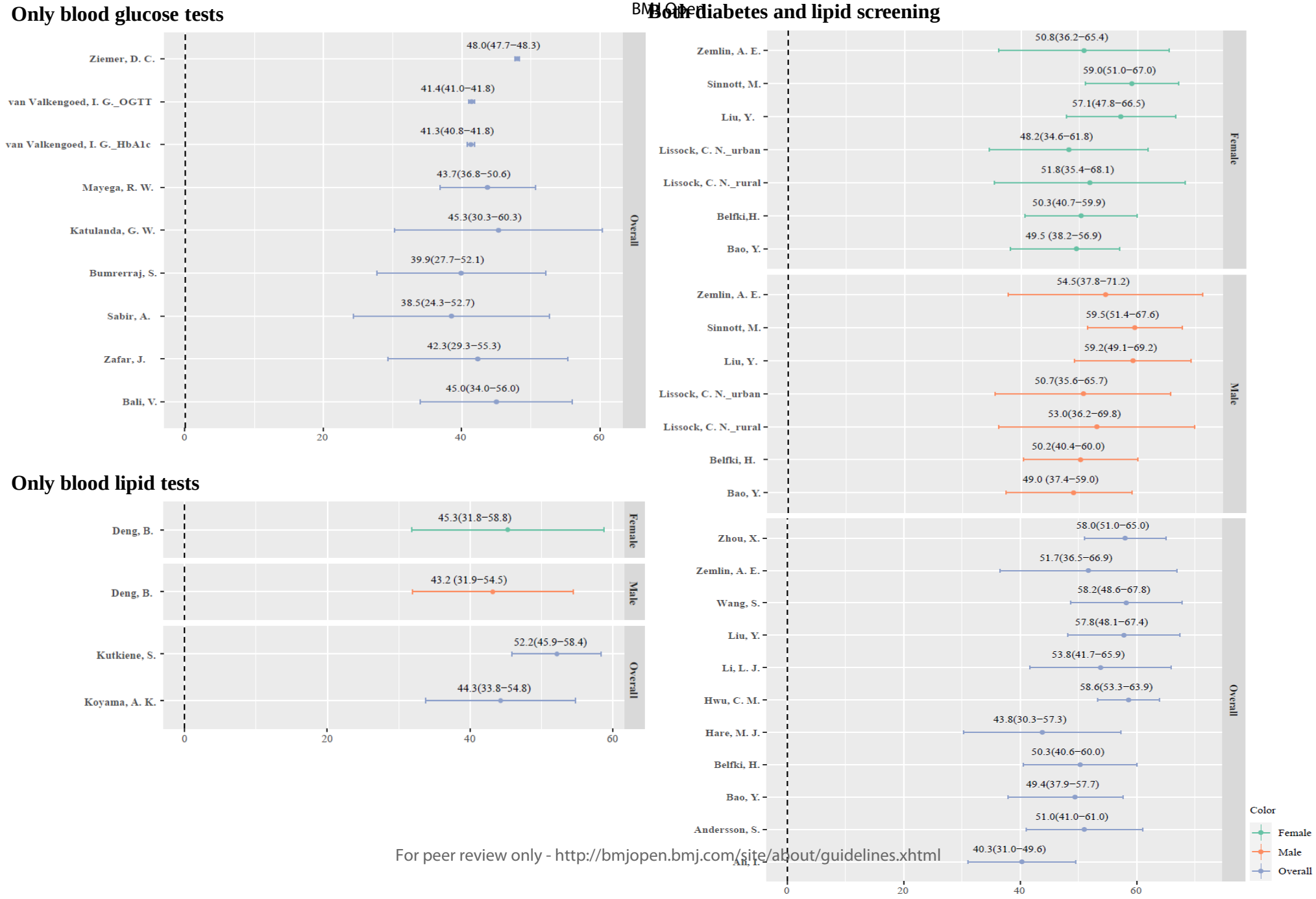
67 Orach CG. Health equity: challenges in low income countries. Afr. Health Sci. 2009;**9 Suppl 2**:S49.

68 Magnani JW, Mujahid MS, Aronow HD, *et al.* Health Literacy and Cardiovascular Disease: Fundamental Relevance to Primary and Secondary Prevention: A Scientific Statement From the American Heart Association. Circulation. 2018;**138**:e48–74. doi:10.1161/CIR.0000000000000579

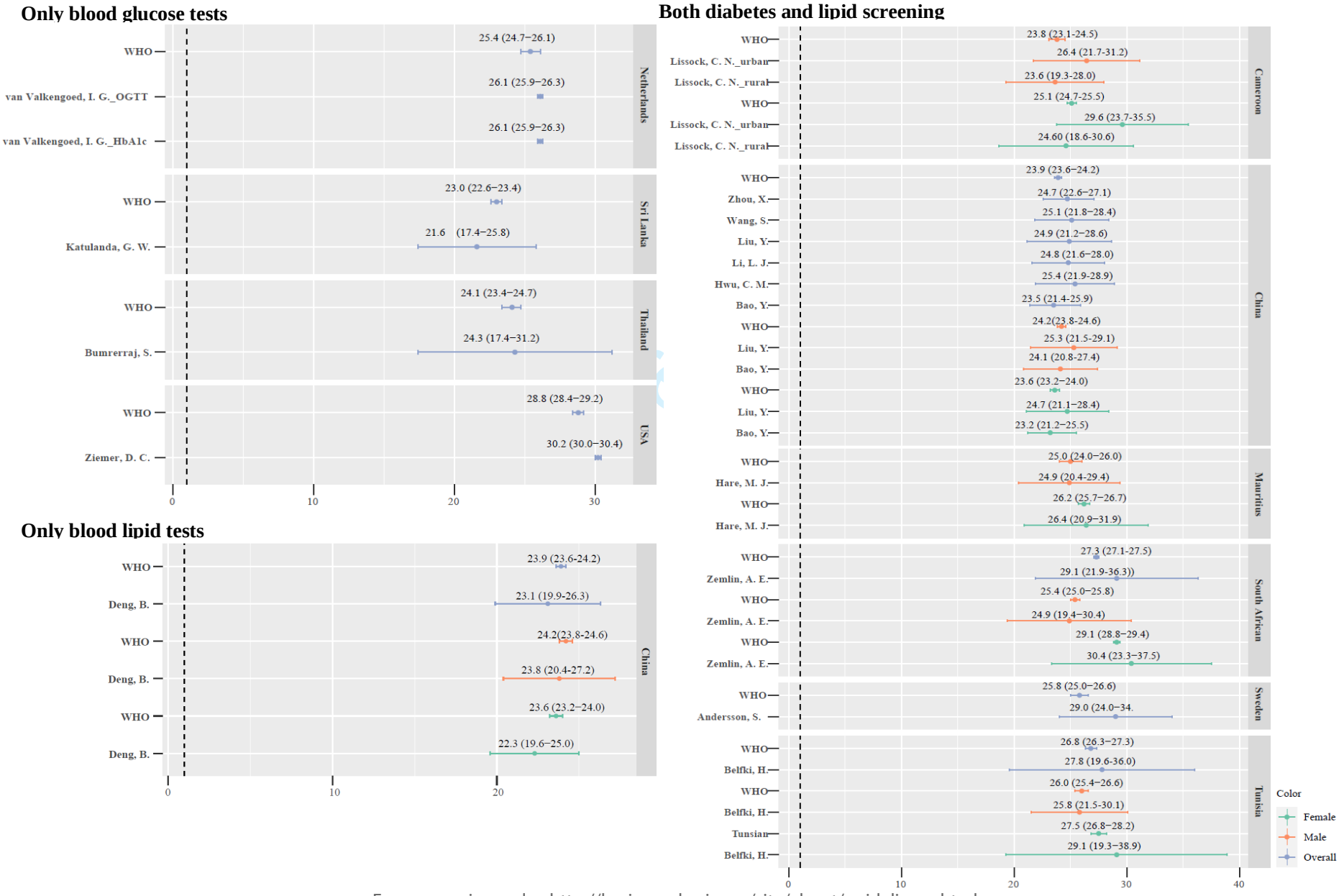
69 American Diabetes Association. Screening for diabetes. *Diabetes Care* 2002;**25**:s21–4. doi:10.2337/diacare.25.2007.s21

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml


Only blood glucose tests
WHO
van Valkengoed, I. G._OGTT
van Valkengoed, I. G._HbA1c
25.4 (24.7−26.1)
26.1 (25.9−26.3)
26.1 (25.9−26.3)
Netherlands
WHO
Katulanda, G. W.
23.0 (22.6−23.4)
21.6 (17.4−25.8)
Sri Lanka
WHO
Bumrerraj, S.
24.1 (23.4−24.7)
24.3 (17.4−31.2)
Thailand
WHO
Ziemer, D. C.
28.8 (28.4−29.2)
30.2 (30.0−30.4)
USA
0
10
20
30
Only blood lipid tests
WHO
Deng, B.
WHO
Deng, B.
WHO
Deng, B.
23.9 (23.6-24.2)
23.1 (19.9-26.3)
24.2(23.8-24.6)
23.8 (20.4-27.2)
23.6 (23.2−24.0)
22.3 (19.6−25.0)
China
0
10
20
Both diabetes and lipid screening
WHO
Lissock, C. N._urban
Lissock, C. N._rural
WHO
Lissock, C. N._urban
Lissock, C. N._rural
23.8 (23.1-24.5)
26.4 (21.7-31.2)
23.6 (19.3-28.0)
25.1 (24.7-25.5)
29.6 (23.7-35.5)
24.60 (18.6-30.6)
Cameroon
WHO
Zhou, X.
Wang, S.
Liu, Y.
Li, L. J.
Hwu, C. M.
Bao, Y.
WHO
Liu, Y.
Bao, Y.
WHO
Liu, Y.
Bao, Y.
23.9 (23.6−24.2)
24.7 (22.6−27.1)
25.1 (21.8−28.4)
24.9 (21.2−28.6)
24.8 (21.6−28.0)
25.4 (21.9-28.9)
23.5 (21.4-25.9)
24.2(23.8-24.6)
25.3 (21.5-29.1)
24.1 (20.8-27.4)
23.6 (23.2−24.0)
24.7 (21.1−28.4)
23.2 (21.2−25.5)
China
WHO
Hare, M. J.
WHO
Hare, M. J.
25.0 (24.0−26.0)
24.9 (20.4-29.4)
26.2 (25.7−26.7)
26.4 (20.9−31.9)
Mauritius
WHO
Zemlin, A. E.
WHO
Zemlin, A. E.
WHO
Zemlin, A. E.
27.3 (27.1-27.5)
29.1 (21.9-36.3))
25.4 (25.0−25.8)
24.9 (19.4−30.4)
29.1 (28.8−29.4)
30.4 (23.3−37.5)
South African
WHO
Andersson, S.
25.8 (25.0−26.6)
29.0 (24.0−34.
Sweden
WHO
Belfki, H.
WHO
Belfki, H.
Tunsian
Belfki, H.
26.8 (26.3−27.3)
27.8 (19.6-36.0)
26.0 (25.4−26.6)
25.8 (21.5-30.1)
27.5 (26.8−28.2)
29.1 (19.3−38.9)
Tunisia
0
10
20
30
40
Color
Female
Male
Overall

Supplementary materials

Supplementary figure 1 Selection of articles for systematic review

Supplementary figure 2 The gender of screening participants

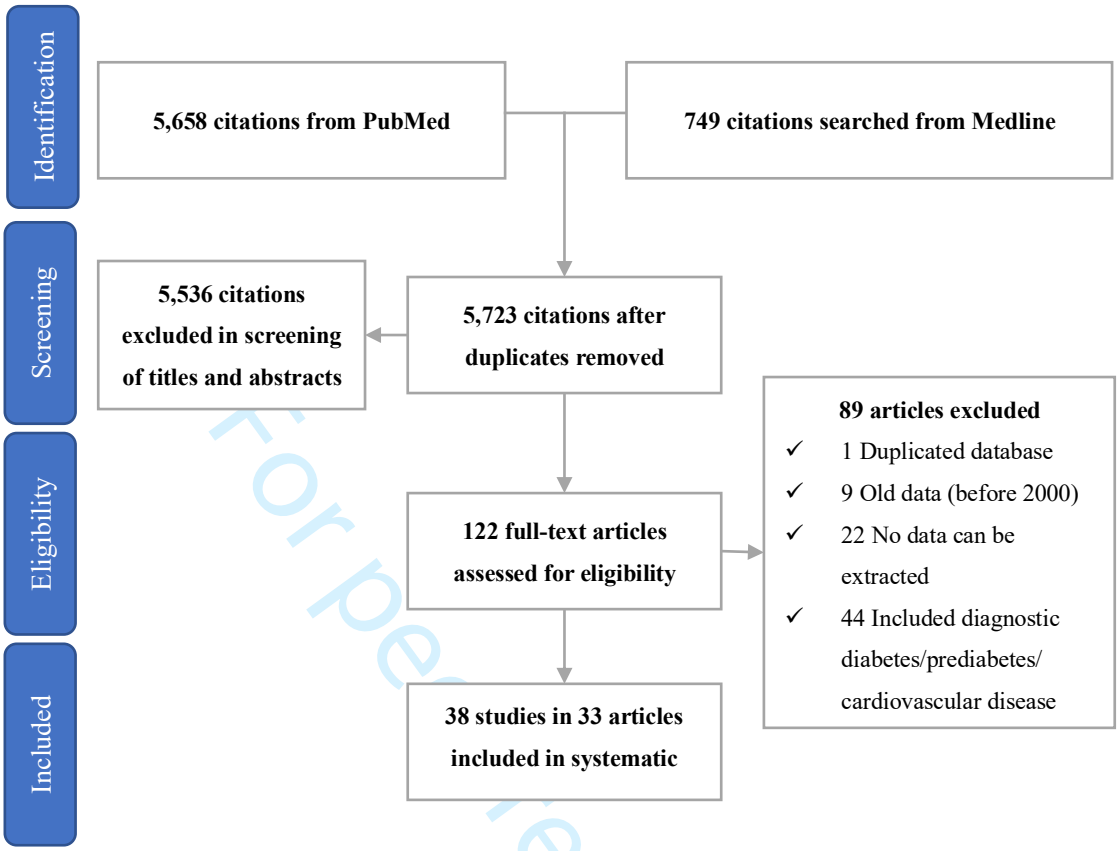
Supplementary table 1 Summary of diabetes and lipid screening guidelines

Supplementary table 2 Quality Assessment of included articles based on Appraisal of Cross-sectional Studies (AXIS)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

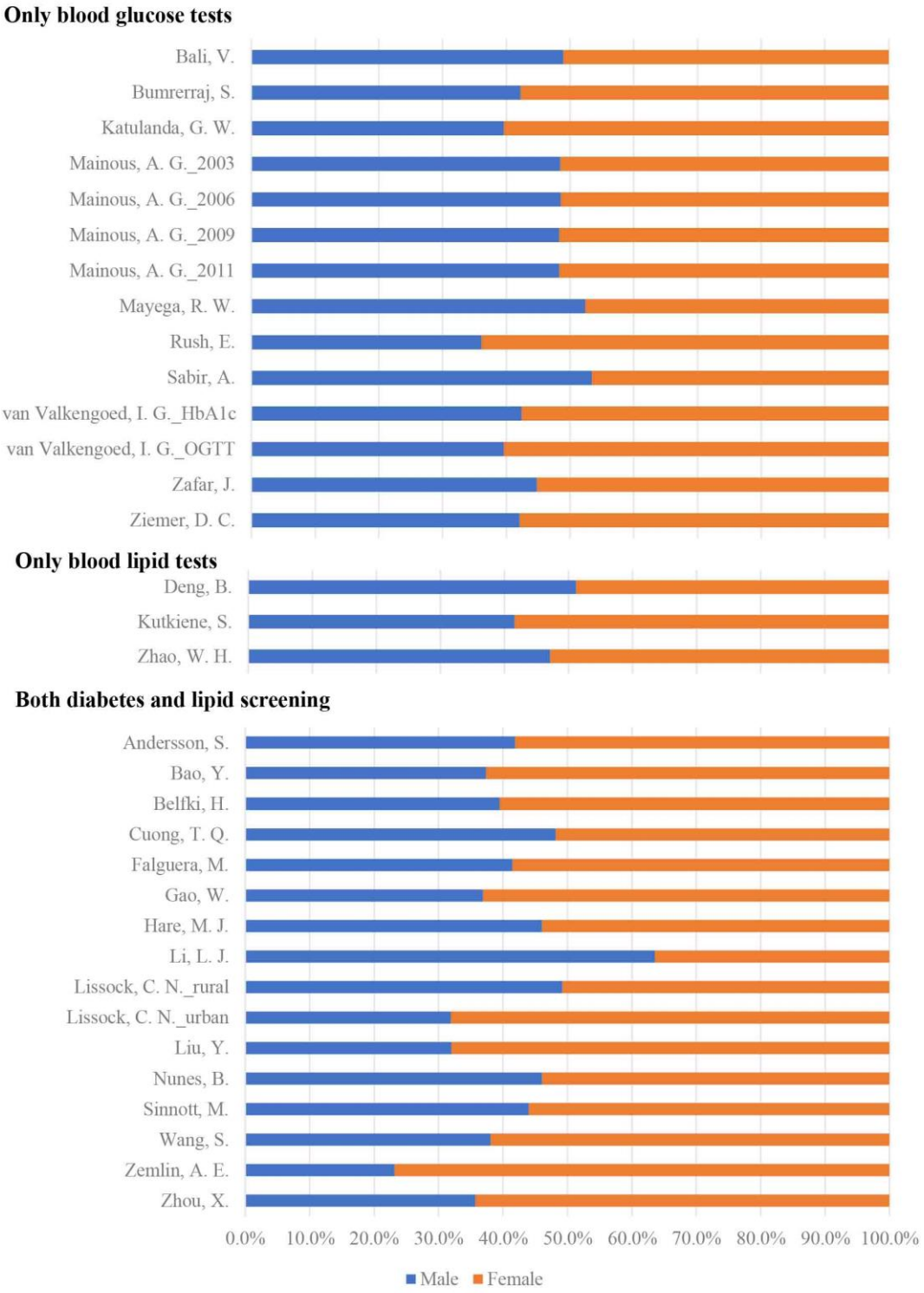
Supplementary figure 1 Selection of articles for systematic review



For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary figure 2 The gender of screening participants



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary table 1 Summary of diabetes and lipid screening guidelines

Country	Diabetes Screening Guidelines	Lipid Screening Guidelines
Australia	Patients should be screened for diabetes risk every three years from 40 years of age using the Australian type 2 diabetes risk assessment tool. Aboriginal and Torres Strait Islander peoples should be screened from 18 years of age. *(RACGP)*	Adults should have their blood lipids assessed every five years starting at 45 years of age. Aboriginal and Torres Strait Islander adults should have lipid tests performed every five years from 35 years of age. *(RACGP)*
Canada	Adults should be tested for type 2 diabetes every 3 years if they are age 40 years or over, or if they are at risk for T2DM. If they have risk factors for T2DM, they should be tested more frequently and/or start regular screening earlier. *(Canada Clinical Practice Guidelines Expert Committee)*	Initiating screening: In patients without CVD (primary prevention), we suggest lipid testing as part of global CVD risk estimation in men at age ≥40 years old and women at age ≥50 years old. Testing can be considered earlier for patients with known traditional CVD risk factors. *(College of Family Physicians of Canada)*
Singapore	Screening for T2DM should begin at 40 years and older; screening should also be considered in adults of any age who have one or more risk factors for diabetes. *(Singapore Ministry of Health)*	Clinicians should routinely screen men and women aged 40 years and older for lipid disorders, and younger adults for lipid disorders if they have other risk factors for coronary artery disease. *(Singapore Ministry of Health)*
United Kingdom	The UK National Screening Committee does not currently recommend screening for T2DM. This is because: there is no evidence that screening is more beneficial than not screening, and there is no evidence that finding T2DM early by screening results in a greater health benefit than usual health care.	National Institute for Health and Care Excellence guideline CG181 recommends (1.1.8) using the QRISK2 risk assessment tool (including ethnicity, age, sex, smoking, blood pressure, lipid profile, BMI, etc.) to assess CVD risk for the primary prevention of CVD in people up to and including age 84 years. There is also a potential need to amend recommendations 1.1.8 and 1.1.10 to advise the use of QRISK3 (including sex, age, BMI, lipid profile, blood pressure, family history, etc.) in place of QRISK2 because QRISK2 is due to be superseded by QRISK3 in 2018.
United States	Adults aged 40 to 70 who are overweight or obese are recommended to screen for abnormal blood glucose. (*USPSTF)*	Identification of dyslipidaemia and calculation of 10-year CVD event risk requires universal lipids screening in adults aged 40 to 75 years. *(USPSTF)*

RACGP: The Royal Australian College of General Practitioners, T2DM: type 2 diabetes mellitus, CVD: cardiovascular disease, USPTF: US preventive services task force recommendation statement

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary table 2 Quality Assessment of included articles based on Appraisal of Cross-sectional Studies (AXIS)

Author	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ali, I.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Andersson, S.	1	1	0	1	1	9	0	1	1	1	1	1	1	1	1	1	1	1	1	1
Bali, V.	1	1	0	1	9	0	0	1	1	0	1	1	0	0	1	1	1	1	9	9
Bao, Y.	1	1	0	1	0	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Belfki, H.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	9	1
Bumrerraj, S.	1	1	0	1	0	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1
Cuong, T. Q.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	9	1
Deng, B.	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	0	1	1
Falguera, M.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Gao, W.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	0	9	1
Hare, M. J.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Hidalgo, L. A.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Hwu, C. M.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Katulanda, G. W.	1	1	1	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Koyama, A. K.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	9	9
Kutkiene, S.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Li, L. J.	1	1	0	1	0	1	0	1	1	1	1	1	0	0	1	1	1	0	1	1
Lin, S.	1	1	0	1	1	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Lissock, C. N.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	0	1	1
Liu, Y.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Mainous, A. G.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	9
Mayega, R. W.	1	1	1	1	1	0	1	1	1	0	1	1	0	1	1	1	1	1	1	1
Nunes, B.	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Rush, E.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	0	1
Sabir, A.	1	1	0	1	0	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Sinnott, M.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	9	1
Valkengoed, I. G.	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1
Wang, S.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Zafar, J.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Zemlin, A. E.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Zhao, W. H.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	1	1
Zhou, X.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Ziemer, D. C.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	9

0: No, 1: Yes, 9: Don't know.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5-6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	-
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6-7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7-8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7-8

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	-
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	-
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8,Supplementary figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-9,Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8,Supplementary table 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-19,Table 1,2,3
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	-
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	-
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	-
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	20-22
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	22
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	22-23
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	24

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



PRISMA 2009 Checklist

For more information, visit: **www.prisma-statement.org**.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

BMJ Open

Characteristic of participants who take up screening tests for diabetes and lipid disorders: a systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-055764.R1
Article Type:	Original research
Date Submitted by the Author:	17-Jan-2022
Complete List of Authors:	Ding, Hanyue; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care Huang, Junjie; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care Deng, Yunyang; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care Tin, Pamela; Our Hong Kong Foundation Wong, Martin Chi-Sang; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care; Peking University, School of Public Health Yeoh, EK; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care; The Chinese University of Hong Kong, Centre for Health Systems and Policy Research
Primary Subject Heading:	Public health
Secondary Subject Heading:	Public health
Keywords:	PREVENTIVE MEDICINE, Lipid disorders < DIABETES & ENDOCRINOLOGY, General diabetes < DIABETES & ENDOCRINOLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Characteristic of participants who take up screening tests for diabetes and lipid disorders: a systematic review

Hanyue Ding[1], Junjie Huang[1], Yunyang Deng[1], Pamela Sze Pui Tin[2], Martin CS Wong[1,3,4], Eng-kiong Yeoh[1,5*]

1. JC School of Public Health and Primary Care, Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong Special Administrative Region, China
2. Our Hong Kong Foundation, Hong Kong Special Administrative Region, China
3. School of Public Health, Peking University, Beijing, China
4. School of Public Health, Chinese Academy of Medical Sciences and Peking Union Hospitals, Beijing, China
5. Centre for Health Systems and Policy Research, The Chinese University of Hong Kong, Hong Kong Special Administrative Region, China

*** Correspondence author**

Professor Eng-kiong Yeoh, GBS, OBE, JP

Director, Centre for Health Systems and Policy Research, JC School of Public Health and Primary Care, Faculty of Medicine, The Chinese University of Hong Kong

Email: yeoh_ek@cuhk.edu.hk; **Telephone number:** 2252-8403; **Fax number:** 2145-8517

Address: Room 202A, 2/F, School of Public Health, Prince of Wales Hospital

(Word count: 3994 words)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

ABSTRACT

Objectives To perform a systematic review on the characteristics of participants who attended screening programs with blood glucose tests, lipid profiles, or a combination of them, respectively.

Design Systematic review following the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) checklist.

Data sources PubMed and Medline database for English literature from 1st Jan 2000 to 1st April 2020.

Eligibility criteria Original observational studies that reported baseline characteristics of apparently healthy, adult screening participants for diabetes and lipid disorders were included in this review.

Data extraction We examined their sociodemographic characteristics, including age, gender, Body Mass Index (BMI), and lifestyle habits. The quality of the included articles was evaluated by the Appraisal of Cross-sectional Studies.

Results A total of 33 articles involving 38 studies in 22 countries were included and analyzed in this systematic review. Overall, there was a higher participation rate among subjects who were female in all screening modalities (female vs. male: 46.6-63.9% vs. 36.1-53.4% for diabetes screening; 48.8-58.4% vs. 41.6-51.2% for lipid screening; and 36.4-76.8% vs. 23.2-63.6% for screening offering both). Compared with the BMI standard from the World Health Organization, participants in lipid screening had lower BMI (males: 23.8 kg/m^2 vs. 24.2 kg/m^2, $p<0.01$; females: 22.3 kg/m^2 vs. 23.6 kg/m^2, $p<0.01$). Furthermore, it is less likely for individuals of lower socioeconomic status to participate in diabetes or lipid screening in developed areas.

Conclusions We identified individuals from lower socioeconomic groups who were less likely to take up programs for diabetes and/or lipid screening in developed areas.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

These populations are also likely to be at higher risk for non-communicable diseases. Future studies should investigate the barriers and facilitators of screening among non-participants, where targeted interventions to enhance their screening uptake are warranted.

(Word count: 283 words)

Keywords: Diabetes Mellitus; Lipid Metabolism Disorders; Mass Screening; Population Characteristics; Systematic Review

Strengths and limitations of this study

- To our knowledge, this is the first systematic review that presented the characteristics of screening participants in diabetes and lipid screening programs.
- Due to the descriptive manner, we cannot quantify the association between patients' characteristics and screening participation.
- We included articles that used different screening tests for diabetes, and this might pose challenges on comparison across studies.
- There are differences in the organization of different screening programs in terms of the tests offered, subsidy amount, and accessibility of the screening centers, and these could influence the screening uptake rate.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

Cardiovascular disease (CVD) represents the major cause of mortality, accounting for 17.9 million deaths or 32% of all global deaths in 2019.[1] It caused more than half of all deaths across the European region, and the death rates were higher in the Russian Federation, Bulgaria, and Romania than in other European regions.[2,3] Most Asian countries had higher age-adjusted mortality than Western countries.[4] Patients with CVD have a high risk of complications, including ischemic heart disease, stroke, and chronic kidney diseases.[5] The modifiable risk factors for CVD included smoking, inadequate physical activity, elevated blood pressure, increased body weight, increased plasma lipid (cholesterol and triglyceride), as well as diabetes.[6] Diabetes is not only the risk factor for CVD but is also associated with significant morbidity and mortality. About 1.6 million deaths were directly caused by diabetes mellitus (DM) in 2016. In addition, diabetes may lead to long-term complications, such as kidney failure, lower limb amputation, blindness, heart attacks, and stroke.[7,8] Worldwide, there were 108 million diabetes patients in 1980, and the number has increased to 422 million in 2014 [8]. The global prevalence of diabetes has been predicted to rise to 10.2% (578 million) by 2030 and 10.9% (700 million) by 2045.[9]

Globally, ischemic heart disease and stroke ranked the first and second causes of disability-adjusted life years (DALYs) in 2016, which were responsible for 12.8% of total DALYs lost.[10] Diabetes caused 65.7 million DALYs lost and 2.5% of the total global DALYs lost in 2016.[10] In addition, dyslipidemia also poses a serious threat to the health of populations. Overall, elevated cholesterol was estimated to contribute to 4.5% of total deaths and 2.0% of total DALYs.[11] Disease screening is regarded as an effective approach for the early detection and prevention of diabetes and dyslipidemia.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

It was demonstrated that cholesterol screening can save 14,300 quality-adjusted life years (QALYs) over 100,000 persons' lifetimes incurring a cost of $33,800 per QALY in the US.[12] Diabetes screening has been shown to be more cost-effective for individuals aged 55 to 75 years as compared to younger subjects.[13]

According to international guidelines for screening **(Supplementary table 1)**, in the United States (US),[14] the United Kingdom,[15] Australia,[16] Canada,[17] and Singapore [18] cholesterol screening is recommended for adults, and the proposed age of onset for screening ranged from 35 to 45 years - or earlier for patients with CVD risk factors. Furthermore, Australia,[16] Canada,[19] and Singapore [20] suggested diabetes screening for average-risk adults above 40 years old, and the US recommended screening for adults aged 40-70 years who are overweight or obese.[21]

However, previous studies have shown that the uptake rate for diabetes and lipid screening was suboptimal. According to a cross-sectional study performed in the US between 2005 and 2012, the diabetes screening rates for "screening-recommended" and "screening-not-recommended" participants were 46.2% and 29.6% respectively.[22] Participation rates of the health examinations for CVD and diabetes ranged between 48% and 67% in Sweden.[23] Although the attendance rate for National Health Service (NHS) health check in England quadrupled from 5.8% in 2010 to 30.1% in 2012, the uptake rates were still relatively low when compared with findings from other studies.[24] To our knowledge, there are no systematic reviews that explored the socioeconomic and demographic characteristics of participants who take up screening programs for diabetes and lipid disorders. Knowing their characteristics could help to target individuals who were less likely to receive screening and improve their

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

participation. These results may help to provide a basis for future studies to examine the enablers and barriers for diabetes and lipid screening. Therefore, we performed a systematic review of the existing literature to evaluate the characteristics of participants who received screening tests for diabetes and lipid disorders.

METHODS

Search strategy

We followed the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) checklist to conduct this systematic review.[25] We searched the PubMed and Medline databases for English literature from 1st Jan 2000 to 1st April 2020. As the characteristics of participants were influenced by health literacy, health development, and health policy, etc., which are changing over time. The older articles showed limited quality to be involved in this study, so we included the studies of recent 20 years to conduct this review. The following keywords were used for the search: 1) AND 2) NOT 3): 1) "fasting plasma glucose test" OR "postprandial plasma glucose" OR "oral glucose tolerance test" OR "random plasma glucose test" OR "glycated hemoglobin" OR "lipid test" OR "cholesterol test" OR "triglyceride test" OR "lipid panel" OR "lipid profile" 2) "uptake" OR "participate" 3) "pediatric" OR "gestational" (**Supplementary table 2**).

Inclusion and exclusion criteria

Studies were included if they: 1) were cross-sectional or reported baseline characteristics of participants in cohort studies; 2) were conducted among adults without a known history of diabetes, prediabetes or CVD; 3) adopted one-step finger stick or blood glucose (including fasting plasma glucose (FPG), oral glucose tolerance

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

tests (OGTT), glycated hemoglobin (HbA1c), and random plasma glucose (RBG)) or blood lipid tests as screening tools; and 4) reported participants' characteristics. The exclusion criteria included: 1) conference abstracts, systematic reviews, or studies published in languages other than English; 2) screening programs for pregnant women, the pediatric population, and/or patients who were previously diagnosed with diabetes, prediabetes or CVD; 3) studies whose target population focused on obese or overweight subjects, people with a family history of diabetes or CVD; 4) studies involved the measurements with many steps/stages; and 5) studies performed before 2000.

Data extraction and quality assessment

Two reviewers (HD and VL) independently searched and screened each title and abstract. Disagreements were solved by discussion. Once the title and abstract of each citation were assessed as eligible, two reviewers reviewed and appraised the full text. We used the Appraisal of Cross-sectional Studies (AXIS) to assess the quality of included articles, which consists of 20 questions and answers recorded "yes", "no" or "don't know".[26] AXIS does not involve a numerical scale that can be used to calculate to produce a quality assessment score. We extracted the following data from all included articles: the first author's name, publication year, country, region/city, project period, study type, subsidization, the age range of the population, number of participants; the screening tests used (finger prick, FPG, OGTT, HbA1c, RBG, fasting lipid profile, waist circumference, hip circumference, blood pressure and other relevant tests); participant characteristics: gender, age, body mass index (BMI), marital status, educational status, occupational status, other indicators of socioeconomic status (SES) (including Household wealth index, social deprivation index, and household income), family history, cigarette smoking, alcohol consumption, physical activity, and the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

reason of non-participation.

Statistical analysis

We summarized and descriptively reported the socioeconomics and demographic characteristics of the populations attending the screening in the included studies. Based on our findings in the initial literature review, we also selected gender as a subgroup to explore the difference in age and BMI of the study participants. Means ± standard deviations of these parameters were obtained from included studies. Figures were plotted using R software version x64 3.6.3 with the ggplot package.

Patient and Public Involvement

It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

RESULTS

From the literature search, 6,407 citations were identified, of which 5,689 were from PubMed and 749 were from Medline **(Supplementary figure 1)**. There were 5,723 citations after the removal of duplicates. We retrieved 122 full-text articles to assess for eligibility after 5,536 citations were excluded during the title or abstract screening with the pre-determined criteria. We excluded 89 articles which were duplicate studies (n=1); performed before the year 2000 (n=9); absence of full text or lack of sufficient data for extraction (n=22); the presence of known diabetes, prediabetes or CVD (n=44); or target population whose subjects reported family history of diabetes, prediabetes or elevated BMI (n=13). Finally, we included 38 studies in 33 articles in the present systematic review.[27–59]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Regarding the AXIS for evaluating each included study, most studies presented with the following limitations: the sample size not being justified (23/33); absence of characterization and categorization of non-participants (29/33); a lack of control for non-response bias (30/33); and absence of description of non-participants (27/33). Overall, all studies (33/33) met the following criteria: appropriate objectives and study design; appropriate risk factors and outcome variables; clear and sufficient description of methods; adequate description of the participants' characteristics; and internally consistent presentation of results (**Supplementary table 3**).

The characteristics of all eligible studies were presented in **Table 1**. Among these thirty-three manuscripts: fifteen were from Asia (China, Vietnam, Sri Lanka, Pakistan, Palestine, Thailand), seven were from Europe (Spain, England, Ireland, Netherlands, Lithuania, Sweden, Portugal), six were from Africa (Mauritius, Cameroon, Uganda, Nigeria, Tunisia, South African), four were from Americas (USA, Ecuador), and one was from Oceania (New Zealand). Ten studies were from developed countries, and the other twenty-three studies were from developing countries. According to screening items, eleven studies only implemented diabetes screening, four studies only conducted lipid tests, the rest of the eighteen studies performed both diabetes and lipid screening.

Diabetes Screening

Eleven studies adopted OGTT, HbA1c, FPG, or RBG to screen for diabetes.[32,36,39–41,43,44,46,48,53,55] OGTT and HbA1c were the commonly used modalities (in seven articles), followed by FPG (in five articles) and RBG (in one article). Two of these studies evaluated combinations of FPG, OGTT, and HbA1c. A total of four studies

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

simultaneously took blood pressure during screening.[41,44,48,53] Except for three studies,[32,36,46] all the others included anthropometry measurements (body weight, height, waist and hip circumference). Nearly half of the studies were from developed countries (five out of eleven).[39,40,43,46,53] Among these eleven articles, one reported findings from workplace screening insured by employers,[39] two were national health surveys and fully subsidized by government organizations,[40,53] five were funded by research institutes or funding agencies,[32,41,44,46,55] one was supported by a pharmaceutical company for free meters and strips,[43] and two did not mention their financial supports.[36,48] (**Table 1**)

The range of participation rate of female and male subjects in the diabetes screening programs was 46.6-63.9% and 36.1-53.4%, respectively. **(Supplementary figure 2)** More female participants participated in screening than males in twelve studies. Only two studies showed more male participants in Uganda (52.4%) and Nigeria (53.4%). The mean age of the participants ranged from 38.5 to 48.0 years in all included studies. **(Figure 1)** The studies varied in terms of participant age, where the average age of most studies were more than 40 years. We extracted the national mean BMI of adults in various jurisdictions reported by the World Health Organization (WHO),[60] and compared these with the participants' average BMI. **(Figure 2)** The average BMI of participants tended to be higher than the national mean, in particular from studies performed in the Netherlands (26.1 kg/m^2 vs. 25.4 kg/m^2, $p<0.01$),[46] Thailand (24.3 kg/m^2 vs. 24.1 kg/m^2, $p=0.39$),[55] and the US (30.2 kg/m^2 vs. 28.8 kg/m^2, $p<0.01$).[53]

The sociodemographic information of the screening participants was shown in **Table 2**. For educational status, different studies adopted different classification methods for

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

educational levels. Mayega *et al* reported that 39.6% of participants had a higher primary level, followed by a lower primary (21.9%), secondary (16.5%), and tertiary (4.5%) educational level in Uganda.[41] Zafar *et al* showed more than half of participants had matriculation level or above (same as the undergraduate degree) (52.5%) and only 4.2% acquired conventional level (same as the postgraduate degree).[48] They also investigated the household income and found that 72.5% of the participants had <30,000 PKR income (approximately 300 USD in 2014). Mainous *et al* reported the trend (2003, 2006, 2009, 2011) of the social deprivation index, which is a composite index based on income, employment, health deprivation and disability, education, skills and training, barriers to housing and services, crime and disorder, and living environment.[40] The most deprived quintile had the lowest participation percentages in all the years (16.5-17.7%), and the other four levels of fewer deprivation quantities presented similar proportions (18.1-23.1%). In addition, the proportions of single, divorced, or widowed status were also lower than that of married participants.[48]

Apart from age, gender, BMI, and sociodemographic information, we also extracted data on family history and lifestyle habits. **(Table 3)** A study from the Netherlands reported that around 70% of participants had a family history of type 2 diabetes,[46] whereas this uptake rate was under 40% in diabetes screening in Pakistan.[48] Of the two studies including smoking status, Sabir *et al* and Zafar *et al* reported low rates of participation among current smokers (9.7% and 13.4%, respectively).[44,48] Sabir *et al* also reported a very low proportion of screening uptake among alcohol drinkers (0.3%) in Nigeria.[44]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Lipid screening

A total of four studies offered lipid screening;[33,34,51,57] one cross-sectional screening survey was covered by research funding,[57] one study was workplace-based screening and insured by employers,[33] and the other two studies were for national screening programs supported by government organizations.[34,51] Only one study was from a developed country.[33] **(Table 1)**

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1 Characteristic of included studies

Author	Published Year	Country	Region/City	Project Period	Study design	Subsidization
Only blood glucose tests						
Bali, V.	2018	USA		Jan 2012-Dec 2014	Workplace screening (Health Advocate)	Self-insured employers
Bumrerraj, S.	2012	Thailand	Khon Kaen	June-Dec 2009	Cross-sectional study	Research funding
Katulanda, G. W.	2019	Sri Lanka		Aug 2005-Sep 2006	Cross-sectional study	Research funding
Lin, S.	2014	China	Guangzhou	Oct 2010-Jan 2011	Cross-sectional survey	Not mentioned
Mainous, A. G.	2014	England		2003, 2006, 2009, 2011	Health Survey for England	Information Centre for Health and Social Care and the Department of Health
Mayega, R. W.	2014	Uganda	Iganga-Mayuge		Cross-sectional survey	Research funding
Rush, E.	2008	New Zealand	Waikato District	May 2004-Mar 2006	Te Wai o Rona Diabetes Prevention Strategy (Cohort)	Roche company and research funding
Sabir, A.	2013	Nigeria	Gumbi&Wamakko		Cross-sectional study	Research funding
Valkengoed, I. G.	2015	Netherlands	Hague	June-Dec 2009 April-Nov 2010	DHIAAN screening	Research funding
Zafar, J.	2016	Pakistan	Rawalpindi	May-Sep 2014	Cross-sectional survey	Not mentioned
Ziemer, D. C.	2010	USA		Jan 2005-Mar 2008	Third National Health and Nutrition Examination Survey	Centers for Disease Control and Prevention
Only blood lipid tests						
Deng, B.	2012	China	Yunnan	July-Oct 2010	Cross-sectional survey	Research funding
Koyama, A. K.	2018	USA		Jan 2012-Dec 2014	Workplace screening (Health Advocate)	Self-insured employers
Kutkiene, S.	2018	Lithuania		2009-2016	Lithuanian High Cardiovascular Risk primary prevention programme	Ministry of Health
Zhao, W. H.	2007	China		2002	Chinese national nutrition and health survey	Chinese Center for Disease Control and Prevention
Combination tests of blood glucose and lipid						
Ali, I.	2019	Palestine	Nablus municipality	Aug 2017-Feb 2018	Cross-sectional study	Not mentioned
Andersson, S.	2016	Sweden		2013	Qualitative and quantitative study	Not mentioned
Bao, Y.	2010	China	Shanghai	May 2007-Aug 2008	Cross-sectional survey	Research funding
Belfki, H.	2013	Tunisia		Apr 2004-Sep 2005	Transition and Health Impact in North Africa project	Research funding
Cuong, T. Q.	2007	Vietnam	Ho Chi Minh City	Mar-Apr 2004	Cross-sectional survey	Not mentioned
Falguera, M.	2020	Spain	Mollerussa	Mar 2011-July 2014	Cohort study	Research funding
Gao, W.	2008	China	Qingdao	Apr-July 2002	Cross-sectional survey	Research funding
Hare, M. J.	2013	Mauritius	Main island& Rodrigues		Mauritius and Rodrigues Non-Communicable Disease Surveys	Ministry of Health and Quality of Life, Mauritius
Hidalgo, L. A.	2006	Ecuador		Dec 2011-June 2012	Cross-sectional study	Research funding
Hwu, C. M.	2008	China	Taiwan		Cross-sectional study	Research funding
Li, L. J.	2012	China	Hainan	May 2010-Aug 2011	Health check up	Not mentioned
Lissock, C. N.	2011	Cameroon	Ngambe&Littoral		Cross-sectional study	Not mentioned
Liu, Y.	2016	China	Changchun	2011-2012	Cohort baseline survey	Not mentioned
Nunes, B.	2019	Portugal		2015	Portuguese National Health Examination Survey	Organizations of Ministry of Health
Sinnott, M.	2015	Ireland	Dublin	Jan 2009-Dec 2012	The Diabetes Mellitus and Vascular health initiative (Cohort study)	Private health insurance provider (for members)
Wang, S.	2020	China	Shanghai	Mar-Aug 2010	Cohort baseline survey	Research funding
Zemlin, A. E.	2015	South African	Cape town	2008	Cross-sectional survey	Research funding
Zhou, X.	2018	China	Shanghai	Jan-July 2013	Cohort baseline survey	Research funding

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1 Characteristic of included studies (Continued)

Author	Age range	Ethnicity	Participants number	FPG	OGTT	HbA1c	RBG	LP	BP	Test Weight	Height	WC	HC	Others
Only blood glucose tests														
Bali, V.		Non-Hispanic White 77-78% African American 2.8-4% Asian/Pacific Islander 1-3% Hispanic 7% Other 2%	22,790	√	-	√	-	-	-	√	√	-	-	-
Bumrerraj, S.			874	-	√	-	-	-	-	√	√	√	-	-
Katulanda, G. W.			4,014	-	√	-	-	-	-	-	-	-	-	-
Lin, S.	>=50		1,312	√	√	√	-	-	-	-	-	-	-	-
Mainous, A. G.	>=16	(2003 2006 2009 2011) White 92.6% 90.5% 92.2% 85.3% South Asian 4.2% 5.7% 3.0% 4.0% Black 1.6% 1.9% 2.9% 3.0% Mixed/other 1.5% 1.8% 1.9% 7.7%	20,139	-	-	√	-	-	-	√	√	-	-	-
Mayega, R. W.	35-60		795	√	-	√	-	-	√	√	√	-	-	-
Rush, E.	>=28		5,264	-	√	-	-	-	-	√	√	√	-	Point-of-care testing
Sabir, A.			393	√	√	-	√	-	√	√	√	√	√	-
Valkengoed, I. G.	18-60		1,092	-	√	√	-	-	-	-	-	-	-	-
Zafar, J.	18-65		404	-	-	√	-	-	√	√	√	√	√	-
Ziemer, D. C.	18-87	non-Hispanic black and white	1,581	√	√	√	-	-	√	√	√	√	-	-
Only blood lipid tests														
Deng, B.	22-84	Han 51.7% Minority 48.3%	1,415	-	-	-	-	√	-	√	√	√	√	Food frequency
Koyama, A. K.			18,993	-	-	-	-	√	-	√	√	-	-	-
Kutkiene, S.	men 40-54 women 50-64		92,373	-	-	-	-	√	-	-	-	-	-	--
Zhao, W. H.	>=18		49,252	-	-	-	-	√	-	-	-	-	-	-
Combination tests of blood glucose and lipid														
Ali, I.	24-60		140	√	-	-	-	√	√	√	√	-	-	--
Andersson, S.	30-75		43	-	√	√	-	√	√	√	√	√	-	-
Bao, Y.	20-79		4,886	√	√	√	-	√	√	√	√	√	-	-
Belfki, H.	35-74		4,654	√	-	-	-	√	√	√	√	√	-	--
Cuong, T. Q.	20-60	Vietnamese 89.1% Chinese 10.8% Other 0.1%	1,488	√	-	-	-	√	√	√	√	√	√	Skinfold thicknesses
Falguera, M.	>=25		583	√	-	√	-	√	√	√	√	√	-	Renal function
Gao, W.	20-74		1,856	-	√	-	-	√	√	√	√	√	√	-
Hare, M. J.	19-78	African (40.7%) South Asian (57.0%) Chinese (2.3%)	6,701	-	√	√	-	√	√	√	√	√	√	-
Hidalgo, L. A.	40-65		204	√	-	-	-	√	√	√	√	√	-	-
Hwu, C. M.			49	√	√	-	-	√	√	√	√	√	-	-
Li, L. J.	27-91		1,337	√	-	√	-	√	√	√	√	-	-	Uric acid/Liver Function
Lissock, C. N.	>=18		452	√	-	-	-	√	√	√	√	√	√	Resting heart rate
Liu, Y.	>=40	Chinese	7,611	√	√	√	-	√	√	√	√	√	√	-
Nunes, B.	25-74		4,911	-	-	√	-	√	√	√	√	√	√	Blood count
Sinnott, M.	45-75		29,144	√	-	-	-	√	√	√	√	√	√	-
Wang, S.	>=40		9,389	√	√	√	-	√	√	√	√	√	-	-
Zemlin, A. E.			667	√	√	√	-	√	√	√	√	√	√	-
Zhou, X.	15-98	Chinese	7,909	√	√	√	-	√	√	√	√	√	-	Heart rate

FPG: fasting plasma glucose, OGTT: oral glucose tolerance tests, HbA1c: glycated hemoglobin, RGB: random plasma glucose, LP: lipid screening, BP: blood pressure, WC: waist circumference, HC: hip circumference.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2 Sociodemographic characteristics of participants in all included studies

Author	Marital status	Educational status	Occupational status	Others
Blood glucose tests only				
Mainous, A. G.	-	-	-	Social deprivation index (2003, 2006, 2009, 2011) Quintile 1 21.6% 20.6% 18.9% 20.1% Quintile 2 20.8% 20.2% 22.7% 23.1% Quintile 3 20.7% 22.3% 21.6% 21.0% Quintile 4 19.8 % 20.4% 20.3% 18.1% Quintile 5 17.0% 16.5% 16.5% 17.7%
Mayega, R. W.	-	None 139 (17.5%) Lower primary 174 (21.9%) Higher primary 315 (39.6%) Secondary 131 (16.5%) Tertiary 36 (4.5%)	Subsistence farmers 500 (62.9%) Traders 164 (20.6%) Formal/salaried 39 (4.9%) Mechanics 92 (11.6%)	-
Valkengoed, I. G.	-	Primary or less 14.7% 16.5% Secondary 11.7% 13.6% Lower vocational 56.8% 53.5%[c] Higher vocational 16.8% 16.5%[c]	Paid work (OGTT, HbA1c) 74.1% 71.5%	-
Zafar, J.	Married 322 (79.7%) Unmarried 61 (15.1%) Widow/Divorced 21 (5.2%)	Illiterate 55 (13.6%) Primary (1-5 grade) 59 (14.6%)[a] Middle (6-9 grade) 61 (15.1%)[b] > Matric 212 (52.5%)[c] Conventional 17 (4.2%)[c]	Govt. Employee 61 (15.1%) Private Employee 61 (15.1%) Self-Employee 80 (19.8%) Un-employed 84 (20.8%) Laborer 12 (3.0%) Any other 106 (26.2%)	Income (PKR) <10000 118 (29.2%) 10,000 - 30,000 175 (43.3%) 31000 - 50,000 72 (17.8%) > 50,000 39 (9.7%)
Plasma lipid tests only				
Deng, B.	Single 94 (6.6%) Married 1,306 (92.3%) Divorced 8 (0.6%) Widow(er) 7 (0.5%)	Never 21 (1.5%) Elementary school 116 (8.2%)[a] Junior-middle school 348 (24.6%)[b] Senior-high school 386 (27.3%)[b] Higher 544 (38.4%)[c]	-	-
Koyama, A. K.	-	Education, median (IQR) Less than high school 9.5% (5.7%-15.1%) High school or equivalent[b] 27.0% (19.5%-33.9%) Some college[c] 29.1% (24.2%-33.5%) College degree[c] 29.2% (19.2%-43.2%)	-	Median income, median (IQR) US$57,622 (US$45,161-US$75,313)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

285 **Table 2 Sociodemographic characteristics of participants in all included studies (Continued)**

Author	Marital status	Educational status	Occupational status	Others
Programs offering both diabetes and lipid tests				
Belfki, H.	Single 117 (2.5%) Married 4,035 (86.7%) Widowed/divorced 502 (10.8%)	Illiterate 2,041(43.9%) Low (<=6yrs) 1,552 (33.3%)[a] Intermediate(7-13yrs) 805 (17.3%)[b] Higher(>=14yrs) 238 (5.1%)[c]	No working/retired 2,479 (56.6%) Employee/worker 1,205 (27.5%) Intermediate 202 (4.6%) Upper 494 (11.3%)	-
Cuong, T. Q.	-	No schooling (1.8%) Primary school (16.3%)[a] Junior high school (33.3%)[b] Senior high school (33.3%)[b] College/University (15.3%)[c]	Teacher, Professional (10.2%) Government officers (14.9%) Small business, Skilled workers (17.9%) Laborers, street or home traders (24.3%) Retired/home maker/students (21.3%) Others (8.5%) No Job (2.9%)	Household wealth index (Male, female, total) Lowest 19.9% 19.9% NA Second 19.8% 20.1% NA Middle 20.5% 19.5% NA Fourth 21.2% 18.8% NA Highest 18.6% 21.7% NA
Falguera, M.	-	High level (>=secondary high school education) 421 (72.2%)	-	-
Hare, M. J.	-	Primary or none 3,317 (49.8%) Secondary 2,832 (42.5%) Tertiary 515 (7.7%)	-	-
Hidalgo, L. A.	Married 93 (45.6%) No 111 (54.4%)	0–6 73 (35.8%)[a] 7–12 68 (33.3%)[b] 13 63 (30.9%)[c]	-	-
Wang, S.	-	High school education or more 6,037 (64.6%)	-	-

286 [a] same as the primary education, b same as secondary education, c same as tertiary education; OGTT: oral glucose tolerance tests, HbA1c: glycated hemoglobin.
287

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 3 Family history and lifestyle habits and of participants

Author	Family history	Smoking	Drinking	Physical activity
Blood glucose tests only				
Sabir, A.		Total 38 (9.7%) Male 38 (18.1%) Female 0 (0%)	Total 1 (0.3%) Male 1 (0.5%) Female 0 (0%)	-
Valkengoed, I. G.	Type 2 diabetes mellitus 74.8% (75-g OGTT group) 68.4% (HbA1c group)			
Zafar, J.	Diabetes Parental history 161 (39.9%) Siblings history 88 (21.8%)	Smokers 54 (13.4%)	-	Exercise 105 (26.0%)
Plasma lipid tests only				
Deng, B.		Current smoking 476 (33.6%)	Current drinking 557 (39.4%)	>2/week /at least 30 mins 806 (57.0%)
Programs offering both diabetes and lipid tests				
Ali, I.	Dyslipidemia 27.9%	Smokers 37.1%	-	-
Andersson, S.		Smoking 19%	-	-
Belfki, H.	Cardiovascular disease 119 (2.6%)	Never 3,246 (71.4%) Smoker 1,301 (28.6%)	-	-
Cuong, T. Q.		Non/ex-smoker (69.3%) Current smoker (30.7%)	-	-
Falguera, M.	Diabetes 180 (30.9%)	Current smoker 148 (25.4%)	-	394 (67.6%)
Hare, M. J.		1,417 (21.2%)	None 3,217 (49.7%) Moderate 2,728 (42.1%) Excessive 531 (8.2%)	Sedentary 694 (10.6%) Insufficient (<150 min/week) 1,883 (28.8%) Sufficient (>=150 min/week) 3,951 (60.5%)
Hidalgo, L. A.		-	-	Sedentary Yes 69 (33.8%) No 135 (66.2%)
Sinnott, M.	Diabetes 9,301 (31.9%)	Ever smoked 11,648 (40.0%)	23,245 (79.8%)	>=5days/week 10,343 (35.5%)
Wang, S.	Diabetes 884 (9.4%)	Current smoking 1,926 (21.2%)	958 (10.5%)	>=600 MET-min/week 6,670 (71.2%)
Zhou, X.	Diabetes 847 (10.71%)	1,746 (22.1%)	1,079 (13.64%)	-

OGTT: oral glucose tolerance tests, HbA1c: glycated hemoglobin, MET: metabolic equivalent of task.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In these four articles, two articles reported body weight and height,[33,57] and one study stated both waist and hip circumference.[57] For participants, the proportion of females ranged from 48.8% to 58.4%. Only one study that targeted rural residents of Yunnan China showed a higher percentage of males (51.2%) among participants. **(Supplementary figure 2)** The mean age of the screening participants was older than 40 years. **(Figure 1)** Turning to mean BMI, however, both male and female subjects were lower than the national mean BMI reported by WHO (males: 23.8 kg/m^2 vs. 24.2 kg/m^2, $p<0.01$; females: 22.3 kg/m^2 vs. 23.6 kg/m^2, $p<0.01$). **(Figure 2)** Studies from Deng *et al* and Koyama *et al* reported the highest proportion of individuals having a higher educational level in China and the US (38.4% and 58.3%, respectively).[33,57] **(Table 2)**

Deng and colleagues reported the smoking, drinking, and physical activity habits of their lipid screening participants in China.[57] A third of them were current smokers. Moreover, 39.4% of subjects reported that they were current drinkers. For physical activity, 57.0% of them took exercise or participated in physical labor activities more than twice per week for at least 30 minutes each. **(Table 3)**

Participants receiving both diabetes and lipid screening

The studies that presented the characteristics of screening participants in programs that offered a screening of both diabetes and lipid profiles were listed in **Table 1**. Fourteen of these papers adopted FPG to evaluate blood glucose.[27,30,31,35,37,38,45,47,49,50,52,54,56,58] Ten studies used HbA1c,[28,29,35,38,42,47,49,50,52,58] nine studies used OGTT,[28,29,31,38,47,49,50,52,59] while five of the included studies offered all three

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

tests.[38,47,49,50,52] Only four studies were from developed countries.[28,42,45,58] Two were organized by the government,[29,42] and one was offered by a private health insurance provider for their members.[45]

For comparison of diabetes and lipid screening participation between males and females, only one study reported having more male than female subjects joining the screening program in Hainan China.[35] **(Supplementary figure 2)** Five studies indicated that male participants were older than females,[37,38,45,49] while the other two studies reported female participants being older.[50,54] **(Figure 1)** Turning to the mean BMI of participants, most of the studies reported a higher number than the national BMI levels, including those conducted in China (24.7 kg/m^2 vs. 23.9 kg/m^2, $p<0.0$;[52] 25.1 kg/m^2 vs. 23.9 kg/m^2, $p<0.01$;[47] 24.9 kg/m^2 vs. 23.9 kg/m^2, $p<0.01$;[38] 24.8 kg/m^2 vs. 23.9 kg/m^2, $p<0.01$;[35] 25.4 kg/m^2 vs. 23.9 kg/m^2, $p<0.01$);[31] South Africa (29.1 kg/m^2 vs. 27.3 kg/m^2, $p<0.01$);[49] Sweden (29.0 kg/m^2 vs. 25.8 kg/m^2, $p<0.01$);[28] and Tunisia (26.8 kg/m^2 vs. 27.8 kg/m^2, $p<0.01$).[54] **(Figure 2)** Overall, females had higher BMI levels than their national average as reported in most studies, whilst only two studies showed a higher BMI among male participants when compared with the WHO's report.[37,38] Some other screening tests were also included in other literature, such as skinfold thicknesses,[56] renal function,[58] uric acid,[35] liver function,[35] blood count,[42] and heart rate.[37,52]

For educational status, we found a substantial difference in participants' educational status across different studies. Belfki *et al* showed that 77.2% of participants had a low educational level in Tunisia.[54] However, Wang and colleagues reported that around 65% of the individuals had an educational level at higher school or above in

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Shanghai.[47] For occupational status, Belfki *et al* indicated more than half (56.6%) of the participants were retired or had no current jobs.[54] The corresponding proportion was only 24.2% in Cuong's study.[56] Cuong and colleagues also measured the household wealth index of participants and showed similar proportions among males (18.6-21.2%) and female participants (18.8-21.7%) across all levels of the index.[56] **(Table 2)**

Of the six studies that reported family history, one study described that 27.9% of the participants had a family history of dyslipidemia,[27] another study indicated the proportion of a family history of CVD was 2.6%,[54] and the other four studies mentioned that 9.4-31.9% of the participants with a family history of diabetes joined the screening programs.[45,47,52,58] **(Table 3)**

Of the ten studies with information on the status of smoking, three reported high rates of current smoking,[27,54,56] which were 37.1%, 28.6%, and 30.7%, respectively. Four studies stated the participants' alcohol consumption, one reported a high rate (79.8%),[45] compared with that reported in two studies (10.5-13.64%).[47,52] Among the five studies which included physical activity, three of them reported that more than half of the participants performed adequate physical activity (60.5-71.2%).[29,47,58] **(Table 3)**

The reasons for non-participation

Four studies mentioned the reasons for non-participation. Two studies on diabetes screening assessed reasons for non-participation. Mayega *et al* reported 5.4% of subjects refused to participate due to travel distance and 4.2% of non-participants

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

declined the blood tests.[41] Valkengoed *et al* described that "no time" and "no interest" were the major reasons among prospective eligible participants.[46] Regarding lipid screening, Deng *et al* explained that 1.0% of subjects declined screening tests due to time constraints.[57] Nunes and colleagues showed that 43.9% of individuals participated in the first Portuguese National Health Examination Survey involving both diabetes and lipid screening.[42] Furthermore, they found that work-related issues (26.6%) and lack of time (26.6%) were the most frequently mentioned reasons for not joining.

DISCUSSION

This systematic review provided socioeconomic and demographic characteristics of participants who received screening tests for diabetes and lipid disorders, including age, gender, BMI, sociodemographic status, family history, and lifestyle habits. We found a higher proportion of females among the screening participants when compared with the male gender in most studies, irrespective of the tests offered. Participants who joined diabetes screening tended to have higher BMI values, whilst those who received lipid screening tended to have lower BMI values. Regarding screening programs offering both diabetes and lipid tests, female participants had higher BMI values and males had lower BMI values than the national means reported by the WHO.

Sargeant and colleagues conducted a stepwise program to screen diabetes, and they reported male gender was significantly associated with lower uptake of RBG testing, which is consistent with our findings on diabetes screening.[61] A serial study of cross-sectional data (1992-2008) evaluated the utilization of preventive health assessments in the UK, and found that the participation rate of lipid tests increased with age, but they

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

reported an absence of association between gender and participation in lipid tests.[62] They also showed that smokers (vs. non-smokers) had significantly lower uptake of lipid tests. In our review, around one-third of participants were current smokers in lipid tests and programs offering both diabetes and lipid tests. A longitudinal study indicated that current smokers had limited health literacy and lower cognitive ability than ever smokers,[63] which may explain their lower screening uptake rates. Interventions to address the health literacy of prospective participants are needed to improve screening participation, which at the same time could also help to change their lifestyle habits for the prevention of chronic diseases.[64]

A postal survey investigated the participation rate of women's health check-ups, including identification of CVD risk factors, as well as examinations for type 2 diabetes and kidney disease in Germany in 2004.[65] Its adjusted model showed that single, divorced, separated or widowed women (OR=0.63, 95% CI: 0.57-0.71) were less likely to receive a preventive health check-up when compared with married women. The proportions of single, divorced, or widowed status were also lower than that of married participants in our review, except for a study performed in Italy. This may be due to peer effects where family members may exert a motivational influence for chronic diseases screening. It is well recognized that SES influenced disease risk, health behavior, and health care of individuals. These health inequities have been affirmed by the WHO.[66] Researchers in Germany found that there was a significant association between women with lower SES (estimated based on educational status, occupational position and household income) (OR=0.82, 95% CI: 0.74-0.92) and lower participation rate of health screening attendance.[65] This is consistent with our finding that health inequities might exist as the most deprived population had the lowest participation rates

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

in England, where only blood glucose tests were provided. Meanwhile, participants of studies in Spain and China (Shanghai) had higher educational status, whereas studies in Tunisia, Mauritius, and Ecuador recruited more participants with lower educational level. Health inequities are more likely to be found in developed areas,[67] yet they also exist in developing countries due to differential health literacy and access to primary care services. For the non-participation of screening programs among those with lower educational levels, health literacy plays an important role in screening participation. The American Heart Association highlighted that low health literacy is a barrier to healthcare screening and basic services for diabetes and CVD.[68] A systematic review included studies that evaluated the effectiveness of health literacy interventions using pre- and post-program assessment, and concluded their positive impact on enhancing screening uptake.[64]

Hypertension is also a risk factor to contribute the CVD, but we did not include it as a target measurement. Because the blood tests had more obstacles than the blood pressure tests, for example, the fear of needles, fear of pain, and cost of test strips and needles.[69] Thus in this study, we focused on the blood glucose tests and lipid profiles to explore the characteristics of participants.

To our knowledge, this is the first systematic review that presented the characteristics of screening participants in diabetes and lipid screening programs. The findings inform population groups where interventions to enhance screening uptake should be targeted. Nevertheless, there are several limitations. Firstly, this is a systematic review of literature presented in a descriptive manner. Therefore, we cannot quantify the association between patients' characteristics and screening participation. In addition,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

we included articles that used different screening tests for diabetes, and this might pose challenges on comparison across studies. The American Diabetes Association proposed that FPG is the best screening test and also a component of the diagnostic test for diabetes.[70] FPG is more convenient and acceptable to patients since it is easier and faster to perform, and is less expensive than OGTT. There are differences in the organization of different screening programs in terms of the tests offered, subsidy amount, and accessibility of the screening centers, and these could influence the screening uptake rate. In addition, the prevalence of risk factors in screening participants who take up the tests might be different from those who do not.

Overall, the present study systematically summarized the characteristics of screening participants in diabetes and lipid screening programs. A higher proportion of female participants was found in diabetes and lipid screening programs than males in most studies. For diabetes screening, participants tended to have higher BMI, while it was the opposite for subjects who participated in lipid screening. Meanwhile, females tended to have higher BMI and males tended to have lower BMI in screening programs offering both diabetes and lipid tests. Participants with lower SES were less likely to undergo screening in developed areas. Around one-third of participants were current smokers in lipid screening and a combination of lipid and diabetes screening programs. The proportion of participants using alcohol and performing regular physical activity varied substantially among included studies. Our findings could inform future studies to investigate the enablers and barriers of screening among participants and non-participants based on their characteristics, and also interventions to enhance their screening uptake.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Acknowledgements: We appreciate the invaluable contributions of Ms. Xiao Chen and Ms. Veeleah Lok.

Competing Interests: Ms. Pamela Sze Pui Tin is from Our Hong Kong Foundation. The other authors declare no competing interests.

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability: All data generated or analysed during this study are included in this published article (and its Supplementary Information files).

Ethics declarations: N/A

Author contributions: HD: Formal analysis, Investigation, Data Curation, Writing - Original Draft; JH: Writing - Review & Editing; YD: Validation, Writing - Review & Editing; PT: Writing - Review & Editing; MCSW: Methodology, Supervision, Project administration; EKY: Conceptualization, Supervision.

FIGURE LEGENDS

Figure 1 The mean age of screening participants

Figure 2 The Body Mass Index (BMI) of screening participants

We compared the national mean BMI of adults, extracted from the World Health Organization (WHO) report, with the participants' average BMI.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

REFERENCE

1 World Health Organization. Cardiovascular diseases (CVDs). World Heal. Organ. 2017.https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds) (accessed 8 May 2020).

2 World Health Organization. WHO/Europe | Cardiovascular diseases - Data and statistics. 2020.http://www.euro.who.int/en/health-topics/noncommunicable-diseases/cardiovascular-diseases/data-and-statistics (accessed 13 May 2020).

3 American Heart Association. Statistical Fact Sheet - International Cardiovascular Disease Statistics. 2004.http://www.sld.cu/galerias/pdf/servicios/hta/international_cardiovascular_disease_statistics.pdf (accessed 13 May 2020).

4 Ohira T, Iso H. Cardiovascular disease epidemiology in Asia - An overview. *Circ J* 2013;**77**:1646–52. doi:10.1253/circj.CJ-13-0702

5 Mendis S, Puska P, Norrving B. Global atlas on cardiovascular disease prevention and control. 2011.

6 Francula-Zaninovic S, Nola IA. Management of Measurable Variable Cardiovascular Disease' Risk Factors. *Curr Cardiol Rev* 2018;**14**:153–63. doi:10.2174/1573403x14666180222102312

7 American Diabetes Association. Diagnosis and classification of diabetes mellitus. *Diabetes Care* 2014;**37**:S81–90. doi:10.2337/dc14-S081

8 World Health Organization. Diabetes. 2020.https://www.who.int/news-room/fact-sheets/detail/diabetes (accessed 11 May 2020).

9 Saeedi P, Petersohn I, Salpea P, *et al.* Global and regional diabetes prevalence estimates for 2019 and projections for 2030 and 2045: Results from the International Diabetes Federation Diabetes Atlas, 9th edition. *Diabetes Res Clin Pract* 2019;**157**:107843. doi:10.1016/j.diabres.2019.107843

10 World Health Organization. Disease burden and mortality estimates. 2020.https://www.who.int/healthinfo/global_burden_disease/estimates/en/index1.html (accessed 9 Jun 2020).

11 World Health Organization. WHO | Raised cholesterol. 2020.https://www.who.int/gho/ncd/risk_factors/cholesterol_text/en/ (accessed 7 Jul 2020).

12 Dehmer SP, Maciosek M V., LaFrance AB, *et al.* Health benefits and cost-effectiveness of asymptomatic screening for hypertension and high cholesterol and aspirin counseling for primary prevention. *Ann Fam Med* 2017;**15**:23–36. doi:10.1370/afm.2015

13 Hoerger TJ, Harris R, Hicks KA, *et al.* Screening for Type 2 Diabetes Mellitus: A Cost-Effectiveness Analysis. *Ann Intern Med* 2004;**140**:689. doi:10.7326/0003-4819-140-9-200405040-00008

14 Bibbins-Domingo K, Grossman DC, Curry SJ, *et al.* Statin use for the primary prevention of cardiovascular disease in adults: US preventive services task force recommendation statement. JAMA - J. Am. Med. Assoc. 2016;**316**:1997–2007. doi:10.1001/jama.2016.15450

15 NICE guideline. Cardiovascular disease: risk assessment and reduction, including lipid modification. Natl. Inst. Heal. Care Excell. 2016.https://www.nice.org.uk/guidance/cg181/resources/surveillance-report-2018-cardiovascular-disease-risk-assessment-and-reduction-including-lipid-modification-2014-nice-guideline-cg181-4724759773/chapter/Surveillance-decision?tab=evidence (accessed 7 Jul 2020).

16 The Royal Australian College of General Practitioners. Guidelines for preventive activities in general practice 9th edition. 2018.https://www.racgp.org.au/download/Documents/Guidelines/Redbook9/17048-Red-Book-9th-Edition.pdf (accessed 9 Jun 2020).

17 Allan GM, Lindblad AJ, Comeau A, *et al.* Simplified lipid guidelines: Prevention and management of cardiovascular disease in primary care. *Can Fam Physician* 2015;**61**:857–67, e439-50.

18 Ministry of Health. LIPIDS EXECUTIVE SUMMARY. 2016.https://www.moh.gov.sg/docs/librariesprovider4/guidelines/moh-lipids-cpg---executive-summary.pdf (accessed 10 Jun 2020).

19 Ekoe JM, Goldenberg R, Katz P. Screening for Diabetes in Adults. *Can J Diabetes* 2018;**42**:S16–9. doi:10.1016/j.jcjd.2017.10.004

20 Goh SY, Ang S Bin, Bee YM, *et al.* Ministry of Health clinical practice guidelines: Diabetes mellitus. *Singapore Med J* 2014;**55**:334–47. doi:10.11622/smedj.2014079

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

21 Siu AL. Screening for abnormal blood glucose and type 2 diabetes mellitus: U.S. preventive services task force recommendation statement. Ann. Intern. Med. 2015;**163**:861–8. doi:10.7326/M15-2345

22 Kiefer MM, Silverman JB, Young BA, *et al.* National Patterns in Diabetes Screening: Data from the National Health and Nutrition Examination Survey (NHANES) 2005–2012. *J Gen Intern Med* 2015;**30**:612–8. doi:10.1007/s11606-014-3147-8

23 Norberg M, Wall S, Boman K, *et al.* The Västerbotten Intervention Programme: background, design and implications. *Glob Health Action* 2010;**3**:4643. doi:10.3402/gha.v3i0.4643

24 Robson J, Dostal I, Sheikh A, *et al.* The NHS Health Check in England: an evaluation of the first 4 years. doi:10.1136/bmjopen-2015-008840

25 Stroup DF, Berlin JA, Morton SC, *et al.* Meta-analysis of observational studies in epidemiology: A proposal for reporting. *J Am Med Assoc* 2000;**283**:2008–12. doi:10.1001/jama.283.15.2008

26 Downes MJ, Brennan ML, Williams HC, *et al.* Development of a critical appraisal tool to assess the quality of cross-sectional studies (AXIS). *BMJ Open* 2016;**6**. doi:10.1136/bmjopen-2016-011458

27 Ali I, Kharma A, Samara M, *et al.* Prevalence of Dyslipidemia in Undiagnosed Palestinian Men: A Cross-Sectional Study. *J Lipids* 2019;**2019**:1–6. doi:10.1155/2019/3473042

28 Andersson S, Karlsson V, Bennet L, *et al.* Attitudes Regarding Participation in a Diabetes Screening Test among an Assyrian Immigrant Population in Sweden. *Nurs Res Pract* 2016;**2016**:1–7. doi:10.1155/2016/1504530

29 Hare MJL, Magliano DJ, Zimmet PZ, *et al.* Glucose-independent ethnic differences in HbA1c in people without known diabetes. *Diabetes Care* 2013;**36**:1534–40. doi:10.2337/dc12-1210

30 Hidalgo LA, Chedraui PA, Morocho N, *et al.* The metabolic syndrome among postmenopausal women in Ecuador. *Gynecol Endocrinol* 2006;**22**:447–54. doi:10.1080/09513590600890272

31 Hwu CM, Lin MW, Liou TL, *et al.* Fasting triglyceride is a major determinant of postprandial triglyceride response in postmenopausal women. *Menopause* 2008;**15**:150–6. doi:10.1097/gme.0b013e31800577856

32 Katulanda GW, Katulanda P, Dematapitiya C, *et al.* Plasma glucose in screening for diabetes and pre-diabetes: How much is too much? Analysis of fasting plasma glucose and oral glucose tolerance test in Sri Lankans. *BMC Endocr Disord* 2019;**19**:11. doi:10.1186/s12902-019-0343-x

33 Koyama AK, Bali V, Yermilov I, *et al.* Identification of Undiagnosed Hyperlipidemia: Do Work Site Screening Programs Work? *Am J Heal Promot* 2018;**32**:971–8. doi:10.1177/0890117116671537

34 Kutkiene S, Petrulioniene Z, Laucevicius A, *et al.* Lipid profile evaluation and severe hypercholesterolaemia screening in the middle-aged population according to nationwide primary prevention programme in Lithuania. *Atherosclerosis* 2018;**277**:267–72. doi:10.1016/j.atherosclerosis.2018.06.008

35 Li LJ, Zhou JX, Chen HT, *et al.* Effect of HbA 1c combined FPG on screening diabetes in health check-up. *Asian Pac J Trop Med* 2012;**5**:472–5. doi:10.1016/S1995-7645(12)60081-9

36 Lin S, Hu L, Li X, *et al.* Glycated haemoglobin A1c for diagnosing diabetes in Chinese subjects over 50 years old: A community-based cross-sectional study. *Clin Endocrinol (Oxf)* 2014;**80**:656–61. doi:10.1111/cen.12202

37 Lissock CNAA, Sobngwi E, Ngassam E, *et al.* Rural and urban differences in metabolic profiles in a Cameroonian population. *Pan Afr Med J* 2011;**10**:1. doi:10.4314/pamj.v10i0.72204

38 Liu Y, Xiao X, Sun C, *et al.* Ideal glycated hemoglobin cut-off points for screening diabetes and prediabetes in a Chinese population. *J Diabetes Investig* 2016;**7**:695–702. doi:10.1111/jdi.12498

39 Bali V, Yermilov I, Koyama A, *et al.* Secondary prevention of diabetes through workplace health screening. *Occup Med (Chic Ill)* 2018;**68**:610–6. doi:10.1093/OCCMED/KQY138

40 Mainous AG, Tanner RJ, Baker R, *et al.* Prevalence of prediabetes in England from 2003 to 2011: population-based, cross-sectional study. *BMJ Open* 2014;**4**:e005002. doi:10.1136/bmjopen-2014-005002

41 Mayega RW, Guwatudde D, Makumbi FE, *et al.* Comparison of fasting plasma glucose and haemoglobin A1c point-of-care tests in screening for diabetes and abnormal glucose regulation in a rural low income setting. *Diabetes Res Clin Pr* 2014;**104**:112–20. doi:10.1016/j.diabres.2013.12.030

42 Nunes B, Barreto M, Gil AP, *et al.* The first Portuguese National Health Examination Survey

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

(2015): design, planning and implementation. *J Public Health (Oxf)* 2019;**41**:511–7. doi:10.1093/pubmed/fdy150

43 Rush E, Crook N, Simmons D. Point-of-care testing as a tool for screening for diabetes and pre-diabetes. *Diabet Med* 2008;**25**:1070–5. doi:10.1111/j.1464-5491.2008.02526.x

44 Sabir A, Ohwovoriole A, Isezuo S, *et al.* Type 2 diabetes mellitus and its risk factors among the rural Fulanis of Northern Nigeria. *Ann Afr Med* 2013;**12**:217–22. doi:10.4103/1596-3519.122689

45 Sinnott M, Kinsley BT, Jackson AD, *et al.* Fasting plasma glucose as initial screening for diabetes and prediabetes in irish adults: The Diabetes Mellitus and Vascular health initiative (DMVhi). *PLoS One* 2015;**10**:e0122704. doi:10.1371/journal.pone.0122704

46 Valkengoed VIG, Vlaar EM, Nierkens V, *et al.* The Uptake of Screening for Type 2 Diabetes and Prediabetes by Means of Glycated Hemoglobin versus the Oral Glucose Tolerance Test among 18 to 60-Year-Old People of South Asian Origin: A Comparative Study. *PLoS One* 2015;**10**:e0136734. doi:10.1371/journal.pone.0136734

47 Wang S, Niu J, Zhao Z, *et al.* Detection of diabetes and prediabetes using glycosylated hemoglobin in Chinese adults living in Shanghai: A prospective analysis. *J Diabetes* Published Online First: 2020. doi:10.1111/1753-0407.13028

48 Zafar J, Nadeem D, Khan SA, *et al.* Prevalence of diabetes and its correlates in urban population of Pakistan: A Cross-sectional survey. *J Pak Med Assoc* 2016;**66**:922–7.

49 Zemlin AE, Matsha TE, Kengne AP, *et al.* Derivation and validation of an HbA1c optimal cutoff for diagnosing prediabetes in a South African mixed ancestry population. *Clin Chim Acta* 2015;**448**:215–9. doi:10.1016/j.cca.2015.06.019

50 Bao Y, Ma X, Li H, *et al.* Glycated haemoglobin A1c for diagnosing diabetes in Chinese population: Cross sectional epidemiological survey. *BMJ* 2010;**340**:1178. doi:10.1136/bmj.c2249

51 Zhao WH, Zhang J, Zhai Y, *et al.* Blood lipid profile and prevalence of dyslipidemia in Chinese adults. *Biomed Env Sci* 2007;**20**:329–35.

52 Zhou X, Ruan X, Hao L, *et al.* Optimal hemoglobin A1C cutoff value for diabetes mellitus and pre-diabetes in Pudong New Area, Shanghai, China. *Prim Care Diabetes* 2018;**12**:238–44. doi:10.1016/j.pcd.2017.12.006

53 Ziemer DC, Kolm P, Weintraub WS, *et al.* Glucose-independent, black-white differences in hemoglobin A1c levels: a cross-sectional analysis of 2 studies. *Ann Intern Med* 2010;**152**:770–7. doi:10.7326/0003-4819-152-12-201006150-00004

54 Belfki H, Ben Ali S, Aounallah-Skhiri H, *et al.* Prevalence and determinants of the metabolic syndrome among Tunisian adults: results of the Transition and Health Impact in North Africa (TAHINA) project. *Public Heal Nutr* 2013;**16**:582–90. doi:10.1017/s1368980012003291

55 Bumrerraj S, Kaczorowski J, Kessomboon P, *et al.* Diagnostic performance of 2 h postprandial capillary and venous glucose as a screening test for abnormal glucose tolerance. *Prim Care Diabetes* 2012;**6**:207–11. doi:10.1016/j.pcd.2012.03.002

56 Cuong TQ, Dibley MJ, Bowe S, *et al.* Obesity in adults: An emerging problem in urban areas of Ho Chi Minh City, Vietnam. *Eur J Clin Nutr* 2007;**61**:673–81. doi:10.1038/sj.ejcn.1602563

57 Deng B, Luo T, Huang Y, *et al.* Prevalence and determinants of hyperlipidemia in moderate altitude areas of the yunnan-kweichow plateau in southwestern China. *High Alt Med Biol* 2012;**13**:13–21. doi:10.1089/ham.2011.1037

58 Falguera M, Vilanova MB, Alcubierre N, *et al.* Prevalence of pre-diabetes and undiagnosed diabetes in the Mollerussa prospective observational cohort study in a semi-rural area of Catalonia. *BMJ Open* 2020;**10**:e033332. doi:10.1136/bmjopen-2019-033332

59 Gao W, Dong Y, Nan H, *et al.* The likelihood of diabetes based on the proposed definitions for impaired fasting glucose. *Diabetes Res Clin Pract* 2008;**79**:151–5. doi:10.1016/j.diabres.2007.07.018

60 Mendis S, Armstrong T, Bettcher D, *et al.* Global status report on noncommunicable diseases 2014. World Health Organisation. 2014. doi:ISBN 9789241564854

61 Sargeant LA, Simmons RK, Barling RS, *et al.* Who attends a UK diabetes screening programme? Findings from the ADDITION-Cambridge study. *Diabet Med* 2010;**27**:995–1003. doi:10.1111/j.1464-5491.2010.03056.x

62 Labeit A, Peinemann F, Baker R. Utilisation of preventative health check-ups in the UK: Findings from individual-level repeated cross-sectional data from 1992 to 2008. *BMJ Open* 2013;**3**. doi:10.1136/bmjopen-2013-003387

63 Fawns-Ritchie C, Starr JM, Deary IJ. Health literacy, cognitive ability and smoking: A cross-sectional analysis of the English Longitudinal Study of Ageing. *BMJ Open* 2018;**8**:e023929.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

doi:10.1136/bmjopen-2018-023929

64 Walters R, Leslie SJ, Polson R, *et al.* Establishing the efficacy of interventions to improve health literacy and health behaviours: A systematic review. *BMC Public Health* 2020;**20**:1040. doi:10.1186/s12889-020-08991-0

65 Schülein S, Taylor KJ, Schriefer D, *et al.* Participation in preventive health check-ups among 19,351 women in Germany. *Prev Med Reports* 2017;**6**:23–6. doi:10.1016/j.pmedr.2017.01.011

66 World Health Organization. Social determinants of health. World Heal. Organ. 2013.https://www.who.int/social_determinants/thecommission/finalreport/key_concepts/en/ (accessed 6 May 2020).

67 Orach CG. Health equity: challenges in low income countries. Afr. Health Sci. 2009;**9 Suppl 2**:S49.

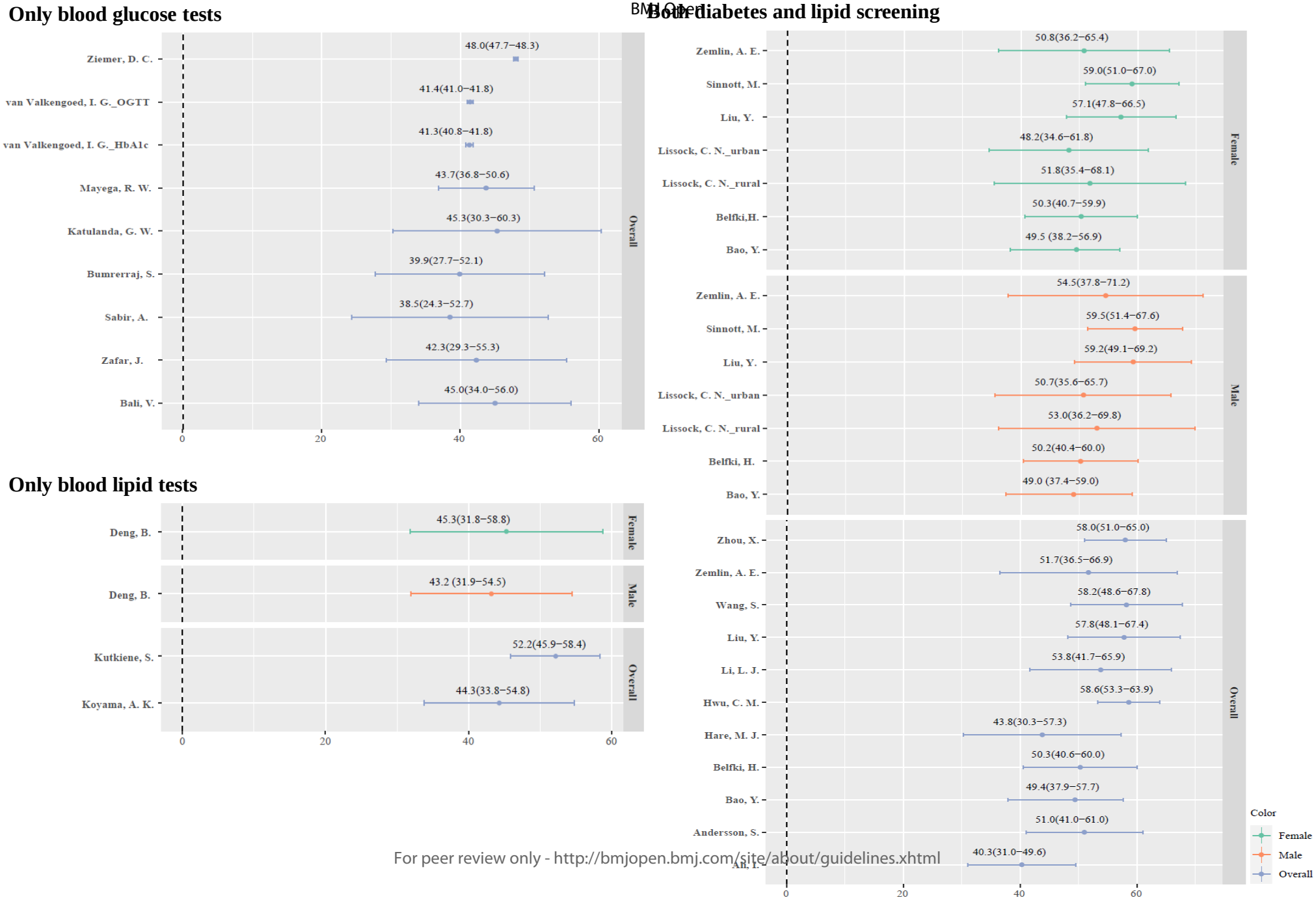
68 Magnani JW, Mujahid MS, Aronow HD, *et al.* Health Literacy and Cardiovascular Disease: Fundamental Relevance to Primary and Secondary Prevention: A Scientific Statement From the American Heart Association. Circulation. 2018;**138**:e48–74. doi:10.1161/CIR.0000000000000579

69 Ong WM, Chua SS, Ng CJ. Barriers and facilitators to self-monitoring of blood glucose in people with type 2 diabetes using insulin: A qualitative study. *Patient Prefer Adherence* 2014;**8**:237–46. doi:10.2147/PPA.S57567

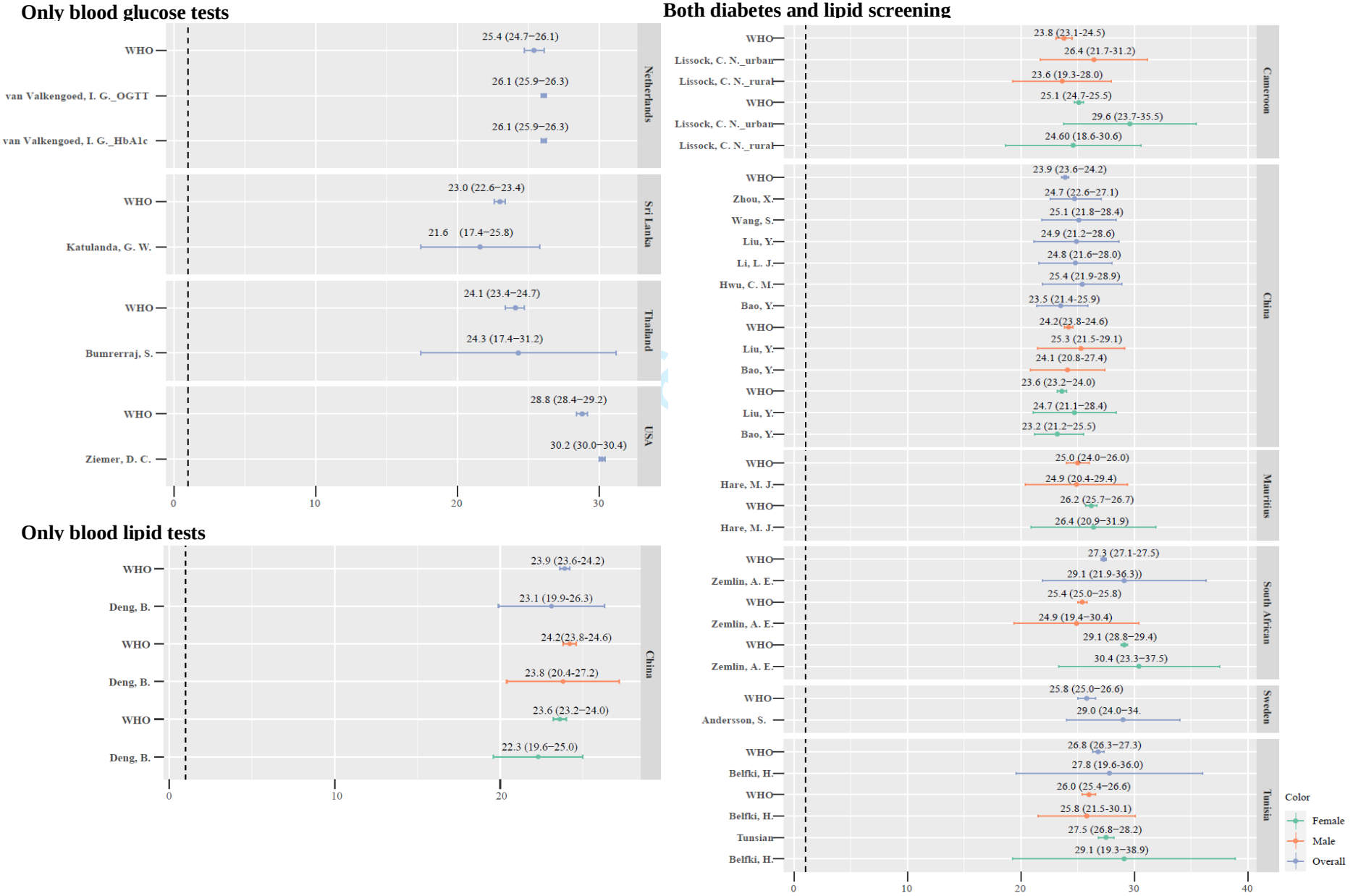
70 American Diabetes Association. Screening for diabetes. *Diabetes Care* 2002;**25**:s21–4. doi:10.2337/diacare.25.2007.s21

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml


Only blood glucose tests
WHO
van Valkengoed, I. G._OGTT
van Valkengoed, I. G._HbA1c
Katulanda, G. W.
Bumrerraj, S.
Ziemer, D. C.
Netherlands
Sri Lanka
Thailand
USA
25.4 (24.7−26.1)
26.1 (25.9−26.3)
26.1 (25.9−26.3)
23.0 (22.6−23.4)
21.6 (17.4−25.8)
24.1 (23.4−24.7)
24.3 (17.4−31.2)
28.8 (28.4−29.2)
30.2 (30.0−30.4)
0
10
20
30
Only blood lipid tests
Deng, B.
China
23.9 (23.6-24.2)
23.1 (19.9-26.3)
24.2(23.8-24.6)
23.8 (20.4-27.2)
23.6 (23.2−24.0)
22.3 (19.6−25.0)
0
10
20
Both diabetes and lipid screening
Lissock, C. N._urban
Lissock, C. N._rural
Cameroon
23.8 (23.1-24.5)
26.4 (21.7-31.2)
23.6 (19.3-28.0)
25.1 (24.7-25.5)
29.6 (23.7-35.5)
24.60 (18.6-30.6)
Zhou, X.
Wang, S.
Liu, Y.
Li, L. J.
Hwu, C. M.
Bao, Y.
China
23.9 (23.6−24.2)
24.7 (22.6−27.1)
25.1 (21.8−28.4)
24.9 (21.2−28.6)
24.8 (21.6−28.0)
25.4 (21.9-28.9)
23.5 (21.4-25.9)
24.2(23.8-24.6)
25.3 (21.5-29.1)
24.1 (20.8-27.4)
23.6 (23.2−24.0)
24.7 (21.1−28.4)
23.2 (21.2−25.5)
Hare, M. J.
Mauritius
25.0 (24.0−26.0)
24.9 (20.4-29.4)
26.2 (25.7−26.7)
26.4 (20.9−31.9)
Zemlin, A. E.
South African
27.3 (27.1-27.5)
29.1 (21.9-36.3))
25.4 (25.0−25.8)
24.9 (19.4−30.4)
29.1 (28.8−29.4)
30.4 (23.3−37.5)
Andersson, S.
Sweden
25.8 (25.0−26.6)
29.0 (24.0−34.
Belfki, H.
Tunsian
Tunisia
26.8 (26.3−27.3)
27.8 (19.6-36.0)
26.0 (25.4−26.6)
25.8 (21.5-30.1)
27.5 (26.8−28.2)
29.1 (19.3−38.9)
0
10
20
30
40
Color
Female
Male
Overall

Supplementary materials

Supplementary figure 1 Selection of articles for systematic review

Supplementary figure 2 The gender of screening participants

Supplementary table 1 Summary of diabetes and lipid screening guidelines

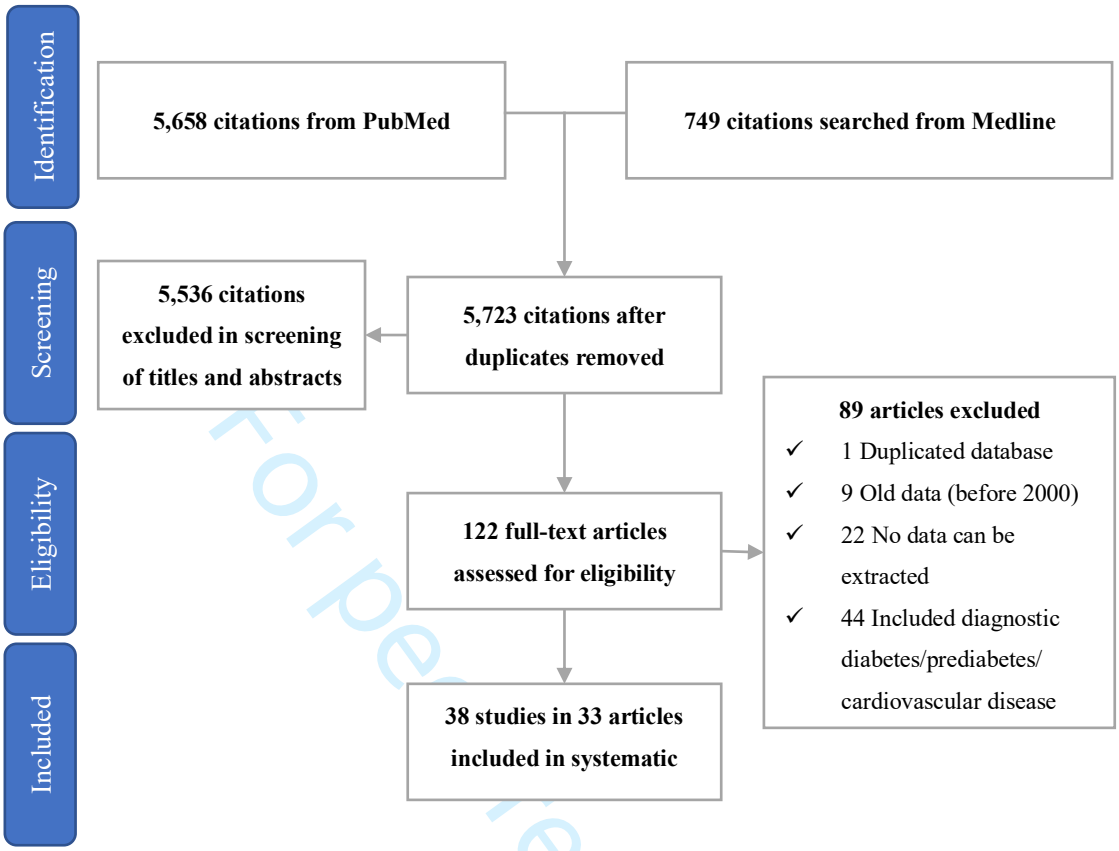
Supplementary table 2 Search strategy for article selection

Supplementary table 3 Quality Assessment of included articles based on Appraisal of Cross-sectional Studies (AXIS)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

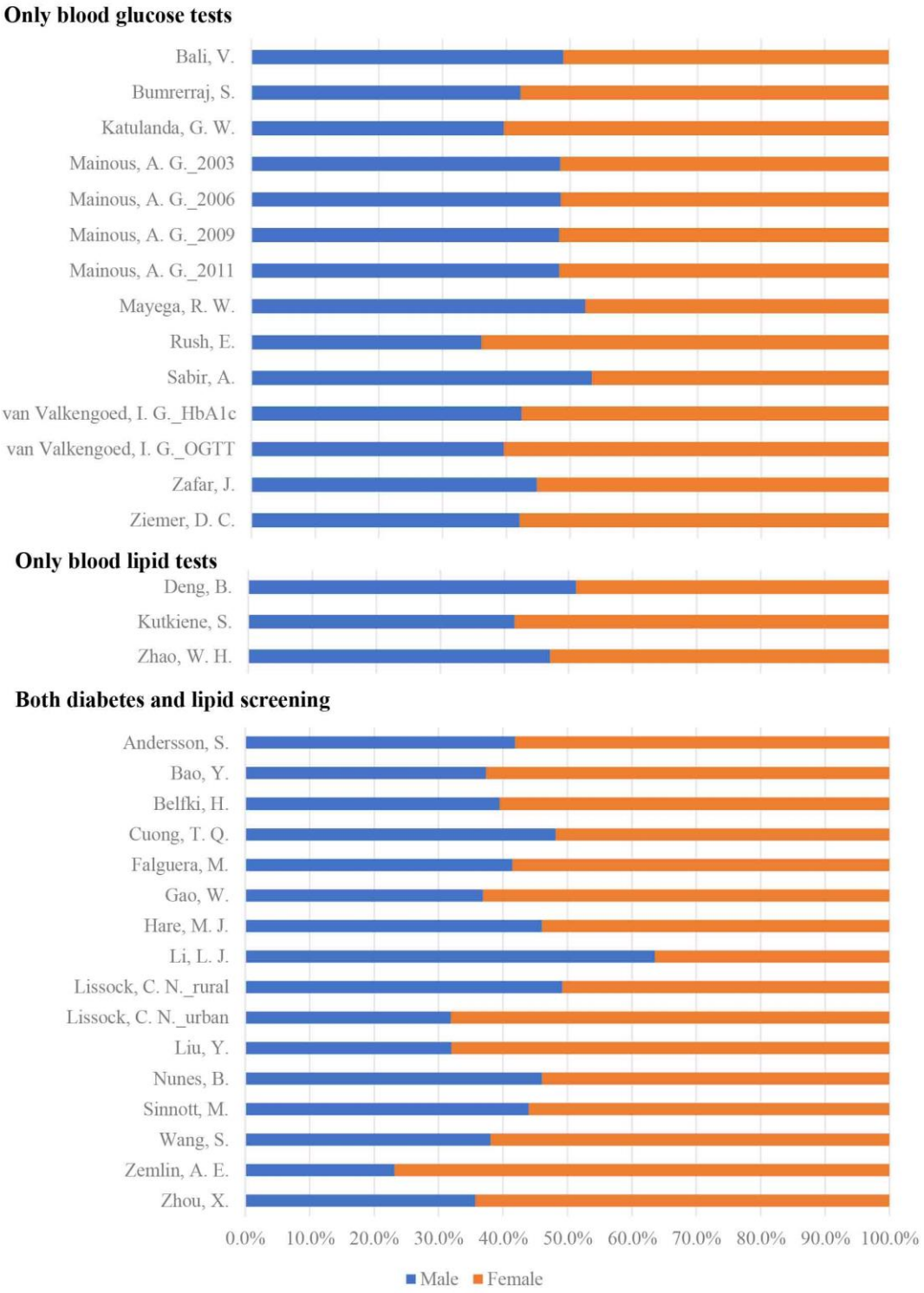
Supplementary figure 1 Selection of articles for systematic review



For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary figure 2 The gender of screening participants



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary table 1 Summary of diabetes and lipid screening guidelines

Country	Diabetes Screening Guidelines	Lipid Screening Guidelines
Australia	Patients should be screened for diabetes risk every three years from 40 years of age using the Australian type 2 diabetes risk assessment tool. Aboriginal and Torres Strait Islander peoples should be screened from 18 years of age. *(RACGP)*	Adults should have their blood lipids assessed every five years starting at 45 years of age. Aboriginal and Torres Strait Islander adults should have lipid tests performed every five years from 35 years of age. *(RACGP)*
Canada	Adults should be tested for type 2 diabetes every 3 years if they are age 40 years or over, or if they are at risk for T2DM. If they have risk factors for T2DM, they should be tested more frequently and/or start regular screening earlier. *(Canada Clinical Practice Guidelines Expert Committee)*	Initiating screening: In patients without CVD (primary prevention), we suggest lipid testing as part of global CVD risk estimation in men at age ≥40 years old and women at age ≥50 years old. Testing can be considered earlier for patients with known traditional CVD risk factors. *(College of Family Physicians of Canada)*
Singapore	Screening for T2DM should begin at 40 years and older; screening should also be considered in adults of any age who have one or more risk factors for diabetes. *(Singapore Ministry of Health)*	Clinicians should routinely screen men and women aged 40 years and older for lipid disorders, and younger adults for lipid disorders if they have other risk factors for coronary artery disease. *(Singapore Ministry of Health)*
United Kingdom	The UK National Screening Committee does not currently recommend screening for T2DM. This is because: there is no evidence that screening is more beneficial than not screening, and there is no evidence that finding T2DM early by screening results in a greater health benefit than usual health care.	National Institute for Health and Care Excellence guideline CG181 recommends (1.1.8) using the QRISK2 risk assessment tool (including ethnicity, age, sex, smoking, blood pressure, lipid profile, BMI, etc.) to assess CVD risk for the primary prevention of CVD in people up to and including age 84 years. There is also a potential need to amend recommendations 1.1.8 and 1.1.10 to advise the use of QRISK3 (including sex, age, BMI, lipid profile, blood pressure, family history, etc.) in place of QRISK2 because QRISK2 is due to be superseded by QRISK3 in 2018.
United States	Adults aged 40 to 70 who are overweight or obese are recommended to screen for abnormal blood glucose. (*USPSTF)*	Identification of dyslipidaemia and calculation of 10-year CVD event risk requires universal lipids screening in adults aged 40 to 75 years. *(USPSTF)*

RACGP: The Royal Australian College of General Practitioners, T2DM: type 2 diabetes mellitus, CVD: cardiovascular disease, USPTF: US preventive services task force recommendation statement

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary table 2 Search strategy for article selection

Database	Search strategy	Result
PubMed	(("fasting plasma glucose" OR "postprandial plasma glucose" OR "oral glucose tolerance" OR "random plasma glucose" OR "glycated haemoglobin" OR "HbA1c" OR "A1C") AND (screening* OR test*) AND (uptake OR participate) NOT (Paediatric OR gestational))	**5,658**
Ovid Medline	(fasting plasma glucose OR postprandial plasma glucose OR oral glucose tolerance OR random plasma glucose OR glycated haemoglobin OR HbA1c OR A1C) AND (screening* OR test*) AND (uptake OR participate) NOT (Paediatric OR gestational)	**749**

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary table 3 Quality Assessment of included articles based on Appraisal of Cross-sectional Studies (AXIS)

Author	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ali, I.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Andersson, S.	1	1	0	1	1	9	0	1	1	1	1	1	1	1	1	1	1	1	1	1
Bali, V.	1	1	0	1	9	0	0	1	1	0	1	1	0	0	1	1	1	1	9	9
Bao, Y.	1	1	0	1	0	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Belfki, H.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	9	1
Bumrerraj, S.	1	1	0	1	0	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1
Cuong, T. Q.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	9	1
Deng, B.	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	0	1	1
Falguera, M.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Gao, W.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	0	9	1
Hare, M. J.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Hidalgo, L. A.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Hwu, C. M.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Katulanda, G. W.	1	1	1	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Koyama, A. K.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	9	9
Kutkiene, S.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Li, L. J.	1	1	0	1	0	1	0	1	1	1	1	1	0	0	1	1	1	0	1	1
Lin, S.	1	1	0	1	1	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Lissock, C. N.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	0	1	1
Liu, Y.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Mainous, A. G.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	9
Mayega, R. W.	1	1	1	1	1	0	1	1	1	0	1	1	0	1	1	1	1	1	1	1
Nunes, B.	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Rush, E.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	0	1
Sabir, A.	1	1	0	1	0	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Sinnott, M.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	9	1
Valkengoed, I. G.	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1
Wang, S.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Zafar, J.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Zemlin, A. E.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Zhao, W. H.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	1	1
Zhou, X.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Ziemer, D. C.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	9

0: No, 1: Yes, 9: Don't know.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

MOOSE Checklist for Meta-analyses of Observational Studies

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	4-6
2	Hypothesis statement	-
3	Description of study outcome(s)	5-6
4	Type of exposure or intervention used	-
5	Type of study designs used	5-6
6	Study population	5-6
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	7,1
8	Search strategy, including time period included in the synthesis and key words	6-7, Supplementary table 2
9	Effort to include all available studies, including contact with authors	6-7
10	Databases and registries searched	6-8
11	Search software used, name and version, including special features used (eg, explosion)	8
12	Use of hand searching (eg, reference lists of obtained articles)	-
13	List of citations located and those excluded, including justification	8, Supplementary figure 1
14	Method of addressing articles published in languages other than English	-
15	Method of handling abstracts and unpublished studies	-
16	Description of any contact with authors	-
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	6-7
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	6-7
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	6-7
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	-
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	7
22	Assessment of heterogeneity	-
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	8
24	Provision of appropriate tables and graphics	Table 1-3, Figure1-2 Supplementary figure 1-2 Supplementary table 1-3
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	-

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

26	Table giving descriptive information for each study included	8-21, Table 1
27	Results of sensitivity testing (eg, subgroup analysis)	-
28	Indication of statistical uncertainty of findings	-

Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	-
30	Justification for exclusion (eg, exclusion of non-English language citations)	23-24
31	Assessment of quality of included studies	10-11, Supplementary table 3
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	21-23
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	24
34	Guidelines for future research	24
35	Disclosure of funding source	25

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

For peer review only

BMJ Open

Characteristic of participants who take up screening tests for diabetes and lipid disorders: a systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-055764.R2
Article Type:	Original research
Date Submitted by the Author:	08-Mar-2022
Complete List of Authors:	Ding, Hanyue; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care Huang, Junjie; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care Deng, Yunyang; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care Tin, Pamela; Our Hong Kong Foundation Wong, Martin Chi-Sang; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care; Peking University, School of Public Health Yeoh, EK; The Chinese University of Hong Kong Faculty of Medicine, JC School of Public Health and Primary Care; The Chinese University of Hong Kong, Centre for Health Systems and Policy Research
Primary Subject Heading:	Public health
Secondary Subject Heading:	Public health
Keywords:	PREVENTIVE MEDICINE, Lipid disorders < DIABETES & ENDOCRINOLOGY, General diabetes < DIABETES & ENDOCRINOLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Characteristic of participants who take up screening tests for diabetes and lipid disorders: a systematic review

Hanyue Ding[1], Junjie Huang[1], Yunyang Deng[1], Pamela Sze Pui Tin[2], Martin CS Wong[1,3,4], Eng-kiong Yeoh[1,5*]

1. JC School of Public Health and Primary Care, Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong Special Administrative Region, China
2. Our Hong Kong Foundation, Hong Kong Special Administrative Region, China
3. School of Public Health, Peking University, Beijing, China
4. School of Public Health, Chinese Academy of Medical Sciences and Peking Union Hospitals, Beijing, China
5. Centre for Health Systems and Policy Research, The Chinese University of Hong Kong, Hong Kong Special Administrative Region, China

*** Correspondence author**

Professor Eng-kiong Yeoh, GBS, OBE, JP

Director, Centre for Health Systems and Policy Research, JC School of Public Health and Primary Care, Faculty of Medicine, The Chinese University of Hong Kong

Email: yeoh_ek@cuhk.edu.hk; **Telephone number:** 2252-8403; **Fax number:** 2145-8517

Address: Room 202A, 2/F, School of Public Health, Prince of Wales Hospital

(Word count: 3983 words)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

ABSTRACT

Objectives To perform a systematic review on the characteristics of participants who attended screening programs with blood glucose tests, lipid profiles, or a combination of them, respectively.

Design Systematic review following the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) checklist.

Data sources PubMed and Medline database for English literature from 1st Jan 2000 to 1st April 2020.

Eligibility criteria Original observational studies that reported baseline characteristics of apparently healthy, adult screening participants for diabetes and lipid disorders were included in this review.

Data extraction We examined their sociodemographic characteristics, including age, gender, Body Mass Index (BMI), and lifestyle habits. The quality of the included articles was evaluated by the Appraisal of Cross-sectional Studies.

Results A total of 33 articles involving 38 studies in 22 countries were included and analyzed in this systematic review. Overall, there was a higher participation rate among subjects who were female in all screening modalities (female vs. male: 46.6-63.9% vs. 36.1-53.4% for diabetes screening; 48.8-58.4% vs. 41.6-51.2% for lipid screening; and 36.4-76.8% vs. 23.2-63.6% for screening offering both). Compared with the BMI standard from the World Health Organization, participants in lipid screening had lower BMI (males: 23.8 kg/m^2 vs. 24.2 kg/m^2, $p<0.01$; females: 22.3 kg/m^2 vs. 23.6 kg/m^2, $p<0.01$). Furthermore, it is less likely for individuals of lower socioeconomic status to participate in diabetes or lipid screening in developed areas.

Conclusions We identified individuals from lower socioeconomic groups were less likely to take up programs for diabetes and/or lipid screening in developed areas. These

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

populations are also likely to be at higher risk for non-communicable diseases. Future studies should investigate the barriers and facilitators of screening among non-participants, where targeted interventions to enhance their screening uptake are warranted.

(Word count: 282 words)

Keywords: Diabetes Mellitus; Lipid Metabolism Disorders; Mass Screening; Population Characteristics; Systematic Review

Strengths and limitations of this study

- To our knowledge, this is the first systematic review that presented the characteristics of screening participants in one-step diabetes and lipid screening programs.
- Due to the descriptive manner, we cannot quantify the association between patients' characteristics and screening participation.
- We included articles that used different screening tests for diabetes, and this might pose challenges on comparison across studies.
- There are differences in the organization of different screening programs in terms of the tests offered, subsidy amount, and accessibility of the screening centers, and these could influence the screening uptake rate.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

Cardiovascular disease (CVD) represents the major cause of mortality, accounting for 17.9 million deaths or 32% of all global deaths in 2019.[1] It caused more than half of all deaths across the European region, and the death rates were higher in the Russian Federation, Bulgaria, and Romania than in other European regions.[2,3] Most Asian countries had higher age-adjusted mortality than Western countries.[4] Patients with CVD have a high risk of complications, including ischemic heart disease, stroke, and chronic kidney diseases.[5] The modifiable risk factors for CVD included smoking, inadequate physical activity, elevated blood pressure, increased body weight, increased plasma lipid (cholesterol and triglyceride), as well as diabetes.[6] Diabetes is not only the risk factor for CVD but is also associated with significant morbidity and mortality. About 1.6 million deaths were directly caused by diabetes mellitus (DM) in 2016. In addition, diabetes may lead to long-term complications, such as kidney failure, lower limb amputation, blindness, heart attacks, and stroke.[7,8] Worldwide, there were 108 million diabetes patients in 1980, and the number has increased to 422 million in 2014 [8]. The global prevalence of diabetes has been predicted to rise to 10.2% (578 million) by 2030 and 10.9% (700 million) by 2045.[9]

Globally, ischemic heart disease and stroke ranked the first and second causes of disability-adjusted life years (DALYs) in 2016, which were responsible for 12.8% of total DALYs lost.[10] Diabetes caused 65.7 million DALYs lost and 2.5% of the total global DALYs lost in 2016.[10] In addition, dyslipidemia also poses a serious threat to the health of populations. Overall, elevated cholesterol was estimated to contribute to 4.5% of total deaths and 2.0% of total DALYs.[11] Disease screening is regarded as an effective approach for the early detection and prevention of diabetes and dyslipidemia.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

It was demonstrated that cholesterol screening can save 14,300 quality-adjusted life years (QALYs) over 100,000 persons' lifetimes incurring a cost of $33,800 per QALY in the US.[12] Diabetes screening has been shown to be more cost-effective for individuals aged 55 to 75 years as compared to younger subjects.[13]

According to international guidelines for screening **(Supplementary table 1)**, in the United States (US),[14] the United Kingdom,[15] Australia,[16] Canada,[17] and Singapore [18] cholesterol screening is recommended for adults, and the proposed age of onset for screening ranged from 35 to 45 years - or earlier for patients with CVD risk factors. Furthermore, Australia,[16] Canada,[19] and Singapore [20] suggested diabetes screening for average-risk adults above 40 years old, and the US recommended screening for adults aged 40-70 years who are overweight or obese.[21]

However, previous studies have shown that the uptake rate for diabetes and lipid screening was suboptimal. According to a cross-sectional study performed in the US between 2005 and 2012, the diabetes screening rates for "screening-recommended" and "screening-not-recommended" participants were 46.2% and 29.6% respectively.[22] Participation rates of the health examinations for CVD and diabetes ranged between 48% and 67% in Sweden.[23] Although the attendance rate for National Health Service (NHS) health check in England quadrupled from 5.8% in 2010 to 30.1% in 2012, the uptake rates were still relatively low when compared with findings from other studies.[24] To our knowledge, there are no systematic reviews that explored the socioeconomic and demographic characteristics of participants who take up screening programs for diabetes and lipid disorders. Knowing their characteristics could help to target individuals who were less likely to receive screening and improve their

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

participation. These results may help to provide a basis for future studies to examine the enablers and barriers for diabetes and lipid screening. Therefore, we performed a systematic review of the existing literature to evaluate the characteristics of participants who received screening tests for diabetes and lipid disorders.

METHODS

Search strategy

We followed the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) checklist to conduct this systematic review.[25] We searched the PubMed and Medline databases for English literature from 1st Jan 2000 to 1st April 2020. As the characteristics of participants were influenced by health literacy, health development, and health policy, etc., which are changing over time. The older articles showed limited quality to be involved in this study, so we included the studies of recent 20 years to conduct this review. The following keywords were used for the search: 1) AND 2) NOT 3): 1) "fasting plasma glucose test" OR "postprandial plasma glucose" OR "oral glucose tolerance test" OR "random plasma glucose test" OR "glycated hemoglobin" OR "lipid test" OR "cholesterol test" OR "triglyceride test" OR "lipid panel" OR "lipid profile" 2) "uptake" OR "participate" 3) "pediatric" OR "gestational" (**Supplementary table 2**).

Inclusion and exclusion criteria

Studies were included if they: 1) were cross-sectional or reported baseline characteristics of participants in cohort studies; 2) were conducted among adults without a known history of diabetes, prediabetes or CVD; 3) adopted one-step finger stick or blood glucose (including fasting plasma glucose (FPG), oral glucose tolerance

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

tests (OGTT), glycated hemoglobin (HbA1c), and random plasma glucose (RBG)) or blood lipid tests as screening tools; and 4) reported participants' characteristics. The exclusion criteria included: 1) conference abstracts, systematic reviews, or studies published in languages other than English; 2) screening programs for pregnant women, the pediatric population, and/or patients who were previously diagnosed with diabetes, prediabetes or CVD; 3) studies whose target population focused on obese or overweight subjects, people with a family history of diabetes or CVD; 4) studies involved the measurements with multiple steps/stages; and 5) studies performed before 2000.

Data extraction and quality assessment

Two reviewers (HD and VL) independently searched and screened each title and abstract. Disagreements were solved by discussion. Once the title and abstract of each citation were assessed as eligible, two reviewers reviewed and appraised the full text. We used the Appraisal of Cross-sectional Studies (AXIS) to assess the quality of included articles, which consists of 20 questions and answers recorded "yes", "no" or "don't know".[26] AXIS does not involve a numerical scale that can be used to calculate to produce a quality assessment score. We extracted the following data from all included articles: the first author's name, publication year, country, region/city, project period, study type, subsidization, the age range of the population, number of participants; the screening tests used (finger prick, FPG, OGTT, HbA1c, RBG, fasting lipid profile, waist circumference, hip circumference, blood pressure and other relevant tests); participant characteristics: gender, age, body mass index (BMI), marital status, educational status, occupational status, other indicators of socioeconomic status (SES) (including Household wealth index, social deprivation index, and household income), family history, cigarette smoking, alcohol consumption, physical activity, and the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

reason of non-participation.

Statistical analysis

We summarized and descriptively reported the socioeconomics and demographic characteristics of the populations attending the screening in the included studies. Based on our findings in the initial literature review, we also selected gender as a subgroup to explore the difference in age and BMI of the study participants. Means ± standard deviations of these parameters were obtained from included studies. Figures were plotted using R software version x64 3.6.3 with the ggplot package.

Patient and Public Involvement

It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

RESULTS

From the literature search, 6,407 citations were identified, of which 5,689 were from PubMed and 749 were from Medline **(Supplementary figure 1)**. There were 5,723 citations after the removal of duplicates. We retrieved 122 full-text articles to assess for eligibility after 5,536 citations were excluded during the title or abstract screening with the pre-determined criteria. We excluded 89 articles which were duplicate studies (n=1); performed before the year 2000 (n=9); absence of full text or lack of sufficient data for extraction (n=22); the presence of known diabetes, prediabetes or CVD (n=44); or target population whose subjects reported family history of diabetes, prediabetes or elevated BMI (n=13). Finally, we included 38 studies in 33 articles in the present systematic review.[27–59]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Regarding the AXIS for evaluating each included study, most studies presented with the following limitations: the sample size not being justified (23/33); absence of characterization and categorization of non-participants (29/33); a lack of control for non-response bias (30/33); and absence of description of non-participants (27/33). Overall, all studies (33/33) met the following criteria: appropriate objectives and study design; appropriate risk factors and outcome variables; clear and sufficient description of methods; adequate description of the participants' characteristics; and internally consistent presentation of results (**Supplementary table 3**).

The characteristics of all eligible studies were presented in **Table 1**. Among these thirty-three manuscripts: fifteen were from Asia (China, Vietnam, Sri Lanka, Pakistan, Palestine, Thailand), seven were from Europe (Spain, England, Ireland, Netherlands, Lithuania, Sweden, Portugal), six were from Africa (Mauritius, Cameroon, Uganda, Nigeria, Tunisia, South African), four were from Americas (USA, Ecuador), and one was from Oceania (New Zealand). Ten studies were from developed countries, and the other twenty-three studies were from developing countries. According to screening items, eleven studies only implemented diabetes screening, four studies only conducted lipid tests, the rest of the eighteen studies performed both diabetes and lipid screening.

Diabetes Screening

Eleven studies adopted OGTT, HbA1c, FPG, or RBG to screen for diabetes.[32,36,39–41,43,44,46,48,53,55] OGTT and HbA1c were the commonly used modalities (in seven articles), followed by FPG (in five articles) and RBG (in one article). Two of these studies evaluated combinations of FPG, OGTT, and HbA1c. A total of four studies

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

simultaneously took blood pressure during screening.[41,44,48,53] Except for three studies,[32,36,46] all the others included anthropometry measurements (body weight, height, waist and hip circumference). Nearly half of the studies were from developed countries (five out of eleven).[39,40,43,46,53] Among these eleven articles, one reported findings from workplace screening insured by employers,[39] two were national health surveys and fully subsidized by government organizations,[40,53] five were funded by research institutes or funding agencies,[32,41,44,46,55] one was supported by a pharmaceutical company for free meters and strips,[43] and two did not mention their financial supports.[36,48] (**Table 1**)

The range of participation rate of female and male subjects in the diabetes screening programs was 46.6-63.9% and 36.1-53.4%, respectively. **(Supplementary figure 2)** More female participants participated in screening than males in twelve studies. Only two studies showed more male participants in Uganda (52.4%) and Nigeria (53.4%). The mean age of the participants ranged from 38.5 to 48.0 years in all included studies. **(Figure 1)** The studies varied in terms of participant age, where the average age of most studies were more than 40 years. We extracted the national mean BMI of adults in various jurisdictions reported by the World Health Organization (WHO),[60] and compared these with the participants' average BMI. **(Figure 2)** The average BMI of participants tended to be higher than the national mean, in particular from studies performed in the Netherlands (26.1 kg/m^2 vs. 25.4 kg/m^2, $p<0.01$),[46] Thailand (24.3 kg/m^2 vs. 24.1 kg/m^2, $p=0.39$),[55] and the US (30.2 kg/m^2 vs. 28.8 kg/m^2, $p<0.01$).[53]

The sociodemographic information of the screening participants was shown in **Table 2**. For educational status, different studies adopted different classification methods for

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

educational levels. Mayega *et al* reported that 39.6% of participants had a higher primary level, followed by a lower primary (21.9%), secondary (16.5%), and tertiary (4.5%) educational level in Uganda.[41] Zafar *et al* showed more than half of participants had matriculation level (same as the 10th grade) or above (52.5%).[48] They also investigated the household income and found that 72.5% of the participants had <30,000 PKR income (approximately 300 USD in 2014). Mainous *et al* reported the trend (2003, 2006, 2009, 2011) of the social deprivation index, which is a composite index based on income, employment, health deprivation and disability, education, skills and training, barriers to housing and services, crime and disorder, and living environment.[40] The most deprived quintile had the lowest participation percentages in all the years (16.5-17.7%), and the other four levels of fewer deprivation quantities presented similar proportions (18.1-23.1%). In addition, the proportions of single, divorced, or widowed status were also lower than that of married participants.[48]

Apart from age, gender, BMI, and sociodemographic information, we also extracted data on family history and lifestyle habits. **(Table 3)** A study from the Netherlands reported that around 70% of participants had a family history of type 2 diabetes,[46] whereas this uptake rate was under 40% in diabetes screening in Pakistan.[48] Of the two studies including smoking status, Sabir *et al* and Zafar *et al* reported low rates of participation among current smokers (9.7% and 13.4%, respectively).[44,48] Sabir *et al* also reported a very low proportion of screening uptake among alcohol drinkers (0.3%) in Nigeria.[44]

Lipid screening

A total of four studies offered lipid screening;[33,34,51,57] one cross-sectional

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

screening survey was covered by research funding,[57] one study was workplace-based screening and insured by employers,[33] and the other two studies were for national screening programs supported by government organizations.[34,51] Only one study was from a developed country.[33] **(Table 1)**

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1 Characteristic of included studies

Author	Published Year	Country	Region/City	Project Period	Study design	Subsidization
Only blood glucose tests						
Bali, V.	2018	USA		Jan 2012-Dec 2014	Workplace screening (Health Advocate)	Self-insured employers
Bumrerraj, S.	2012	Thailand	Khon Kaen	June-Dec 2009	Cross-sectional study	Research funding
Katulanda, G. W.	2019	Sri Lanka		Aug 2005-Sep 2006	Cross-sectional study	Research funding
Lin, S.	2014	China	Guangzhou	Oct 2010-Jan 2011	Cross-sectional survey	Not mentioned
Mainous, A. G.	2014	England		2003, 2006, 2009, 2011	Health Survey for England	Information Centre for Health and Social Care and the Department of Health
Mayega, R. W.	2014	Uganda	Iganga-Mayuge		Cross-sectional survey	Research funding
Rush, E.	2008	New Zealand	Waikato District	May 2004-Mar 2006	Te Wai o Rona Diabetes Prevention Strategy (Cohort)	Roche company and research funding
Sabir, A.	2013	Nigeria	Gumbi&Wamakko		Cross-sectional study	Research funding
Valkengoed, I. G.	2015	Netherlands	Hague	June-Dec 2009 April-Nov 2010	DHIAAN screening	Research funding
Zafar, J.	2016	Pakistan	Rawalpindi	May-Sep 2014	Cross-sectional survey	Not mentioned
Ziemer, D. C.	2010	USA		Jan 2005-Mar 2008	Third National Health and Nutrition Examination Survey	Centers for Disease Control and Prevention
Only blood lipid tests						
Deng, B.	2012	China	Yunnan	July-Oct 2010	Cross-sectional survey	Research funding
Koyama, A. K.	2018	USA		Jan 2012-Dec 2014	Workplace screening (Health Advocate)	Self-insured employers
Kutkiene, S.	2018	Lithuania		2009-2016	Lithuanian High Cardiovascular Risk primary prevention programme	Ministry of Health
Zhao, W. H.	2007	China		2002	Chinese national nutrition and health survey	Chinese Center for Disease Control and Prevention
Combination tests of blood glucose and lipid						
Ali, I.	2019	Palestine	Nablus municipality	Aug 2017-Feb 2018	Cross-sectional study	Not mentioned
Andersson, S.	2016	Sweden		2013	Qualitative and quantitative study	Not mentioned
Bao, Y.	2010	China	Shanghai	May 2007-Aug 2008	Cross-sectional survey	Research funding
Belfki, H.	2013	Tunisia		Apr 2004-Sep 2005	Transition and Health Impact in North Africa project	Research funding
Cuong, T. Q.	2007	Vietnam	Ho Chi Minh City	Mar-Apr 2004	Cross-sectional survey	Not mentioned
Falguera, M.	2020	Spain	Mollerussa	Mar 2011-July 2014	Cohort study	Research funding
Gao, W.	2008	China	Qingdao	Apr-July 2002	Cross-sectional survey	Research funding
Hare, M. J.	2013	Mauritius	Main island& Rodrigues		Mauritius and Rodrigues Non-Communicable Disease Surveys	Ministry of Health and Quality of Life, Mauritius
Hidalgo, L. A.	2006	Ecuador		Dec 2011-June 2012	Cross-sectional study	Research funding
Hwu, C. M.	2008	China	Taiwan		Cross-sectional study	Research funding
Li, L. J.	2012	China	Hainan	May 2010-Aug 2011	Health check up	Not mentioned
Lissock, C. N.	2011	Cameroon	Ngambe&Littoral		Cross-sectional study	Not mentioned
Liu, Y.	2016	China	Changchun	2011-2012	Cohort baseline survey	Not mentioned
Nunes, B.	2019	Portugal		2015	Portuguese National Health Examination Survey	Organizations of Ministry of Health
Sinnott, M.	2015	Ireland	Dublin	Jan 2009-Dec 2012	The Diabetes Mellitus and Vascular health initiative (Cohort study)	Private health insurance provider (for members)
Wang, S.	2020	China	Shanghai	Mar-Aug 2010	Cohort baseline survey	Research funding
Zemlin, A. E.	2015	South African	Cape town	2008	Cross-sectional survey	Research funding
Zhou, X.	2018	China	Shanghai	Jan-July 2013	Cohort baseline survey	Research funding

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1 Characteristic of included studies (Continued)

Author	Age range	Ethnicity	Participants number	FPG	OGTT	HbA1c	RBG	LP	BP	Test Weight	Height	WC	HC	Others
Only blood glucose tests														
Bali, V.		Non-Hispanic White 77-78% African American 2.8-4% Asian/Pacific Islander 1-3% Hispanic 7% Other 2%	22,790	√	-	√	-	-	-	√	√	-	-	-
Bumrerraj, S.			874	-	√	-	-	-	-	√	√	√	-	-
Katulanda, G. W.			4,014	-	√	-	-	-	-	-	-	-	-	-
Lin, S.	>=50		1,312	√	√	√	-	-	-	-	-	-	-	-
Mainous, A. G.	>=16	(2003 2006 2009 2011) White 92.6% 90.5% 92.2% 85.3% South Asian 4.2% 5.7% 3.0% 4.0% Black 1.6% 1.9% 2.9% 3.0% Mixed/other 1.5% 1.8% 1.9% 7.7%	20,139	-	-	√	-	-	-	√	√	-	-	-
Mayega, R. W.	35-60		795	√	-	√	-	-	√	√	√	-	-	-
Rush, E.	>=28		5,264	-	√	-	-	-	-	√	√	√	-	Point-of-care testing
Sabir, A.			393	√	√	-	√	-	√	√	√	√	√	-
Valkengoed, I. G.	18-60		1,092	-	√	√	-	-	-	-	-	-	-	-
Zafar, J.	18-65		404	-	-	√	-	-	√	√	√	√	√	-
Ziemer, D. C.	18-87	non-Hispanic black and white	1,581	√	√	√	-	-	√	√	√	√	-	-
Only blood lipid tests														
Deng, B.	22-84	Han 51.7% Minority 48.3%	1,415	-	-	-	-	√	-	√	√	√	√	Food frequency
Koyama, A. K.			18,993	-	-	-	-	√	-	√	√	-	-	-
Kutkiene, S.	men 40-54 women 50-64		92,373	-	-	-	-	√	-	-	-	-	-	--
Zhao, W. H.	>=18		49,252	-	-	-	-	√	-	-	-	-	-	-
Combination tests of blood glucose and lipid														
Ali, I.	24-60		140	√	-	-	-	√	√	√	√	-	-	--
Andersson, S.	30-75		43	-	√	√	-	√	√	√	√	√	-	-
Bao, Y.	20-79		4,886	√	√	√	-	√	√	√	√	√	-	-
Belfki, H.	35-74		4,654	√	-	-	-	√	√	√	√	√	-	--
Cuong, T. Q.	20-60	Vietnamese 89.1% Chinese 10.8% Other 0.1%	1,488	√	-	-	-	√	√	√	√	√	√	Skinfold thicknesses
Falguera, M.	>=25		583	√	-	√	-	√	√	√	√	√	-	Renal function
Gao, W.	20-74		1,856	-	√	-	-	√	√	√	√	√	√	-
Hare, M. J.	19-78	African (40.7%) South Asian (57.0%) Chinese (2.3%)	6,701	-	√	√	-	√	√	√	√	√	√	-
Hidalgo, L. A.	40-65		204	√	-	-	-	√	√	√	√	√	-	-
Hwu, C. M.			49	√	√	-	-	√	√	√	√	√	-	-
Li, L. J.	27-91		1,337	√	-	√	-	√	√	√	√	-	-	Uric acid/Liver Function
Lissock, C. N.	>=18		452	√	-	-	-	√	√	√	√	√	√	Resting heart rate
Liu, Y.	>=40	Chinese	7,611	√	√	√	-	√	√	√	√	√	√	-
Nunes, B.	25-74		4,911	-	-	√	-	√	√	√	√	√	√	Blood count
Sinnott, M.	45-75		29,144	√	-	-	-	√	√	√	√	√	√	-
Wang, S.	>=40		9,389	√	√	√	-	√	√	√	√	√	-	-
Zemlin, A. E.			667	√	√	√	-	√	√	√	√	√	√	-
Zhou, X.	15-98	Chinese	7,909	√	√	√	-	√	√	√	√	√	-	Heart rate

FPG: fasting plasma glucose, OGTT: oral glucose tolerance tests, HbA1c: glycated hemoglobin, RGB: random plasma glucose, LP: lipid screening, BP: blood pressure, WC: waist circumference, HC: hip circumference.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2 Sociodemographic characteristics of participants in all included studies

Author	Marital status	Educational status	Occupational status	Others
Blood glucose tests only				
Mainous, A. G.	-	-	-	Social deprivation index (2003, 2006, 2009, 2011) Quintile 1 21.6% 20.6% 18.9% 20.1% Quintile 2 20.8% 20.2% 22.7% 23.1% Quintile 3 20.7% 22.3% 21.6% 21.0% Quintile 4 19.8 % 20.4% 20.3% 18.1% Quintile 5 17.0% 16.5% 16.5% 17.7%
Mayega, R. W.	-	None 139 (17.5%) Lower primary 174 (21.9%) Higher primary 315 (39.6%) Secondary 131 (16.5%) Tertiary 36 (4.5%)	Subsistence farmers 500 (62.9%) Traders 164 (20.6%) Formal/salaried 39 (4.9%) Mechanics 92 (11.6%)	-
Valkengoed, I. G.	-	Primary or less 14.7% 16.5% Secondary 11.7% 13.6% Lower vocational 56.8% 53.5%[c] Higher vocational 16.8% 16.5%[c]	Paid work (OGTT, HbA1c) 74.1% 71.5%	-
Zafar, J.	Married 322 (79.7%) Unmarried 61 (15.1%) Widow/Divorced 21 (5.2%)	Illiterate 55 (13.6%) Primary (1-5 grade) 59 (14.6%)[a] Middle (6-8 grade) 61 (15.1%)[b] > Matric (9-10 grade)212 (52.5%)[b] Conventional 17 (4.2%)	Govt. Employee 61 (15.1%) Private Employee 61 (15.1%) Self-Employee 80 (19.8%) Un-employed 84 (20.8%) Laborer 12 (3.0%) Any other 106 (26.2%)	Income (PKR) <10000 118 (29.2%) 10,000 - 30,000 175 (43.3%) 31000 - 50,000 72 (17.8%) > 50,000 39 (9.7%)
Plasma lipid tests only				
Deng, B.	Single 94 (6.6%) Married 1,306 (92.3%) Divorced 8 (0.6%) Widow(er) 7 (0.5%)	Never 21 (1.5%) Elementary school 116 (8.2%)[a] Junior-middle school 348 (24.6%)[b] Senior-high school 386 (27.3%)[b] Higher 544 (38.4%)[c]	-	-
Koyama, A. K.	-	Education, median (IQR) Less than high school 9.5% (5.7%-15.1%) High school or equivalent[b] 27.0% (19.5%-33.9%) Some college[c] 29.1% (24.2%-33.5%) College degree[c] 29.2% (19.2%-43.2%)	-	Median income, median (IQR) US$57,622 (US$45,161-US$75,313)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2 Sociodemographic characteristics of participants in all included studies (Continued)

Author	Marital status	Educational status	Occupational status	Others
Programs offering both diabetes and lipid tests				
Belfki, H.	Single 117 (2.5%) Married 4,035 (86.7%) Widowed/divorced 502 (10.8%)	Illiterate 2,041(43.9%) Low (<=6yrs) 1,552 (33.3%)[a] Intermediate(7-13yrs) 805 (17.3%)[b] Higher(>=14yrs) 238 (5.1%)[c]	No working/retired 2,479 (56.6%) Employee/worker 1,205 (27.5%) Intermediate 202 (4.6%) Upper 494 (11.3%)	-
Cuong, T. Q.	-	No schooling (1.8%) Primary school (16.3%)[a] Junior high school (33.3%)[b] Senior high school (33.3%)[b] College/University (15.3%)[c]	Teacher, Professional (10.2%) Government officers (14.9%) Small business, Skilled workers (17.9%) Laborers, street or home traders (24.3%) Retired/home maker/students (21.3%) Others (8.5%) No Job (2.9%)	Household wealth index (Male, female, total) Lowest 19.9% 19.9% NA Second 19.8% 20.1% NA Middle 20.5% 19.5% NA Fourth 21.2% 18.8% NA Highest 18.6% 21.7% NA
Falguera, M.	-	High level (>=secondary high school education) 421 (72.2%)	-	-
Hare, M. J.	-	Primary or none 3,317 (49.8%) Secondary 2,832 (42.5%) Tertiary 515 (7.7%)	-	-
Hidalgo, L. A.	Married 93 (45.6%) No 111 (54.4%)	0–6 73 (35.8%)[a] 7–12 68 (33.3%)[b] 13 63 (30.9%)[c]	-	-
Wang, S.	-	High school education or more 6,037 (64.6%)	-	-

[a] same as the primary education, b same as secondary education, c same as tertiary education; OGTT: oral glucose tolerance tests, HbA1c: glycated hemoglobin.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 3 Family history and lifestyle habits and of participants

Author	Family history	Smoking	Drinking	Physical activity
Blood glucose tests only				
Sabir, A.		Total 38 (9.7%) Male 38 (18.1%) Female 0 (0%)	Total 1 (0.3%) Male 1 (0.5%) Female 0 (0%)	-
Valkengoed, I. G.	Type 2 diabetes mellitus 74.8% (75-g OGTT group) 68.4% (HbA1c group)			
Zafar, J.	Diabetes Parental history 161 (39.9%) Siblings history 88 (21.8%)	Smokers 54 (13.4%)	-	Exercise 105 (26.0%)
Plasma lipid tests only				
Deng, B.		Current smoking 476 (33.6%)	Current drinking 557 (39.4%)	>2/week /at least 30 mins 806 (57.0%)
Programs offering both diabetes and lipid tests				
Ali, I.	Dyslipidemia 27.9%	Smokers 37.1%	-	-
Andersson, S.		Smoking 19%	-	-
Belfki, H.	Cardiovascular disease 119 (2.6%)	Never 3,246 (71.4%) Smoker 1,301 (28.6%)	-	-
Cuong, T. Q.		Non/ex-smoker (69.3%) Current smoker (30.7%)	-	-
Falguera, M.	Diabetes 180 (30.9%)	Current smoker 148 (25.4%)	-	394 (67.6%)
Hare, M. J.		1,417 (21.2%)	None 3,217 (49.7%) Moderate 2,728 (42.1%) Excessive 531 (8.2%)	Sedentary 694 (10.6%) Insufficient (<150 min/week) 1,883 (28.8%) Sufficient (>=150 min/week) 3,951 (60.5%)
Hidalgo, L. A.		-	-	Sedentary Yes 69 (33.8%) No 135 (66.2%)
Sinnott, M.	Diabetes 9,301 (31.9%)	Ever smoked 11,648 (40.0%)	23,245 (79.8%)	>=5days/week 10,343 (35.5%)
Wang, S.	Diabetes 884 (9.4%)	Current smoking 1,926 (21.2%)	958 (10.5%)	>=600 MET-min/week 6,670 (71.2%)
Zhou, X.	Diabetes 847 (10.71%)	1,746 (22.1%)	1,079 (13.64%)	-

OGTT: oral glucose tolerance tests, HbA1c: glycated hemoglobin, MET: metabolic equivalent of task.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

In these four articles, two articles reported body weight and height,[33,57] and one study stated both waist and hip circumference.[57] For participants, the proportion of females ranged from 48.8% to 58.4%. Only one study that targeted rural residents of Yunnan China showed a higher percentage of males (51.2%) among participants. **(Supplementary figure 2)** The mean age of the screening participants was older than 40 years. **(Figure 1)** Turning to mean BMI, however, both male and female subjects were lower than the national mean BMI reported by WHO (males: 23.8 kg/m^2 vs. 24.2 kg/m^2, $p<0.01$; females: 22.3 kg/m^2 vs. 23.6 kg/m^2, $p<0.01$). **(Figure 2)** Studies from Deng *et al* and Koyama *et al* reported the highest proportion of individuals having a higher educational level in China and the US (38.4% and 58.3%, respectively).[33,57] **(Table 2)**

Deng and colleagues reported the smoking, drinking, and physical activity habits of their lipid screening participants in China.[57] A third of them were current smokers. Moreover, 39.4% of subjects reported that they were current drinkers. For physical activity, 57.0% of them took exercise or participated in physical labor activities more than twice per week for at least 30 minutes each. **(Table 3)**

Participants receiving both diabetes and lipid screening

The studies that presented the characteristics of screening participants in programs that offered a screening of both diabetes and lipid profiles were listed in **Table 1**. Fourteen of these papers adopted FPG to evaluate blood glucose.[27,30,31,35,37,38,45,47,49,50,52,54,56,58] Ten studies used HbA1c,[28,29,35,38,42,47,49,50,52,58] nine studies used OGTT,[28,29,31,38,47,49,50,52,59] while five of the included studies offered all three

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

tests.[38,47,49,50,52] Only four studies were from developed countries.[28,42,45,58] Two were organized by the government,[29,42] and one was offered by a private health insurance provider for their members.[45]

For comparison of diabetes and lipid screening participation between males and females, only one study reported having more male than female subjects joining the screening program in Hainan China.[35] **(Supplementary figure 2)** Five studies indicated that male participants were older than females,[37,38,45,49] while the other two studies reported female participants being older.[50,54] **(Figure 1)** Turning to the mean BMI of participants, most of the studies reported a higher number than the national BMI levels, including those conducted in China (24.7 kg/m^2 vs. 23.9 kg/m^2, $p<0.0$;[52] 25.1 kg/m^2 vs. 23.9 kg/m^2, $p<0.01$;[47] 24.9 kg/m^2 vs. 23.9 kg/m^2, $p<0.01$;[38] 24.8 kg/m^2 vs. 23.9 kg/m^2, $p<0.01$;[35] 25.4 kg/m^2 vs. 23.9 kg/m^2, $p<0.01$);[31] South Africa (29.1 kg/m^2 vs. 27.3 kg/m^2, $p<0.01$);[49] Sweden (29.0 kg/m^2 vs. 25.8 kg/m^2, $p<0.01$);[28] and Tunisia (26.8 kg/m^2 vs. 27.8 kg/m^2, $p<0.01$).[54] **(Figure 2)** Overall, females had higher BMI levels than their national average as reported in most studies, whilst only two studies showed a higher BMI among male participants when compared with the WHO's report.[37,38] Some other screening tests were also included in other literature, such as skinfold thicknesses,[56] renal function,[58] uric acid,[35] liver function,[35] blood count,[42] and heart rate.[37,52]

For educational status, we found a substantial difference in participants' educational status across different studies. Belfki *et al* showed that 77.2% of participants had a low educational level in Tunisia.[54] However, Wang and colleagues reported that around 65% of the individuals had an educational level at higher school or above in

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Shanghai.[47] For occupational status, Belfki *et al* indicated more than half (56.6%) of the participants were retired or had no current jobs.[54] The corresponding proportion was only 24.2% in Cuong's study.[56] Cuong and colleagues also measured the household wealth index of participants and showed similar proportions among males (18.6-21.2%) and female participants (18.8-21.7%) across all levels of the index.[56] **(Table 2)**

Of the six studies that reported family history, one study described that 27.9% of the participants had a family history of dyslipidemia,[27] another study indicated the proportion of a family history of CVD was 2.6%,[54] and the other four studies mentioned that 9.4-31.9% of the participants with a family history of diabetes joined the screening programs.[45,47,52,58] **(Table 3)**

Of the ten studies with information on the status of smoking, three reported high rates of current smoking,[27,54,56] which were 37.1%, 28.6%, and 30.7%, respectively. Four studies stated the participants' alcohol consumption, one reported a high rate (79.8%),[45] compared with that reported in two studies (10.5-13.64%).[47,52] Among the five studies which included physical activity, three of them reported that more than half of the participants performed adequate physical activity (60.5-71.2%).[29,47,58] **(Table 3)**

The reasons for non-participation

Four studies mentioned the reasons for non-participation. Two studies on diabetes screening assessed reasons for non-participation. Mayega *et al* reported 5.4% of subjects refused to participate due to travel distance and 4.2% of non-participants

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

declined the blood tests.[41] Valkengoed *et al* described that "no time" and "no interest" were the major reasons among prospective eligible participants.[46] Regarding lipid screening, Deng *et al* explained that 1.0% of subjects declined screening tests due to time constraints.[57] Nunes and colleagues showed that 43.9% of individuals participated in the first Portuguese National Health Examination Survey involving both diabetes and lipid screening.[42] Furthermore, they found that work-related issues (26.6%) and lack of time (26.6%) were the most frequently mentioned reasons for not joining.

DISCUSSION

This systematic review provided socioeconomic and demographic characteristics of participants who received screening tests for diabetes and lipid disorders, including age, gender, BMI, sociodemographic status, family history, and lifestyle habits. We found a higher proportion of females among the screening participants when compared with the male gender in most studies, irrespective of the tests offered. Participants who joined diabetes screening tended to have higher BMI values, whilst those who received lipid screening tended to have lower BMI values. Regarding screening programs offering both diabetes and lipid tests, female participants had higher BMI values and males had lower BMI values than the national means reported by the WHO.

Sargeant and colleagues conducted a stepwise program to screen diabetes, and they reported male gender was significantly associated with lower uptake of RBG testing, which is consistent with our findings on diabetes screening.[61] A serial study of cross-sectional data (1992-2008) evaluated the utilization of preventive health assessments in the UK, and found that the participation rate of lipid tests increased with age, but they

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

reported an absence of association between gender and participation in lipid tests.[62] They also showed that smokers (vs. non-smokers) had significantly lower uptake of lipid tests. In our review, around one-third of participants were current smokers in lipid tests and programs offering both diabetes and lipid tests. A longitudinal study indicated that current smokers had limited health literacy and lower cognitive ability than ever smokers,[63] which may explain their lower screening uptake rates. Interventions to address the health literacy of prospective participants are needed to improve screening participation, which at the same time could also help to change their lifestyle habits for the prevention of chronic diseases.[64]

A postal survey investigated the participation rate of women's health check-ups, including identification of CVD risk factors, as well as examinations for type 2 diabetes and kidney disease in Germany in 2004.[65] Its adjusted model showed that single, divorced, separated or widowed women (OR=0.63, 95% CI: 0.57-0.71) were less likely to receive a preventive health check-up when compared with married women. The proportions of single, divorced, or widowed status were also lower than that of married participants in our review, except for a study performed in Italy. This may be due to peer effects where family members may exert a motivational influence for chronic diseases screening. It is well recognized that SES influenced disease risk, health behavior, and health care of individuals. These health inequities have been affirmed by the WHO.[66] Researchers in Germany found that there was a significant association between women with lower SES (estimated based on educational status, occupational position and household income) (OR=0.82, 95% CI: 0.74-0.92) and lower participation rate of health screening attendance.[65] This is consistent with our finding that health inequities might exist as the most deprived population had the lowest participation rates

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

in England, where only blood glucose tests were provided. Meanwhile, participants of studies in Spain and China (Shanghai) had higher educational status, whereas studies in Tunisia, Mauritius, and Ecuador recruited more participants with lower educational level. Health inequities are more likely to be found in developed areas,[67] yet they also exist in developing countries due to differential health literacy and access to primary care services. For the non-participation of screening programs among those with lower educational levels, health literacy plays an important role in screening participation. The American Heart Association highlighted that low health literacy is a barrier to healthcare screening and basic services for diabetes and CVD.[68] A systematic review included studies that evaluated the effectiveness of health literacy interventions using pre- and post-program assessment, and concluded their positive impact on enhancing screening uptake.[64]

Hypertension is also a risk factor to contribute the CVD, but we did not include it as a target measurement, because the blood tests had more obstacles than the blood pressure tests, for example, the fear of needles, fear of pain, and cost of test strips and needles.[69] Thus in this study, we focused on the blood glucose tests and lipid profiles to explore the characteristics of participants.

To our knowledge, this is the first systematic review that presented the characteristics of screening participants in diabetes and lipid screening programs. The findings inform population groups where interventions to enhance screening uptake should be targeted. Nevertheless, there are several limitations. Firstly, this is a systematic review of literature presented in a descriptive manner. Therefore, we cannot quantify the association between patients' characteristics and screening participation. In addition,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

we included articles that used different screening tests for diabetes, and this might pose challenges on comparison across studies. The American Diabetes Association proposed that FPG is the best screening test and also a component of the diagnostic test for diabetes.[70] FPG is more convenient and acceptable to patients since it is easier and faster to perform, and is less expensive than OGTT. There are differences in the organization of different screening programs in terms of the tests offered, subsidy amount, and accessibility of the screening centers, and these could influence the screening uptake rate. In addition, the prevalence of risk factors in screening participants who take up the tests might be different from those who do not.

Overall, the present study systematically summarized the characteristics of screening participants in diabetes and lipid screening programs. A higher proportion of female participants was found in diabetes and lipid screening programs than males in most studies. For diabetes screening, participants tended to have higher BMI, while it was the opposite for subjects who participated in lipid screening. Meanwhile, females tended to have higher BMI and males tended to have lower BMI in screening programs offering both diabetes and lipid tests. Participants with lower SES were less likely to undergo screening in developed areas. Around one-third of participants were current smokers in lipid screening and a combination of lipid and diabetes screening programs. The proportion of participants using alcohol and performing regular physical activity varied substantially among included studies. Our findings could inform future studies to investigate the enablers and barriers of screening among participants and non-participants based on their characteristics, and also interventions to enhance their screening uptake.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Acknowledgements: We appreciate the invaluable contributions of Ms. Xiao Chen and Ms. Veeleah Lok.

Competing Interests: Ms. Pamela Sze Pui Tin is from Our Hong Kong Foundation. The other authors declare no competing interests.

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability: All data generated or analysed during this study are included in this published article (and its Supplementary Information files).

Ethics declarations: N/A

Author contributions: HD: Formal analysis, Investigation, Data Curation, Writing - Original Draft; JH: Writing - Review & Editing; YD: Validation, Writing - Review & Editing; PT: Writing - Review & Editing; MCSW: Methodology, Supervision, Project administration; EKY: Conceptualization, Supervision.

FIGURE LEGENDS

Figure 1 The mean age of screening participants

Figure 2 The Body Mass Index (BMI) of screening participants

We compared the national mean BMI of adults, extracted from the World Health Organization (WHO) report, with the participants' average BMI.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

REFERENCE

1 World Health Organization. Cardiovascular diseases (CVDs). World Heal. Organ. 2017.https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds) (accessed 8 May 2020).

2 World Health Organization. WHO/Europe | Cardiovascular diseases - Data and statistics. 2020.http://www.euro.who.int/en/health-topics/noncommunicable-diseases/cardiovascular-diseases/data-and-statistics (accessed 13 May 2020).

3 American Heart Association. Statistical Fact Sheet - International Cardiovascular Disease Statistics. 2004.http://www.sld.cu/galerias/pdf/servicios/hta/international_cardiovascular_disease_statistics.pdf (accessed 13 May 2020).

4 Ohira T, Iso H. Cardiovascular disease epidemiology in Asia - An overview. *Circ J* 2013;**77**:1646–52. doi:10.1253/circj.CJ-13-0702

5 Mendis S, Puska P, Norrving B. Global atlas on cardiovascular disease prevention and control. 2011.

6 Francula-Zaninovic S, Nola IA. Management of Measurable Variable Cardiovascular Disease' Risk Factors. *Curr Cardiol Rev* 2018;**14**:153–63. doi:10.2174/1573403x14666180222102312

7 American Diabetes Association. Diagnosis and classification of diabetes mellitus. *Diabetes Care* 2014;**37**:S81–90. doi:10.2337/dc14-S081

8 World Health Organization. Diabetes. 2020.https://www.who.int/news-room/fact-sheets/detail/diabetes (accessed 11 May 2020).

9 Saeedi P, Petersohn I, Salpea P, *et al.* Global and regional diabetes prevalence estimates for 2019 and projections for 2030 and 2045: Results from the International Diabetes Federation Diabetes Atlas, 9th edition. *Diabetes Res Clin Pract* 2019;**157**:107843. doi:10.1016/j.diabres.2019.107843

10 World Health Organization. Disease burden and mortality estimates. 2020.https://www.who.int/healthinfo/global_burden_disease/estimates/en/index1.html (accessed 9 Jun 2020).

11 World Health Organization. WHO | Raised cholesterol. 2020.https://www.who.int/gho/ncd/risk_factors/cholesterol_text/en/ (accessed 7 Jul 2020).

12 Dehmer SP, Maciosek M V., LaFrance AB, *et al.* Health benefits and cost-effectiveness of asymptomatic screening for hypertension and high cholesterol and aspirin counseling for primary prevention. *Ann Fam Med* 2017;**15**:23–36. doi:10.1370/afm.2015

13 Hoerger TJ, Harris R, Hicks KA, *et al.* Screening for Type 2 Diabetes Mellitus: A Cost-Effectiveness Analysis. *Ann Intern Med* 2004;**140**:689. doi:10.7326/0003-4819-140-9-200405040-00008

14 Bibbins-Domingo K, Grossman DC, Curry SJ, *et al.* Statin use for the primary prevention of cardiovascular disease in adults: US preventive services task force recommendation statement. JAMA - J. Am. Med. Assoc. 2016;**316**:1997–2007. doi:10.1001/jama.2016.15450

15 NICE guideline. Cardiovascular disease: risk assessment and reduction, including lipid modification. Natl. Inst. Heal. Care Excell. 2016.https://www.nice.org.uk/guidance/cg181/resources/surveillance-report-2018-cardiovascular-disease-risk-assessment-and-reduction-including-lipid-modification-2014-nice-guideline-cg181-4724759773/chapter/Surveillance-decision?tab=evidence (accessed 7 Jul 2020).

16 The Royal Australian College of General Practitioners. Guidelines for preventive activities in general practice 9th edition. 2018.https://www.racgp.org.au/download/Documents/Guidelines/Redbook9/17048-Red-Book-9th-Edition.pdf (accessed 9 Jun 2020).

17 Allan GM, Lindblad AJ, Comeau A, *et al.* Simplified lipid guidelines: Prevention and management of cardiovascular disease in primary care. *Can Fam Physician* 2015;**61**:857–67, e439-50.

18 Ministry of Health. LIPIDS EXECUTIVE SUMMARY. 2016.https://www.moh.gov.sg/docs/librariesprovider4/guidelines/moh-lipids-cpg---executive-summary.pdf (accessed 10 Jun 2020).

19 Ekoe JM, Goldenberg R, Katz P. Screening for Diabetes in Adults. *Can J Diabetes* 2018;**42**:S16–9. doi:10.1016/j.jcjd.2017.10.004

20 Goh SY, Ang S Bin, Bee YM, *et al.* Ministry of Health clinical practice guidelines: Diabetes mellitus. *Singapore Med J* 2014;**55**:334–47. doi:10.11622/smedj.2014079

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

21 Siu AL. Screening for abnormal blood glucose and type 2 diabetes mellitus: U.S. preventive services task force recommendation statement. Ann. Intern. Med. 2015;**163**:861–8. doi:10.7326/M15-2345

22 Kiefer MM, Silverman JB, Young BA, *et al.* National Patterns in Diabetes Screening: Data from the National Health and Nutrition Examination Survey (NHANES) 2005–2012. *J Gen Intern Med* 2015;**30**:612–8. doi:10.1007/s11606-014-3147-8

23 Norberg M, Wall S, Boman K, *et al.* The Västerbotten Intervention Programme: background, design and implications. *Glob Health Action* 2010;**3**:4643. doi:10.3402/gha.v3i0.4643

24 Robson J, Dostal I, Sheikh A, *et al.* The NHS Health Check in England: an evaluation of the first 4 years. doi:10.1136/bmjopen-2015-008840

25 Stroup DF, Berlin JA, Morton SC, *et al.* Meta-analysis of observational studies in epidemiology: A proposal for reporting. *J Am Med Assoc* 2000;**283**:2008–12. doi:10.1001/jama.283.15.2008

26 Downes MJ, Brennan ML, Williams HC, *et al.* Development of a critical appraisal tool to assess the quality of cross-sectional studies (AXIS). *BMJ Open* 2016;**6**. doi:10.1136/bmjopen-2016-011458

27 Ali I, Kharma A, Samara M, *et al.* Prevalence of Dyslipidemia in Undiagnosed Palestinian Men: A Cross-Sectional Study. *J Lipids* 2019;**2019**:1–6. doi:10.1155/2019/3473042

28 Andersson S, Karlsson V, Bennet L, *et al.* Attitudes Regarding Participation in a Diabetes Screening Test among an Assyrian Immigrant Population in Sweden. *Nurs Res Pract* 2016;**2016**:1–7. doi:10.1155/2016/1504530

29 Hare MJL, Magliano DJ, Zimmet PZ, *et al.* Glucose-independent ethnic differences in HbA1c in people without known diabetes. *Diabetes Care* 2013;**36**:1534–40. doi:10.2337/dc12-1210

30 Hidalgo LA, Chedraui PA, Morocho N, *et al.* The metabolic syndrome among postmenopausal women in Ecuador. *Gynecol Endocrinol* 2006;**22**:447–54. doi:10.1080/09513590600890272

31 Hwu CM, Lin MW, Liou TL, *et al.* Fasting triglyceride is a major determinant of postprandial triglyceride response in postmenopausal women. *Menopause* 2008;**15**:150–6. doi:10.1097/gme.0b013e31800577856

32 Katulanda GW, Katulanda P, Dematapitiya C, *et al.* Plasma glucose in screening for diabetes and pre-diabetes: How much is too much? Analysis of fasting plasma glucose and oral glucose tolerance test in Sri Lankans. *BMC Endocr Disord* 2019;**19**:11. doi:10.1186/s12902-019-0343-x

33 Koyama AK, Bali V, Yermilov I, *et al.* Identification of Undiagnosed Hyperlipidemia: Do Work Site Screening Programs Work? *Am J Heal Promot* 2018;**32**:971–8. doi:10.1177/0890117116671537

34 Kutkiene S, Petrulioniene Z, Laucevicius A, *et al.* Lipid profile evaluation and severe hypercholesterolaemia screening in the middle-aged population according to nationwide primary prevention programme in Lithuania. *Atherosclerosis* 2018;**277**:267–72. doi:10.1016/j.atherosclerosis.2018.06.008

35 Li LJ, Zhou JX, Chen HT, *et al.* Effect of HbA 1c combined FPG on screening diabetes in health check-up. *Asian Pac J Trop Med* 2012;**5**:472–5. doi:10.1016/S1995-7645(12)60081-9

36 Lin S, Hu L, Li X, *et al.* Glycated haemoglobin A1c for diagnosing diabetes in Chinese subjects over 50 years old: A community-based cross-sectional study. *Clin Endocrinol (Oxf)* 2014;**80**:656–61. doi:10.1111/cen.12202

37 Lissock CNAA, Sobngwi E, Ngassam E, *et al.* Rural and urban differences in metabolic profiles in a Cameroonian population. *Pan Afr Med J* 2011;**10**:1. doi:10.4314/pamj.v10i0.72204

38 Liu Y, Xiao X, Sun C, *et al.* Ideal glycated hemoglobin cut-off points for screening diabetes and prediabetes in a Chinese population. *J Diabetes Investig* 2016;**7**:695–702. doi:10.1111/jdi.12498

39 Bali V, Yermilov I, Koyama A, *et al.* Secondary prevention of diabetes through workplace health screening. *Occup Med (Chic Ill)* 2018;**68**:610–6. doi:10.1093/OCCMED/KQY138

40 Mainous AG, Tanner RJ, Baker R, *et al.* Prevalence of prediabetes in England from 2003 to 2011: population-based, cross-sectional study. *BMJ Open* 2014;**4**:e005002. doi:10.1136/bmjopen-2014-005002

41 Mayega RW, Guwatudde D, Makumbi FE, *et al.* Comparison of fasting plasma glucose and haemoglobin A1c point-of-care tests in screening for diabetes and abnormal glucose regulation in a rural low income setting. *Diabetes Res Clin Pr* 2014;**104**:112–20. doi:10.1016/j.diabres.2013.12.030

42 Nunes B, Barreto M, Gil AP, *et al.* The first Portuguese National Health Examination Survey

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

(2015): design, planning and implementation. *J Public Health (Oxf)* 2019;**41**:511–7. doi:10.1093/pubmed/fdy150

43 Rush E, Crook N, Simmons D. Point-of-care testing as a tool for screening for diabetes and pre-diabetes. *Diabet Med* 2008;**25**:1070–5. doi:10.1111/j.1464-5491.2008.02526.x

44 Sabir A, Ohwovoriole A, Isezuo S, *et al.* Type 2 diabetes mellitus and its risk factors among the rural Fulanis of Northern Nigeria. *Ann Afr Med* 2013;**12**:217–22. doi:10.4103/1596-3519.122689

45 Sinnott M, Kinsley BT, Jackson AD, *et al.* Fasting plasma glucose as initial screening for diabetes and prediabetes in irish adults: The Diabetes Mellitus and Vascular health initiative (DMVhi). *PLoS One* 2015;**10**:e0122704. doi:10.1371/journal.pone.0122704

46 Valkengoed VIG, Vlaar EM, Nierkens V, *et al.* The Uptake of Screening for Type 2 Diabetes and Prediabetes by Means of Glycated Hemoglobin versus the Oral Glucose Tolerance Test among 18 to 60-Year-Old People of South Asian Origin: A Comparative Study. *PLoS One* 2015;**10**:e0136734. doi:10.1371/journal.pone.0136734

47 Wang S, Niu J, Zhao Z, *et al.* Detection of diabetes and prediabetes using glycosylated hemoglobin in Chinese adults living in Shanghai: A prospective analysis. *J Diabetes* Published Online First: 2020. doi:10.1111/1753-0407.13028

48 Zafar J, Nadeem D, Khan SA, *et al.* Prevalence of diabetes and its correlates in urban population of Pakistan: A Cross-sectional survey. *J Pak Med Assoc* 2016;**66**:922–7.

49 Zemlin AE, Matsha TE, Kengne AP, *et al.* Derivation and validation of an HbA1c optimal cutoff for diagnosing prediabetes in a South African mixed ancestry population. *Clin Chim Acta* 2015;**448**:215–9. doi:10.1016/j.cca.2015.06.019

50 Bao Y, Ma X, Li H, *et al.* Glycated haemoglobin A1c for diagnosing diabetes in Chinese population: Cross sectional epidemiological survey. *BMJ* 2010;**340**:1178. doi:10.1136/bmj.c2249

51 Zhao WH, Zhang J, Zhai Y, *et al.* Blood lipid profile and prevalence of dyslipidemia in Chinese adults. *Biomed Env Sci* 2007;**20**:329–35.

52 Zhou X, Ruan X, Hao L, *et al.* Optimal hemoglobin A1C cutoff value for diabetes mellitus and pre-diabetes in Pudong New Area, Shanghai, China. *Prim Care Diabetes* 2018;**12**:238–44. doi:10.1016/j.pcd.2017.12.006

53 Ziemer DC, Kolm P, Weintraub WS, *et al.* Glucose-independent, black-white differences in hemoglobin A1c levels: a cross-sectional analysis of 2 studies. *Ann Intern Med* 2010;**152**:770–7. doi:10.7326/0003-4819-152-12-201006150-00004

54 Belfki H, Ben Ali S, Aounallah-Skhiri H, *et al.* Prevalence and determinants of the metabolic syndrome among Tunisian adults: results of the Transition and Health Impact in North Africa (TAHINA) project. *Public Heal Nutr* 2013;**16**:582–90. doi:10.1017/s1368980012003291

55 Bumrerraj S, Kaczorowski J, Kessomboon P, *et al.* Diagnostic performance of 2 h postprandial capillary and venous glucose as a screening test for abnormal glucose tolerance. *Prim Care Diabetes* 2012;**6**:207–11. doi:10.1016/j.pcd.2012.03.002

56 Cuong TQ, Dibley MJ, Bowe S, *et al.* Obesity in adults: An emerging problem in urban areas of Ho Chi Minh City, Vietnam. *Eur J Clin Nutr* 2007;**61**:673–81. doi:10.1038/sj.ejcn.1602563

57 Deng B, Luo T, Huang Y, *et al.* Prevalence and determinants of hyperlipidemia in moderate altitude areas of the yunnan-kweichow plateau in southwestern China. *High Alt Med Biol* 2012;**13**:13–21. doi:10.1089/ham.2011.1037

58 Falguera M, Vilanova MB, Alcubierre N, *et al.* Prevalence of pre-diabetes and undiagnosed diabetes in the Mollerussa prospective observational cohort study in a semi-rural area of Catalonia. *BMJ Open* 2020;**10**:e033332. doi:10.1136/bmjopen-2019-033332

59 Gao W, Dong Y, Nan H, *et al.* The likelihood of diabetes based on the proposed definitions for impaired fasting glucose. *Diabetes Res Clin Pract* 2008;**79**:151–5. doi:10.1016/j.diabres.2007.07.018

60 Mendis S, Armstrong T, Bettcher D, *et al.* Global status report on noncommunicable diseases 2014. World Health Organisation. 2014. doi:ISBN 9789241564854

61 Sargeant LA, Simmons RK, Barling RS, *et al.* Who attends a UK diabetes screening programme? Findings from the ADDITION-Cambridge study. *Diabet Med* 2010;**27**:995–1003. doi:10.1111/j.1464-5491.2010.03056.x

62 Labeit A, Peinemann F, Baker R. Utilisation of preventative health check-ups in the UK: Findings from individual-level repeated cross-sectional data from 1992 to 2008. *BMJ Open* 2013;**3**. doi:10.1136/bmjopen-2013-003387

63 Fawns-Ritchie C, Starr JM, Deary IJ. Health literacy, cognitive ability and smoking: A cross-sectional analysis of the English Longitudinal Study of Ageing. *BMJ Open* 2018;**8**:e023929.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

doi:10.1136/bmjopen-2018-023929

64 Walters R, Leslie SJ, Polson R, *et al.* Establishing the efficacy of interventions to improve health literacy and health behaviours: A systematic review. *BMC Public Health* 2020;**20**:1040. doi:10.1186/s12889-020-08991-0

65 Schülein S, Taylor KJ, Schriefer D, *et al.* Participation in preventive health check-ups among 19,351 women in Germany. *Prev Med Reports* 2017;**6**:23–6. doi:10.1016/j.pmedr.2017.01.011

66 World Health Organization. Social determinants of health. World Heal. Organ. 2013.https://www.who.int/social_determinants/thecommission/finalreport/key_concepts/en/ (accessed 6 May 2020).

67 Orach CG. Health equity: challenges in low income countries. Afr. Health Sci. 2009;**9 Suppl 2**:S49.

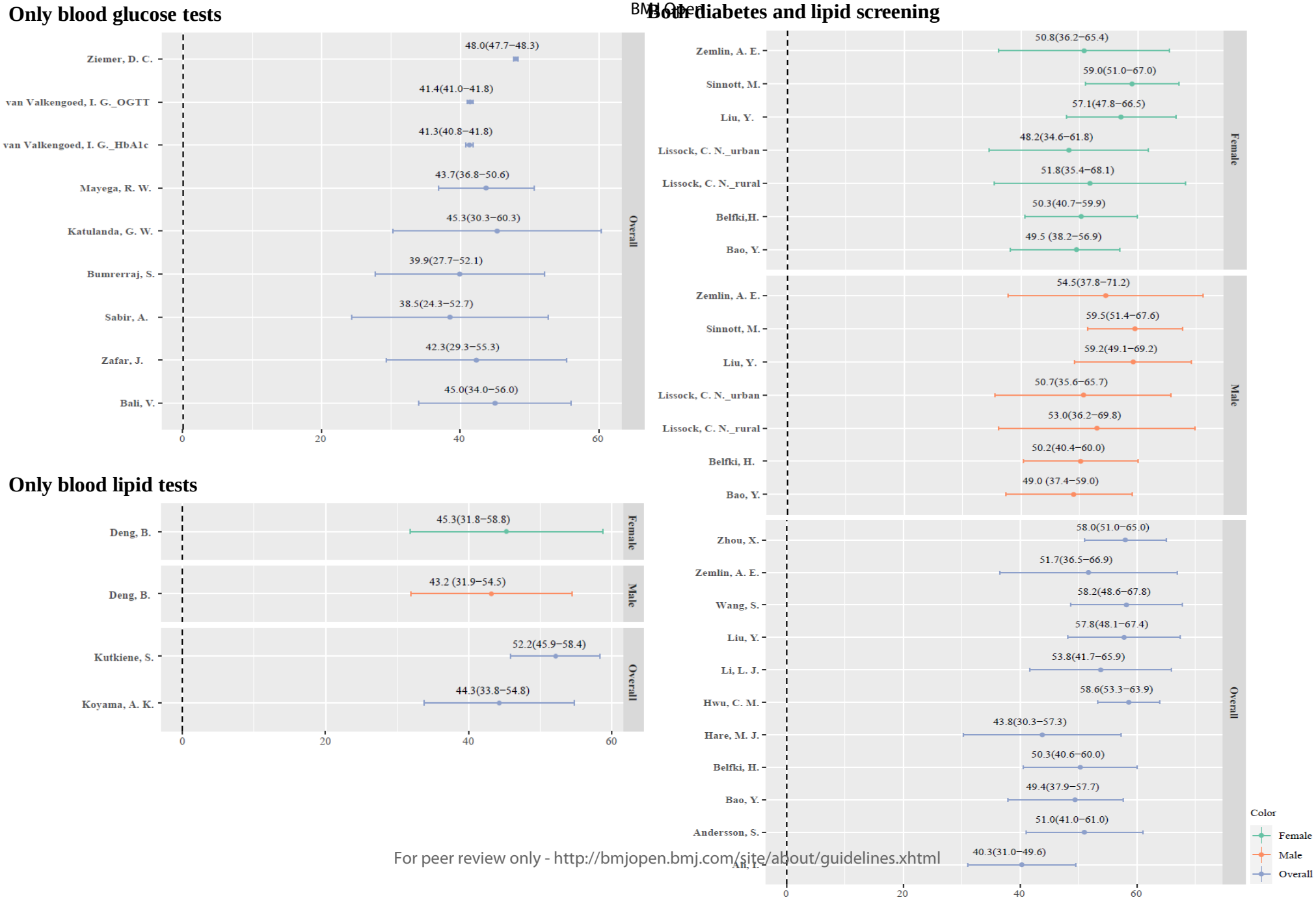
68 Magnani JW, Mujahid MS, Aronow HD, *et al.* Health Literacy and Cardiovascular Disease: Fundamental Relevance to Primary and Secondary Prevention: A Scientific Statement From the American Heart Association. Circulation. 2018;**138**:e48–74. doi:10.1161/CIR.0000000000000579

69 Ong WM, Chua SS, Ng CJ. Barriers and facilitators to self-monitoring of blood glucose in people with type 2 diabetes using insulin: A qualitative study. *Patient Prefer Adherence* 2014;**8**:237–46. doi:10.2147/PPA.S57567

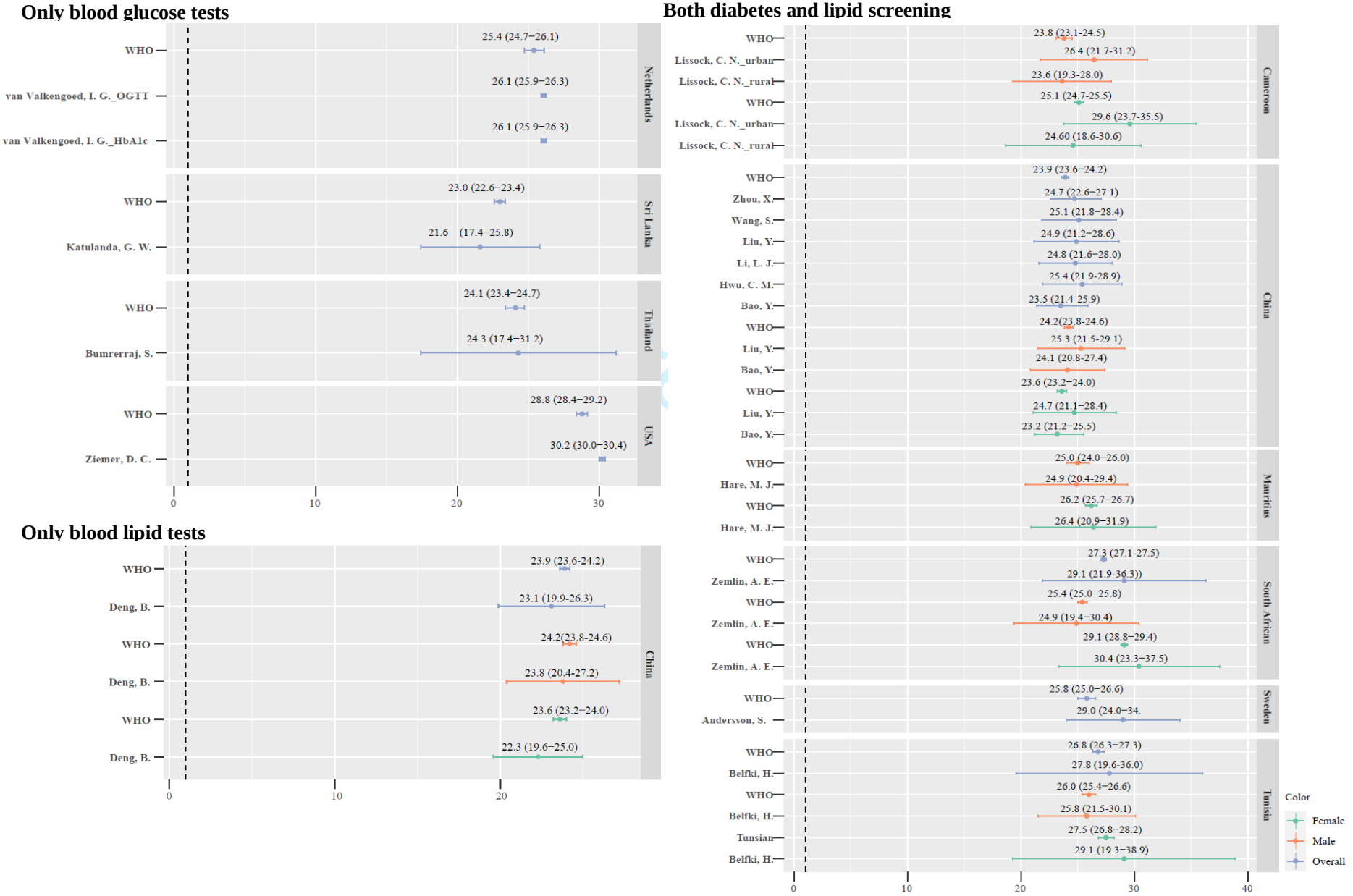
70 American Diabetes Association. Screening for diabetes. *Diabetes Care* 2002;**25**:s21–4. doi:10.2337/diacare.25.2007.s21

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml


Only blood glucose tests
WHO
van Valkengoed, I. G._OGTT
van Valkengoed, I. G._HbA1c
Katulanda, G. W.
Bumrerraj, S.
Ziemer, D. C.
Netherlands
Sri Lanka
Thailand
USA
25.4 (24.7−26.1)
26.1 (25.9−26.3)
26.1 (25.9−26.3)
23.0 (22.6−23.4)
21.6 (17.4−25.8)
24.1 (23.4−24.7)
24.3 (17.4−31.2)
28.8 (28.4−29.2)
30.2 (30.0−30.4)
0
10
20
30
Only blood lipid tests
Deng, B.
China
23.9 (23.6-24.2)
23.1 (19.9-26.3)
24.2(23.8-24.6)
23.8 (20.4-27.2)
23.6 (23.2−24.0)
22.3 (19.6−25.0)
0
10
20
Both diabetes and lipid screening
Lissock, C. N._urban
Lissock, C. N._rural
Cameroon
23.8 (23.1-24.5)
26.4 (21.7-31.2)
23.6 (19.3-28.0)
25.1 (24.7-25.5)
29.6 (23.7-35.5)
24.60 (18.6-30.6)
Zhou, X.
Wang, S.
Liu, Y.
Li, L. J.
Hwu, C. M.
Bao, Y.
China
23.9 (23.6−24.2)
24.7 (22.6−27.1)
25.1 (21.8−28.4)
24.9 (21.2−28.6)
24.8 (21.6−28.0)
25.4 (21.9-28.9)
23.5 (21.4-25.9)
24.2(23.8-24.6)
25.3 (21.5-29.1)
24.1 (20.8-27.4)
23.6 (23.2−24.0)
24.7 (21.1−28.4)
23.2 (21.2−25.5)
Hare, M. J.
Mauritius
25.0 (24.0−26.0)
24.9 (20.4-29.4)
26.2 (25.7−26.7)
26.4 (20.9−31.9)
Zemlin, A. E.
South African
27.3 (27.1-27.5)
29.1 (21.9-36.3))
25.4 (25.0−25.8)
24.9 (19.4−30.4)
29.1 (28.8−29.4)
30.4 (23.3−37.5)
Andersson, S.
Sweden
25.8 (25.0−26.6)
29.0 (24.0−34.
Belfki, H.
Tunsian
Tunisia
26.8 (26.3−27.3)
27.8 (19.6-36.0)
26.0 (25.4−26.6)
25.8 (21.5-30.1)
27.5 (26.8−28.2)
29.1 (19.3−38.9)
0
10
20
30
40
Color
Female
Male
Overall

Supplementary materials

Supplementary figure 1 Selection of articles for systematic review

Supplementary figure 2 The gender of screening participants

Supplementary table 1 Summary of diabetes and lipid screening guidelines

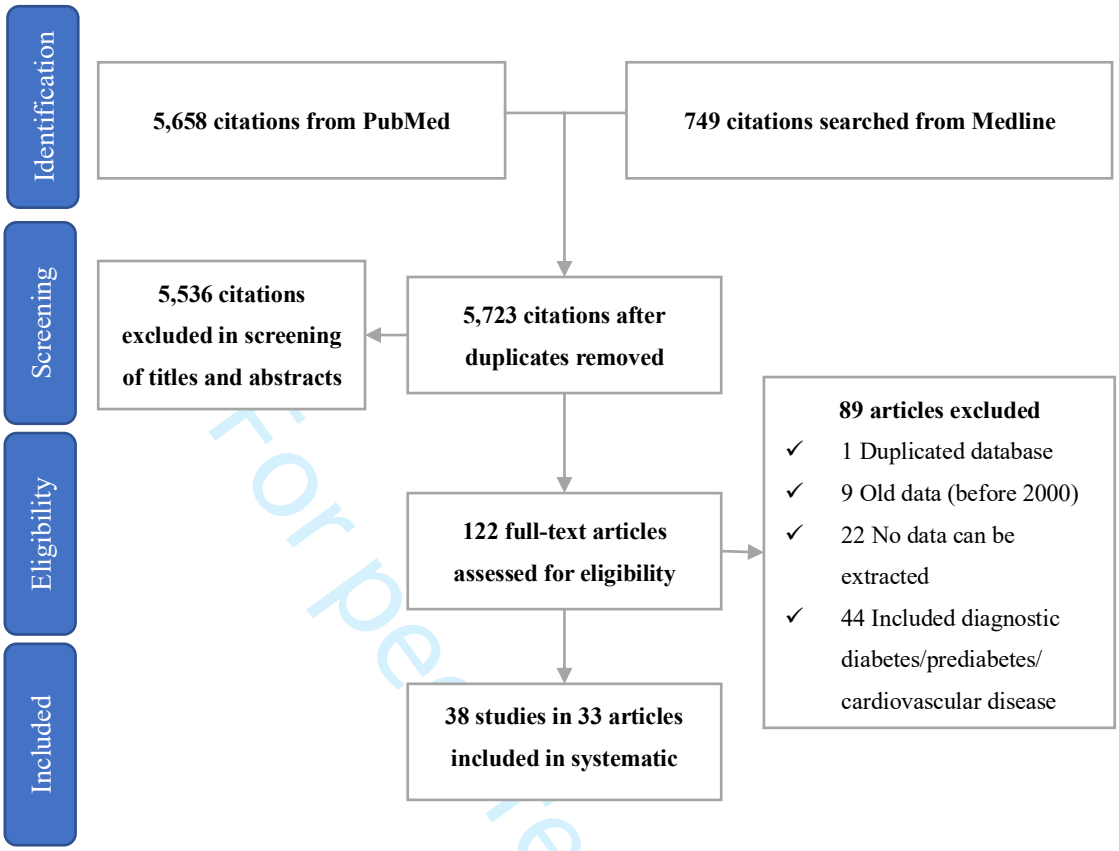
Supplementary table 2 Search strategy for article selection

Supplementary table 3 Quality Assessment of included articles based on Appraisal of Cross-sectional Studies (AXIS)

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

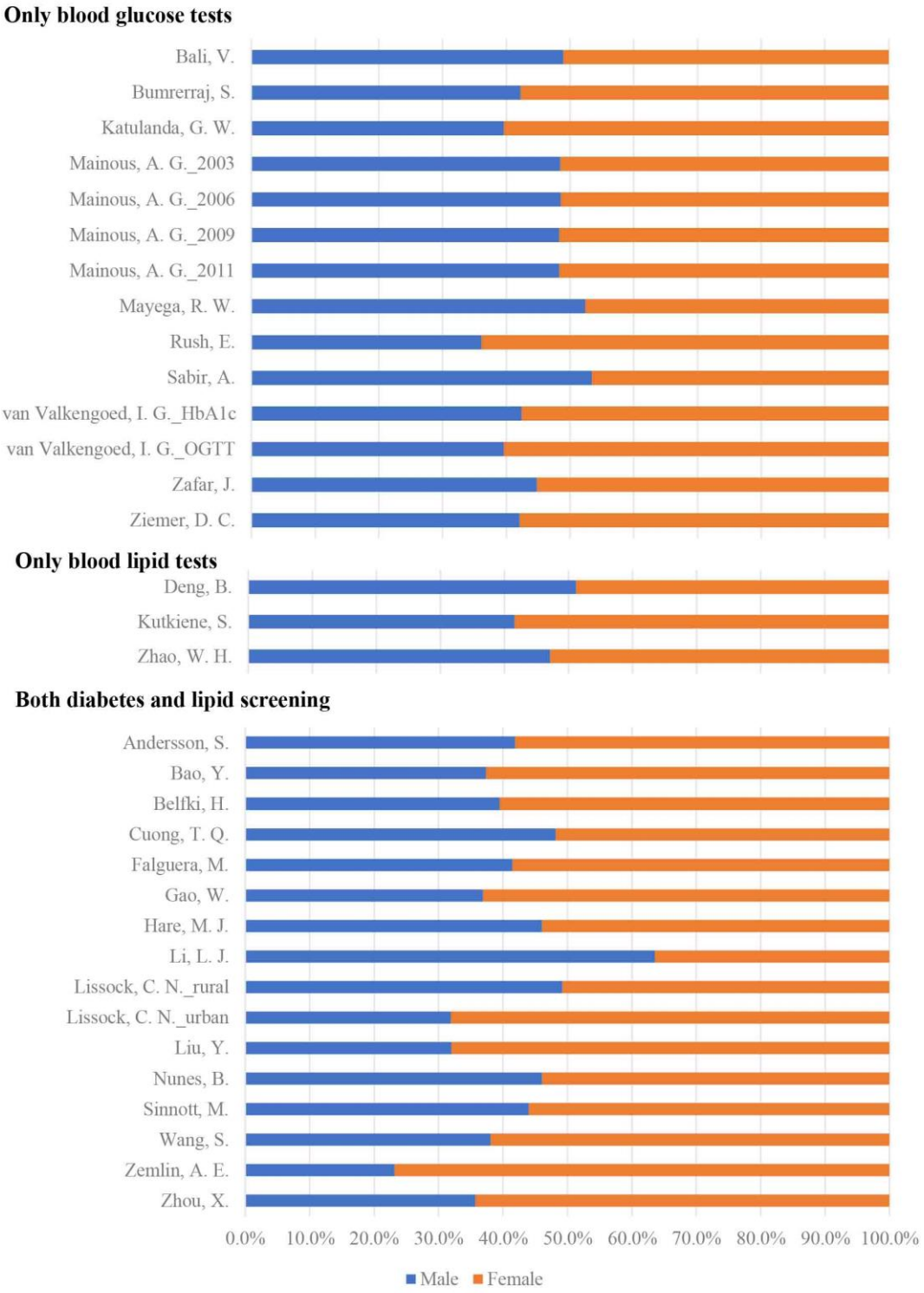
Supplementary figure 1 Selection of articles for systematic review



For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary figure 2 The gender of screening participants



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary table 1 Summary of diabetes and lipid screening guidelines

Country	Diabetes Screening Guidelines	Lipid Screening Guidelines
Australia	Patients should be screened for diabetes risk every three years from 40 years of age using the Australian type 2 diabetes risk assessment tool. Aboriginal and Torres Strait Islander peoples should be screened from 18 years of age. *(RACGP)*	Adults should have their blood lipids assessed every five years starting at 45 years of age. Aboriginal and Torres Strait Islander adults should have lipid tests performed every five years from 35 years of age. *(RACGP)*
Canada	Adults should be tested for type 2 diabetes every 3 years if they are age 40 years or over, or if they are at risk for T2DM. If they have risk factors for T2DM, they should be tested more frequently and/or start regular screening earlier. *(Canada Clinical Practice Guidelines Expert Committee)*	Initiating screening: In patients without CVD (primary prevention), we suggest lipid testing as part of global CVD risk estimation in men at age ≥40 years old and women at age ≥50 years old. Testing can be considered earlier for patients with known traditional CVD risk factors. *(College of Family Physicians of Canada)*
Singapore	Screening for T2DM should begin at 40 years and older; screening should also be considered in adults of any age who have one or more risk factors for diabetes. *(Singapore Ministry of Health)*	Clinicians should routinely screen men and women aged 40 years and older for lipid disorders, and younger adults for lipid disorders if they have other risk factors for coronary artery disease. *(Singapore Ministry of Health)*
United Kingdom	The UK National Screening Committee does not currently recommend screening for T2DM. This is because: there is no evidence that screening is more beneficial than not screening, and there is no evidence that finding T2DM early by screening results in a greater health benefit than usual health care.	National Institute for Health and Care Excellence guideline CG181 recommends (1.1.8) using the QRISK2 risk assessment tool (including ethnicity, age, sex, smoking, blood pressure, lipid profile, BMI, etc.) to assess CVD risk for the primary prevention of CVD in people up to and including age 84 years. There is also a potential need to amend recommendations 1.1.8 and 1.1.10 to advise the use of QRISK3 (including sex, age, BMI, lipid profile, blood pressure, family history, etc.) in place of QRISK2 because QRISK2 is due to be superseded by QRISK3 in 2018.
United States	Adults aged 40 to 70 who are overweight or obese are recommended to screen for abnormal blood glucose. (*USPSTF)*	Identification of dyslipidaemia and calculation of 10-year CVD event risk requires universal lipids screening in adults aged 40 to 75 years. *(USPSTF)*

RACGP: The Royal Australian College of General Practitioners, T2DM: type 2 diabetes mellitus, CVD: cardiovascular disease, USPTF: US preventive services task force recommendation statement

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary table 2 Search strategy for article selection

Database	Search strategy	Result
PubMed	(("fasting plasma glucose" OR "postprandial plasma glucose" OR "oral glucose tolerance" OR "random plasma glucose" OR "glycated haemoglobin" OR "HbA1c" OR "A1C") AND (screening* OR test*) AND (uptake OR participate) NOT (Paediatric OR gestational))	**5,658**
Ovid Medline	(fasting plasma glucose OR postprandial plasma glucose OR oral glucose tolerance OR random plasma glucose OR glycated haemoglobin OR HbA1c OR A1C) AND (screening* OR test*) AND (uptake OR participate) NOT (Paediatric OR gestational)	**749**

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary table 3 Quality Assessment of included articles based on Appraisal of Cross-sectional Studies (AXIS)

Author	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ali, I.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Andersson, S.	1	1	0	1	1	9	0	1	1	1	1	1	1	1	1	1	1	1	1	1
Bali, V.	1	1	0	1	9	0	0	1	1	0	1	1	0	0	1	1	1	1	9	9
Bao, Y.	1	1	0	1	0	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Belfki, H.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	9	1
Bumrerraj, S.	1	1	0	1	0	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1
Cuong, T. Q.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	9	1
Deng, B.	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	0	1	1
Falguera, M.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Gao, W.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	0	9	1
Hare, M. J.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Hidalgo, L. A.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Hwu, C. M.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Katulanda, G. W.	1	1	1	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Koyama, A. K.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	9	9
Kutkiene, S.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Li, L. J.	1	1	0	1	0	1	0	1	1	1	1	1	0	0	1	1	1	0	1	1
Lin, S.	1	1	0	1	1	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Lissock, C. N.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	0	1	1
Liu, Y.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Mainous, A. G.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	9
Mayega, R. W.	1	1	1	1	1	0	1	1	1	0	1	1	0	1	1	1	1	1	1	1
Nunes, B.	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Rush, E.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	0	1
Sabir, A.	1	1	0	1	0	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Sinnott, M.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	9	1
Valkengoed, I. G.	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1
Wang, S.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Zafar, J.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Zemlin, A. E.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Zhao, W. H.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	1	1
Zhou, X.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Ziemer, D. C.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	9

0: No, 1: Yes, 9: Don't know.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

MOOSE Checklist for Meta-analyses of Observational Studies

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	4-6
2	Hypothesis statement	-
3	Description of study outcome(s)	5-6
4	Type of exposure or intervention used	-
5	Type of study designs used	5-6
6	Study population	5-6
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	7,1
8	Search strategy, including time period included in the synthesis and key words	6-7, Supplementary table 2
9	Effort to include all available studies, including contact with authors	6-7
10	Databases and registries searched	6-8
11	Search software used, name and version, including special features used (eg, explosion)	8
12	Use of hand searching (eg, reference lists of obtained articles)	-
13	List of citations located and those excluded, including justification	8, Supplementary figure 1
14	Method of addressing articles published in languages other than English	-
15	Method of handling abstracts and unpublished studies	-
16	Description of any contact with authors	-
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	6-7
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	6-7
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	6-7
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	-
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	7
22	Assessment of heterogeneity	-
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	8
24	Provision of appropriate tables and graphics	Table 1-3, Figure1-2 Supplementary figure 1-2 Supplementary table 1-3
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	-

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

26	Table giving descriptive information for each study included	8-21, Table 1
27	Results of sensitivity testing (eg, subgroup analysis)	-
28	Indication of statistical uncertainty of findings	-

Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	-
30	Justification for exclusion (eg, exclusion of non-English language citations)	23-24
31	Assessment of quality of included studies	10-11, Supplementary table 3
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	21-23
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	24
34	Guidelines for future research	24
35	Disclosure of funding source	25

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

For peer review only